

In this data transfer is done as per patterns so there is a switch that will keep changing as per requirement and connect as per need like

We want 1 to 3 so at MUR it will connect with 1 and Demox will connected with 3rd slot and data will be transferred as per required

Main advantage is we will get rid of cross points.

Data-Link Layer

In case of physical link all links are logical

The internet is combination of networks.

If a packets is to travel from a host to another host it needs to pass through these networks

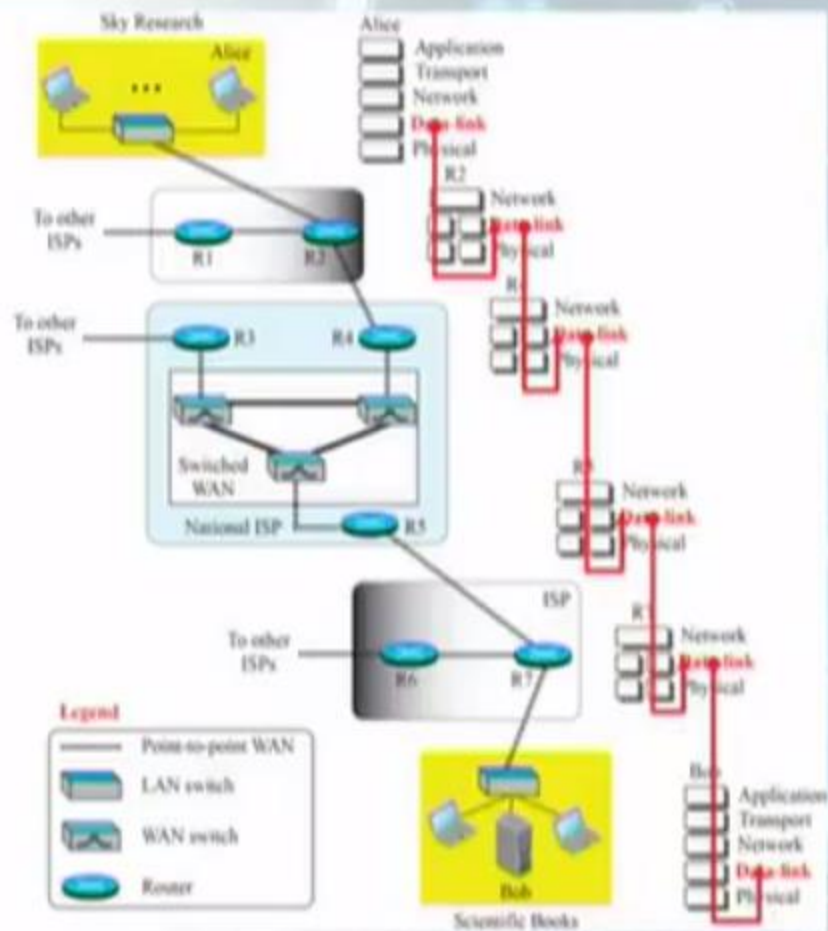
Data Link Layer controls Node-to-Node Communication

Data link exist btw physical layer and network layer.

It provide services to network layer and take services from physical layer.

Communication at the Data Link Layer

Communication at the Data-Link Layer



Let us have sending which is Alice. It will connect with the Data link and that data link will communicate with R2 and like these routers will communicate with each other with the help of data links.

Nodes and Links in Data Link Layer:

As in Data link layer communication is node to node.

Data unit from one point needs to pass through multiple networks (LAN and WANS) to reach another point or Host.

Hosts and in connecting devices are nodes and these nodes are connected through the links.

Hosts and all the connecting devices are nodes.

Links = Network that are connecting these nodes are links

Services That are provided by Data-Link Layer.

It provides services to network layer and receive from physical layers.

Services:

- Framing: (Data Link receive data in form of Packets, then First task of data link is to divide these packets into frames)
- Flow Control: (Transferring needs flow control to manage speed of data creation and data consumption)
- Error Control: It consist of Error Detection and then Error Correction
- Congestion Control: It happens at Transfer of the data.

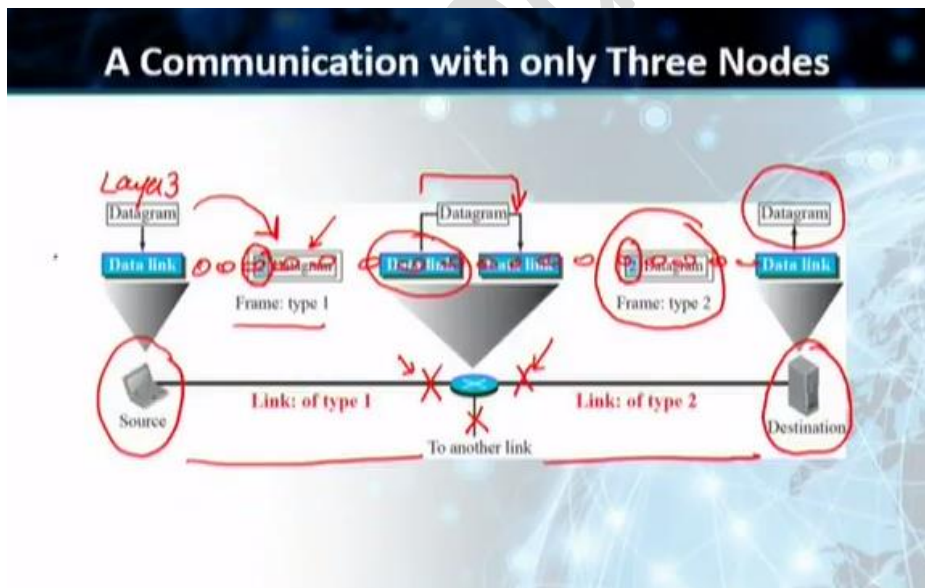
Process in Framing:

Encapsulation and Decapsulation

Data Link Receives data into packets and then it encapsulates into multiple frames. And then at receiving end it decapsulate the frames into packet and transfer it to next layer.

Example: A Communication with only Three Nodes.

Our Datagram or Frames do have Header which is 2 in the following photo



Flow Control:

As frames are sent then to manage amount of frames sent and amount of receive. To prevent overflow of frames.

Flow control once the buffer overflow it will waste all the overflowed frames but this method isn't recommended

Another method is: once the buffer get full it will stop sending the encapsulation or transfer until the buffer get level once its normal it will send notification to start transfer again.

Error Control:

When we send electromagnetic signals there will be loss of energy and that can cause error so we need to detect and fix that error which is done by error control method.

Categories of Links

Two nodes are physically connected by a transmission medium such as cable or air.

Data link layer control how these media are going to be used.

- Data Link can use whole capacity
- Or it can use part of whole capacity

Link Types:

1 – Point to Point Link

2 – Broadcast link

Point to Point link: If we made a phone call on telephone and we are talking and that time the link which is generated while we are talking is point to point link

Broadcast link: If the same communication is done face to face then the air is medium which is being used as transmission medium then that would be Broadcast Medium as small part of air is being used.

Two Sublayers of Data-Link Layer:

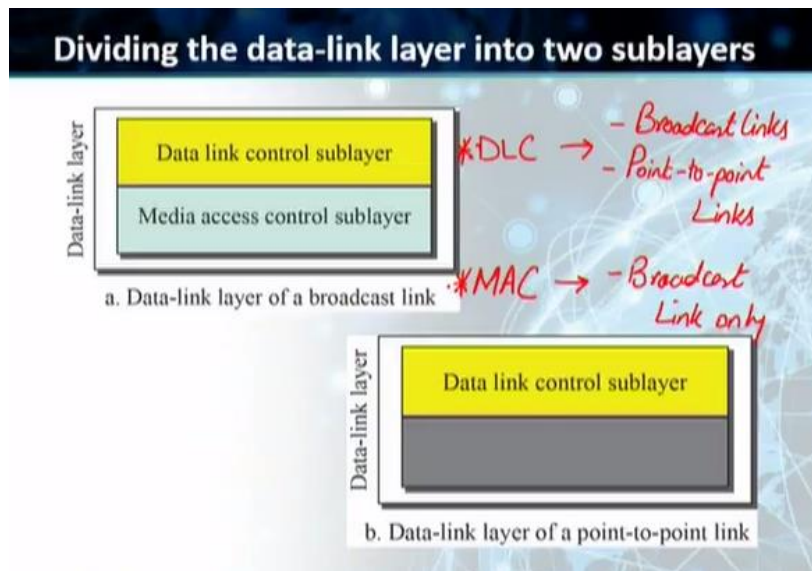
1 – Data Link Control (DLC)

2 – Media Access Control (MAC)

Here in photo we have data link layer of broadcast link and other one point to point link

DLC deals with the broadcast Links and Point to Point links as well.

MAC deals with the issues related to broadcast link only.



Link Layer Addressing:

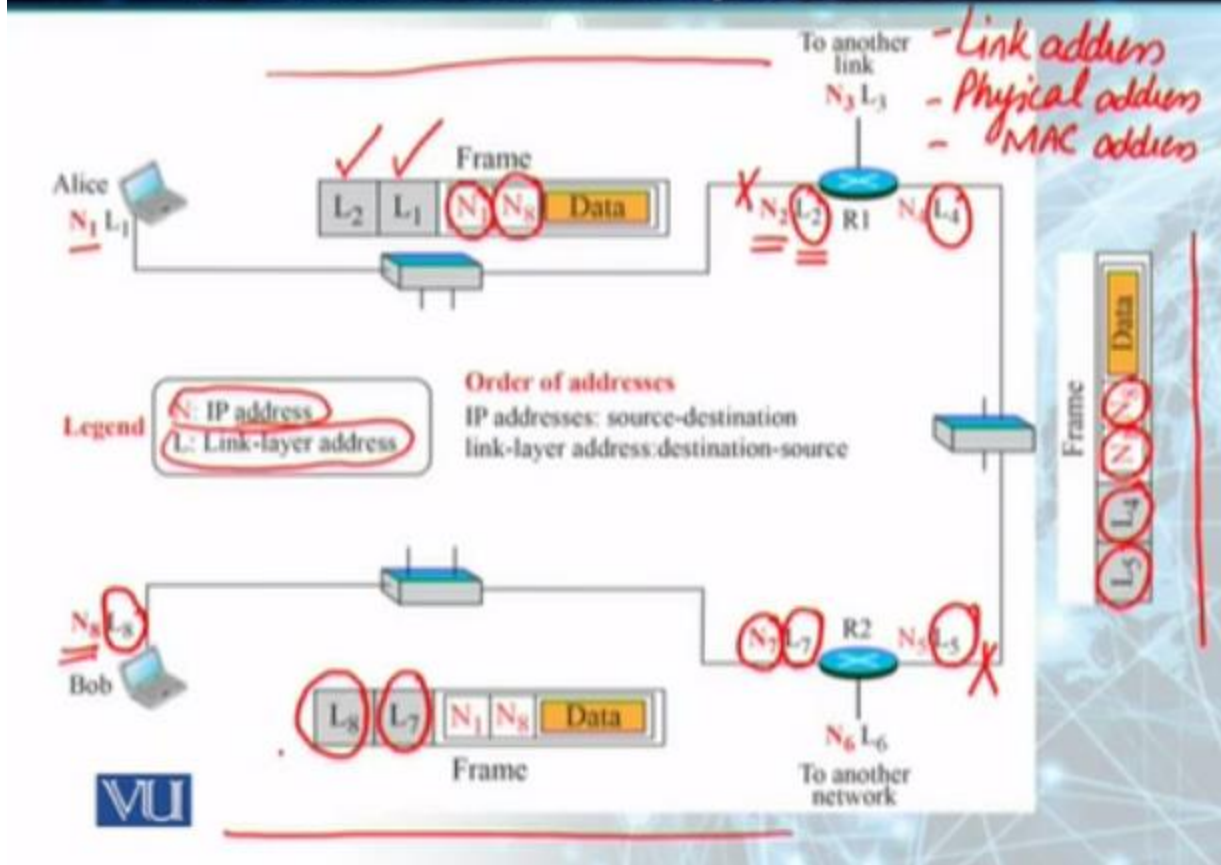
IP Addresses are the identifiers at the network layer (its layer 3 concept)

In internet we can't make packets to reach its destination with the help of IP Addresses only

It will only define the source and destination but can't define which links the packet will take.

Example:

IP addresses & Link-Layer Addresses



Link Layer Addresses also called Link Address, Physical Addresses, and MAC Addresses as well.

Here in example we have 3 links and 2 routers. Total nodes = 4

Sender is Alice and receiver is Bob

Alice IP is N and Link Layer address is Address

So every frame will have the source IP address and Destination IP address as well

Once the reach to the router it will have the Link Layer Addresses of source and destination. It will also have ip address to next link and that link will have addresses to further or receiving end data.

Finally, we will get link layer address and then data transfer takes place.

Why Link-Layer Addressing?

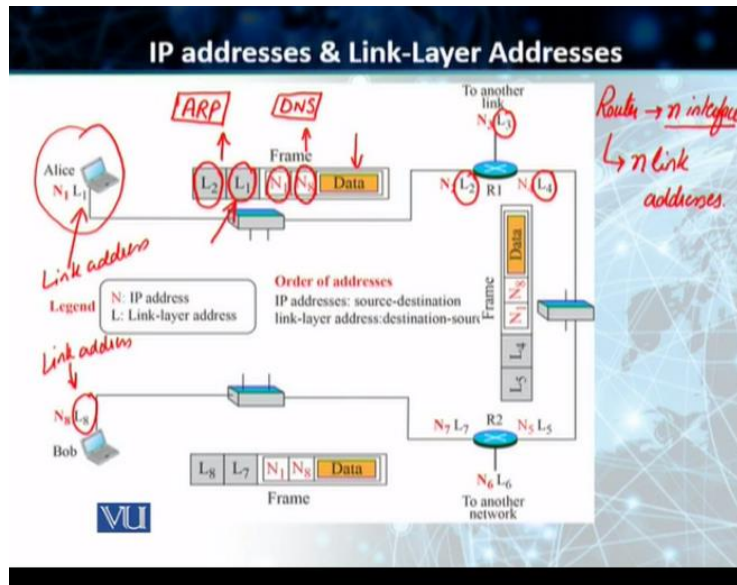
DNS is the reason how sender gets the IP address of receiver.

Sender will have its own Link Layer address How it will find next link layer address?

Ans: sender will get net Link Layer Addresses with ARP = Address Resolution protocol.

Q3: Sending and Reciving node will have link address but how routers get the links address?

Basically, the Routers which are connected trough the link address it will have physical layer addresses of all links that are connected with network and IP address as well.



Types of Addresses

1. Unicast
2. Multicast
3. Broadcast

Every host means nodes which include source hosts, destination host as well as connecting devices assigns **unicast address**. With **unicast** address we can do 1 to 1 communication.

Multicast Address: multicast means one to many. It is kind of broadcast but within the link.

Broadcast Address: When a host sends packets using broadcast address then that information is sent to everyone available on the network.

Example:

Unicast Example of Ethernet

Example

The unicast link-layer addresses in the most common LAN, Ethernet, are 48 bits (six bytes) that are presented as 12 hexadecimal digits separated by colons; for example, the following is a link-layer address of a computer. The second digit needs to be an odd number.

A3:34:45:11:92:F1

*48 bits
↳ 6 Bytes
12 Hex digits
separated by
colons*

VU

It consist of 48 bits = 6 bytes

Consist of 12 Hexadigital separated by colons

2nd digit of first address set should be odd number like here we have 3

Multicast Example:

Example

The multicast link-layer addresses in the most common LAN, Ethernet, are 48 bits (six bytes) that are presented as 12 hexadecimal digits separated by colons. The second digit, however, needs to be an even number in hexadecimal. The following shows a multicast address:

A2:34:45:11:92:F1

even → multicast address

It even has 48 bits and 6 bytes

And in this case we have 2nd digit is a even number

Broadcast:

It is 48 bits, all 1s are represented as 12 hexadecimals

Address Resolution Protocol (ARP):

Anytime when a node need to send something to another node source will have ip address of destination in a link.

Ip address is not enough to convey message

Ip address of the next node is not helpful in moving a frame through a link; we need the link layer address of the next node.

So in this situation ARP is needed.

Position of ARP is existing on the Network Layer.

ARP is axulary protocol. This protocol is converting IP address to Pysical Address a part of Data link layer.

ARP accept the IP Packet which is coming down from layer 3 then ARP convert that IP to Link-Layer Address and then this link layer address pass to Data link layer.

ARP Operation:

We have System A sender and System B reciver

ARP request is always broadcast

ARP Reply is always unicast

System got IP N1 and L1 want to send to N2 and L2

Looking for a link layer address of a node with IP Address N2

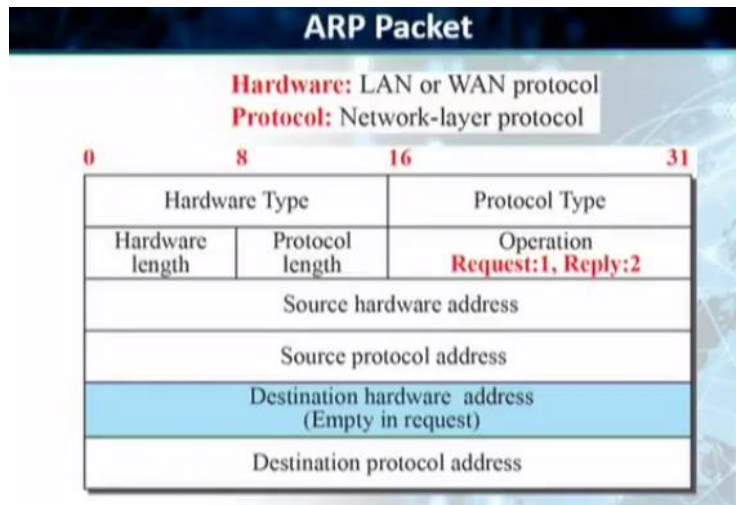
Request is sent as a broadcast to all links in the network to look for the N2

Once system B recive the request and it identify the ip address and it reply back directly back to System A and this ARP reply will consist of Physical Address of System B. This is how ARP works.

Caching: store data once its received this process is used to remove need of ARP request all the time so to prevent that caching keep the Physical address.

After caching all the communication will be unicast.

Structure of ARP Packet:



1 – Hardware Type Define the type of Ethernet Protocol

2 – We have protocol Type.

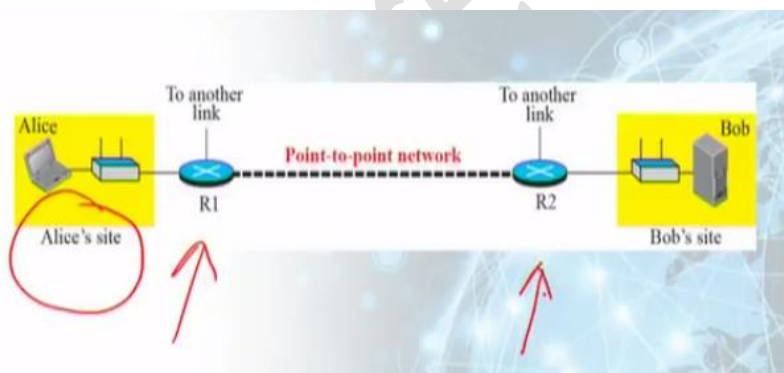
Then we will have Source Hardware address:

Source Protocol Address:

Destination hardware address

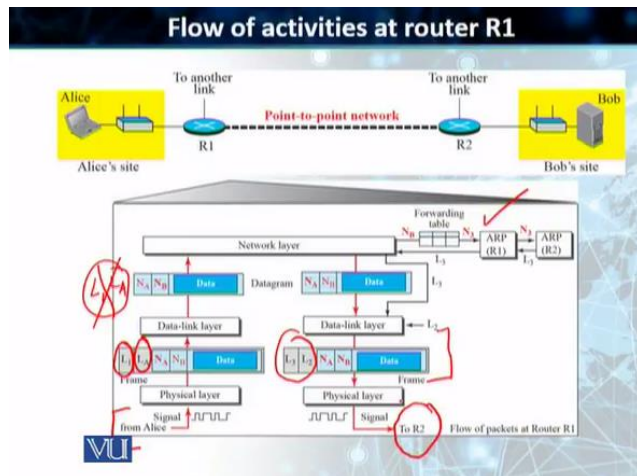
Destination protocol Address.

Example:



Flow of Packets

Data from Network layer comes down to Data link layer that time ARP request is generated.



Here we have R1 we have source and destination IP addresses and Link address L1 and LA(data link layer)

These Frames will go to Data link Layer and Decapsulation will happen that time L1 and LA will be removed

Data link will give it to Network Layer. Network Layer will encapsulate it and then ARP happens then it's again encapsulated at L2 and L3

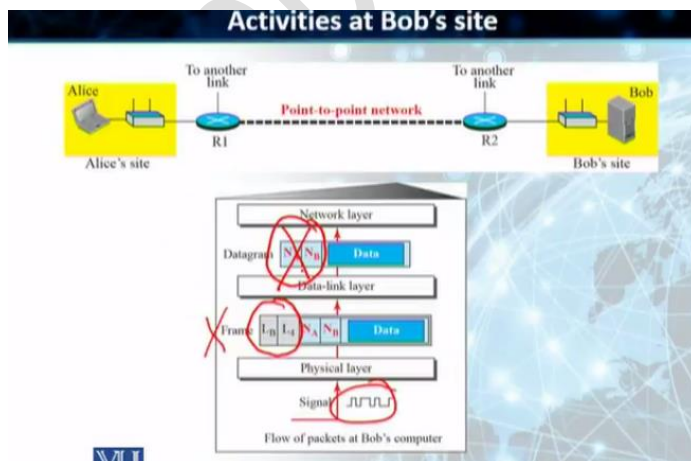
This was the process at Process R1

And the same process will happen for Router R2

And in this ARP for Bob will be used

At the BOB Site:

Signals are received it is encapsulated and then it is sent to Data Link Layer afterward it is forwarded to Network Layer where the header is removed



Types of Errors:

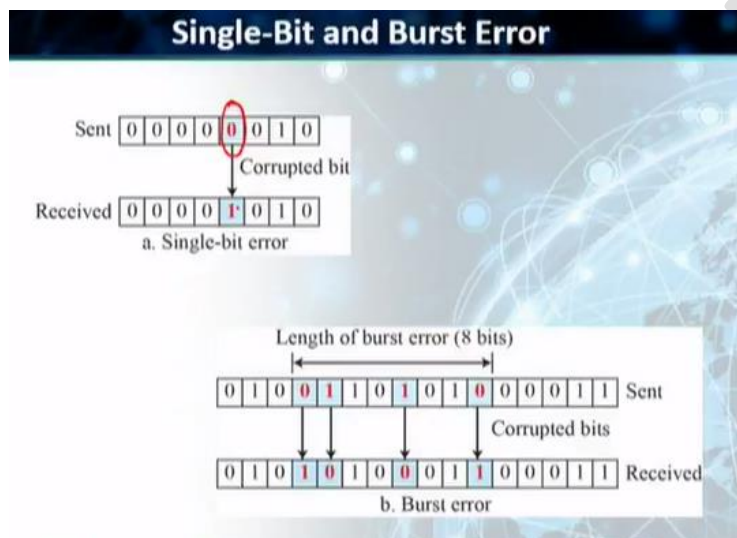
Data Transmission Suffers unpredictable changes because of Interference these interferences can change the shape of Signals.

Types of Errors:

1 – Single Bit Error: One bit is changed from the signal that is being transferred. In which one-bit difference is produced like 1 is changed to 0 or 0 to 1.

Burst Error: in which 2 or more than 2 bits are affected because of Interference.

Example of Single and Burst Error:

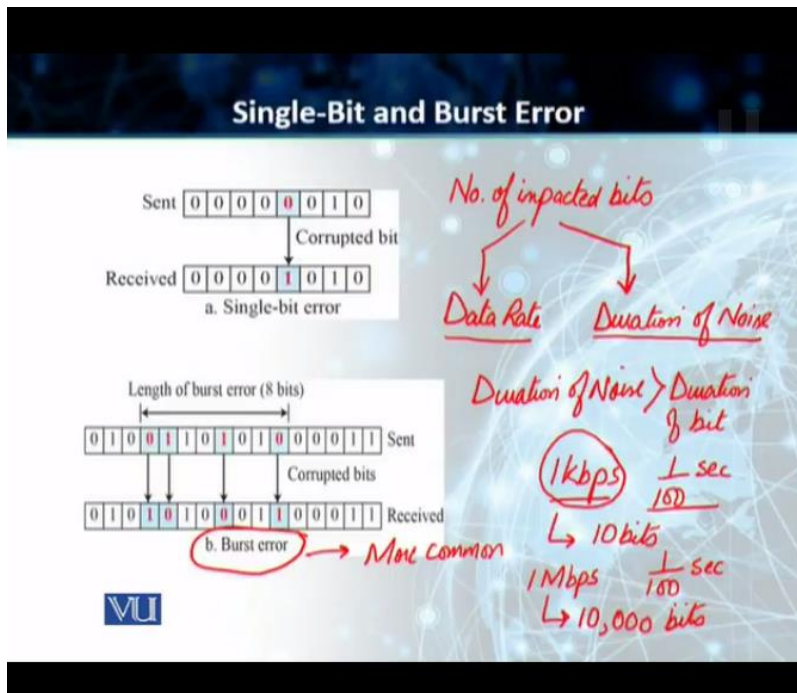


Length of Burst Error (8 bits)

No of bits impacted bits it depends on Data Rate and Duration of Noise

Higher the Data Rate more bits could impact and higher duration could impact higher number of bits

Duration of Noise is greater than duration of Bits



Here 1kbps at duration of 1/100 sec

Will get 10 bits

And 1 mbps with same duration

Will get 10,000 bits

Like data rate is increasing no of bits are getting impacted will increase.

Redundancy:

Central Concept in Detecting or correcting Errors is redundancy

To able to detect and correct errors we send some extra bit with our data.

Presence of these redundant bits allows the receiver to detect or correct corrupted bits.

Error Detection Versus Correction:

Error correction is more difficult than detection.

In error detection, we are only looking to see if any error has occurred or not.

In error detection we are not interested in the number of corrupted bits in detection

Single bit error is same as burst error.

Error Correction:

The Exact number of bits that are corrupted.

Location of corrupted bits in the message.

If we have message with 8 bits than we must know at which bit there is error

Coding:

Redundancy is achieved through various coding schemes

Coding allows us to add redundant bits at the sending end through a process that creates a relationship between redundant bits and the actual data bits.

The receiver checks the relationships between the two sets of bits to detect errors. If the result is appropriate after checking relationship then there is not error if its not appropriate then there is error.

The Ratio of redundant bits to data bits and robustness of the checking this process in which the coding scheme is depends upon.

Minimum redundant bit means more efficient coding scheme.

Types of Coding Schemes:

1. Block Coding
2. Convolution Coding (not normally used)

Block Coding:

In case block coding we divide our message into blocks into k bits each that is called datawords.

Then we add 'r' redundant bits to each block we get n bits which is equal to $n = k + r$

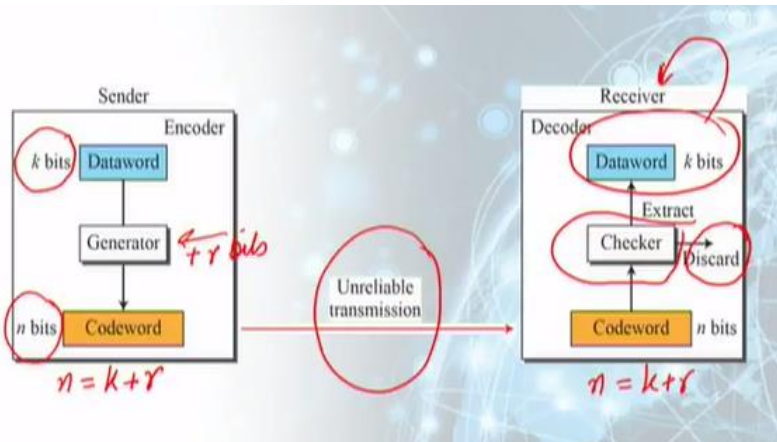
The resulting n-bit blocks are called codewords.

Block Coding in Error Detection:

Works if following conditions are met.

- If the Receiver has (or can find) a list of all valid codewords
- The original codework has changed to an invalid one.

Process of Error Detection in Block Coding



Here Sender divided message into k codewords

Then there is generator where we add r bits

At codeword we will have $n = k + r$

That will be sent to receiver codeword

Afterward that message will be sent to check and same as then to Receiver's Dataword

If there is a error and checker find it but still its somehow a valid code then checker will allow that piece of code if its matching a valid codeword.

Example of Block Coding:

Example

Let us assume that $k = 2$ and $n = 3$. Table below shows the list of datawords and codewords. Later, we will see how to derive a codeword from a dataword.

$$2^n = 2^3 = 8$$

$$2^k = 2^2 = 4$$

$$n = k + r$$

red bits

Datawords	Codewords	Datawords	Codewords
1). 00	000	3). 10	101
2). 01	011	4). 11	110

- ① 011 → valid codeword 01 → extracted
- ② 111 → Corrupted codeword 111 → discarded
- ③ 000 → Right (2 bits) are corrupted → Receiver accept as valid → 00



Here we got the value of $K = 2$ and $n = 3$

We need $n = K + r$

Here we have table with list of dataword and codewords.

2^n = possible number of data word

If receives 011 then it's a valid codeword if 01 is extracted by received then its passed to receiver.

If we get 011 but we get 111 corrupted codeword. As its not a codeword then it will not be processed by receiver's checker

If 2 bits are corrupted we get 000 instead of 011 but still there is codeword 000 so receiver will consider it as valid but it was corrupted yet it was present in our codeword list.

Harming Distance:

Hamming distance between two words of the same sized is the number of differences between the corresponding bits of these words

Lets consider two words x and y is $d(x,y)$

Hamming distance btw received codeword and send codeword is number of bits corrupted.

How to Find Hamming distance

Example

Let us find the Hamming distance between two pairs of words.

1. $d(000, 011) \rightarrow d(x, y) = d(000, 011) = 2$

2. $d(10101, 11110) \rightarrow d(x, y) = d(10101, 11110) = 3$

Here we have 2 pairs

Both have x and y

We will compare both x and y to find hamming distance

Like: 1 $d(000, 011) \rightarrow = 2$

There is difference of two values mean two values 2nd and 3rd values are changing.

Hamming distance tells us how many bits are corrupted

If resultant isn't equal to 0 than some bits are corrupted.

Example

Let us find the Hamming distance between two pairs of words.

1. $d(000, 011) \rightarrow d(x, y) = d(000, 011) = 2$

2. $d(10101, 11110) \rightarrow d(x, y) = d(10101, 11110) = 3$

$d(x, y) \neq 0 \rightarrow$ corrupted (error)

XOR $\oplus$ $000 \oplus 011 \rightarrow$ Count no. of 1s in result $\rightarrow 2$

Minimum Hamming Distance:

The smallest hamming distance between all possible pairs of codewords

$$D_{\min} = s + 1$$

$S \rightarrow$ no of detectable errors

$d_{\min}$ = is minimum hamming distance

Example

Example

A code scheme has a Hamming distance $d_{\min} = 4$. This code guarantees the detection of up to how many errors?

$d_{\min} = 4$
 $d_{\min} = S + 1$
 $S = d_{\min} - 1$
 $= 4 - 1$
 $S = 3$

If block code have minimum hamming distance of 4 then it grants error detection for upto 3 bits in codewords.

Linear Block Codes:

It is the subset of block codes

In which the If we do Exclusive OR of two valid codewords then it will creates another 3rd valid codeword.

Example:

Example

The code below is a linear block code because the result of XORing any codeword with any other codeword is a valid codeword. For example, the XORing of the second and third codewords creates the fourth one.

Datawords	Codewords	Datawords	Codewords
00	000	10	101
01	011	11	110

$011 \oplus 101 = 110 \rightarrow$ Linear block code

The numbers of 1s in the nonzero codewords are 2, 2, and 2. So the minimum Hamming distance is $d_{\min} = 2$.

$d_{\min} = 2$

Example of Linear block code

We can check by doing Exclusive Or then we will get 3rd term

Minimum number of 1s in all non zero codeword are 2,2 and 2, so Minimum hamming distance is 2

Parity Check Code:

Most common Error detecting code

The Extra parity bit is selected in order to make total number of 1s in even.

If total number is already even then extra parity bit will be zero.

Simple Parity Check Code: C(5,4)

Simple parity-check code C(5, 4)

$2^4 = 16$ $2^5 = 32$ $k=4, n=5$

Datawords	Codewords	Datawords	Codewords
0000	00000	1000	10001
0001	00010	1001	10010
0010	00100	1010	10100
0011	00110	1011	10111
0100	01000	1100	11000
0101	01010	1101	11011
0110	01100	1110	11101
0111	01111	1111	11110

$n > k \rightarrow \text{no. of codewords} > \text{no. of datawords}$
 $2^n - 2^k \rightarrow (\text{codewords}) \rightarrow \text{invalid}$

Here number of codes $k = 4$ we added an extra parity bit then n will be equal to 5

As in first case we have $k = 4$ so total number of data combination will be 16 as $2^4 = 16 = \text{dataword}$

So as $n^5 = 32$

N (codewords) are always greater than K (datawords)

In first case there is no one so we zero

For 2^{nd} there is 1 we add 1

Where we have 2 (1's) that mean its even so we will add zero

When there is only 1 (1's) then we will add 1

We can see in above example clearly

Encoder and Decoder for Simple Parity check Code

R_0 = Total 1's is even

Parity bit is added with the help of generator.

On the receiving side

Let suppose transmission is corrupt receiver will receive 5 bits corrupt

Checker will do same thing that is done by generator

Syndrome: If we do modulation over 5 bits that we have received at the receiving end then we get syndrome which is one bit.

Syndrome is zero when no of 1's is even

Syndrome is 1 when no of 1s is odd

Syndrome = S_0

Syndrome is passed to next step if syndrome is zero then there is no error if its non zero then value will be discard.

Cyclic Codes:

Here we have linear block code with additional property

If a codeword is shifted in a cyclic manner (rotated) the result will be another code word.

Rotation of codeword result another valid codeword

As we have code 1011000 is a codeword

Here we are cyclic shifting 1st digit to the last then we get 0110001 it's a codeword as well.

Cyclic Redundancy Check CRC

It is a subset of Cyclic codes.

It is used in Local areas network and WANs

A CRC Code with C(7,4)

A CRC code with C(7, 4)

$n-k=3$ $k=4, n=7$

Dataword	Codeword	Dataword	Codeword
0000	0000000	1000	1000101
0001	0001011	1001	1001110
0010	0010110	1010	1010011
0011	0011101	1011	1011000
0100	0100111	1100	1100010
0101	0101100	1101	1101001
0110	0110001	1110	1110100
0111	0111010	1111	1111111

$K = 4, n = 7$

Amount of bit to add is calculated by $n-k$

CRC Encoder and Decoder:

Let we have $k = 4$

And Codeword is equal to $n = 7$

So as $n-k = 3$

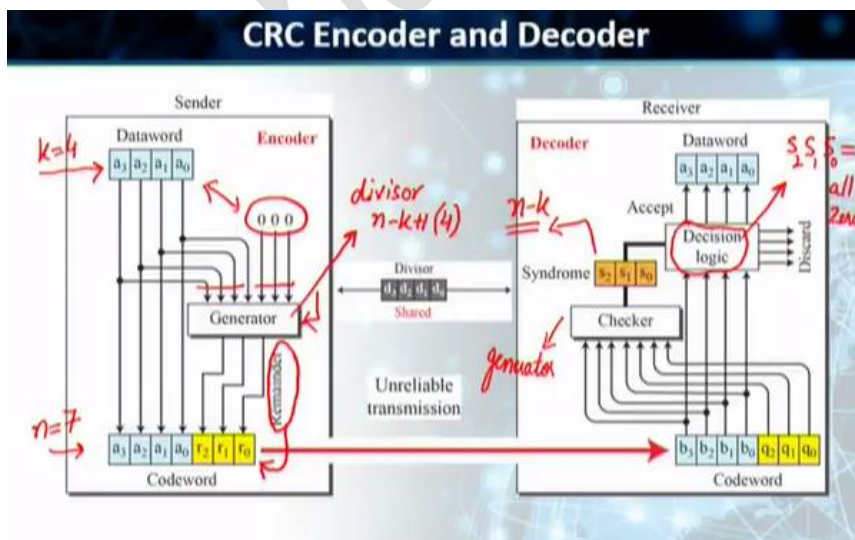
So we will fed 3 zeros in our generator. And generator will use divisor which got a size of $N-k+1$

And generator will add 3 bit reminder to our codeword.

On the receiving side let suppose we got a corrupted code

We will put code in checker and the result by checker is syndrome and it will fed out the bits.

As we know if all vales of syndrome is equal to zero then its acceptable otherwise we will discard it.



Cyclic Code Analysis using Polynomials

Let suppose we have Dataword = $d(x)$, Codeword = $c(x)$, Generator = $g(x)$, Syndrome = $S(x)$, error = $e(x)$

We saw two situations

When $S(x)$ is not equal to zero $\rightarrow$ data is corrupted $\rightarrow$ discard:

If $S(x) = 0 \rightarrow$ No Bit is corrupted And another situation

There is corruption $\rightarrow$ decoder has not detected it.

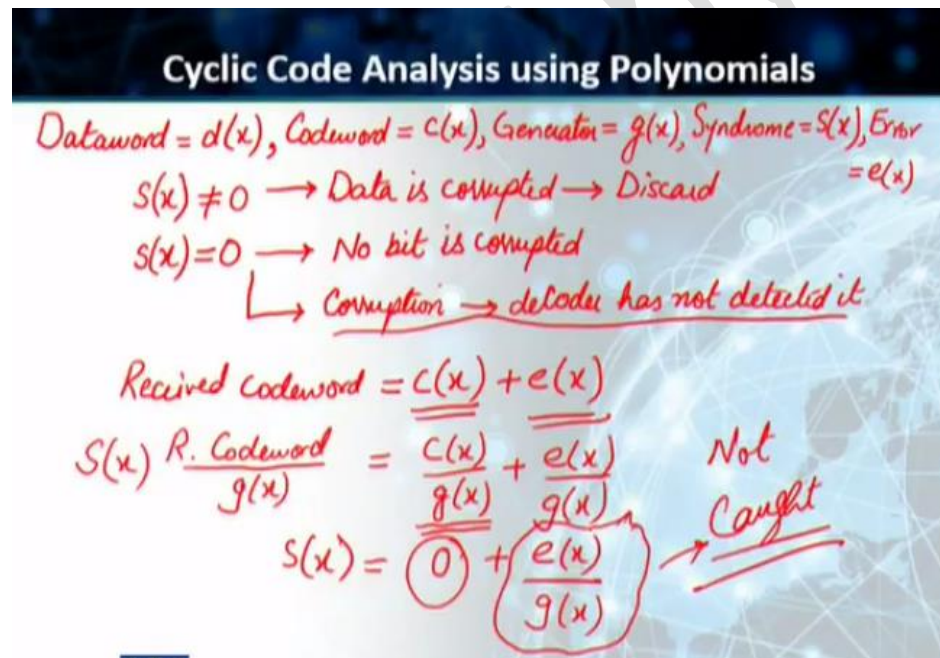
Let we received codeword = $c(x) + e(x)$ (with the error that is introduced)

We will divide term with $G(x)$

= $c(x)/g(x) + e(x)/g(x)$ which is equal to $S(x)$

= $0 + e(x)/g(x)$

Value of codeword is got zero but its not wo worry about but $e(x)/g(x)$ which are error that are not caught.



Cyclic Code Analysis using Polynomials

Dataword = $d(x)$, Codeword = $c(x)$, Generator = $g(x)$, Syndrome = $S(x)$, Error = $e(x)$

$S(x) \neq 0 \rightarrow$ Data is corrupted $\rightarrow$ Discard

$S(x) = 0 \rightarrow$ No bit is corrupted

$\rightarrow$ Corruption $\rightarrow$ decoder has not detected it.

Received Codeword = $\underline{c(x)} + \underline{e(x)}$

$S(x) \frac{R. \text{ Codeword}}{g(x)} = \frac{\underline{c(x)}}{g(x)} + \frac{\underline{e(x)}}{g(x)}$

$S(x) = \underline{0} + \frac{e(x)}{g(x)} \rightarrow$ Not Caught

Advantage:

Good Performance in Detection

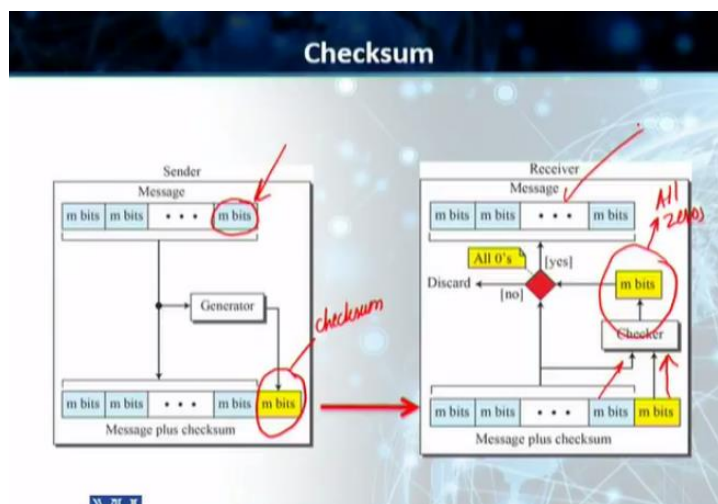
- Single Bit Error

- Double Error
- Odd Numbers of Error
- Burst Error
- Easy Implementation
- Fast Implementation

Checksum:

It is a error detection technique which can be applied to a message of any length.

It the mostly deal with network and transport layer rather than the data-link layer.



In Sender we got M bits

And generator creates same number of m bits (checksum)

Then we are sending checksum to the receiving end

And their Checker will create another M-Bit checksum

If the new checksum is all zero then message is acceptable else is rejected.

Example:

we got message with a list of 5 4-bits numbers we want to send it to destination.

In addition to sending these numbers we need to send sum of the numbers as well.

Example

Suppose the message is a list of five 4-bit numbers that we want to send to a destination. In addition to sending these numbers, we send the sum of the numbers.

Set of numbers is (7, 11, 12, 0, 6)

$$7+11+12+0+6 = 36$$

$$\begin{array}{c} (7, 11, 12, 0, 6, 36) \\ \text{4-bit word (each < 15)} \\ \text{36} \end{array}$$

Let we add all numbers which we have

$$7+11+12+0+16 = 36$$

What we need to send is = $7+11+12+0+6+36$

All 5 numbers can be written in 4 bits number as its less than 15

But our checksum can't be written in it.

Example:

Here the 36 is written in the binary

We add the extra left most bit to the right four bits as shown in figure.

We write number in 2 forms

We will have two values by splitting it.

$$(10) + (0100) = (0110)$$

$$= 6$$

Receiver will add the result in one's complement if the result is 6 then code is acceptable else rejected

Forward Error Correction:

Retransmission of Corrupted and lost packets is not usefull all the time mostly when its case for any multimedia (audio or video)

For this situation we need to correct error and reproduce it immediately.

Several techniques developed are commonly called forward error correction techniques that will be used to fix error immediately

Using Hamming Distance:

1st technique to fix error immediately

For correction we need bigger hamming distance

So we will get

$$D_{min} = 2t + 1$$

So if we want to fix 10 bits then we have to make minimum hamming distance of 21 bits

$$2(10) + 1 = 21 \text{ (number of redundant bits we will send with our dataword)}$$

Lots of redundant bits need to be sent.

Using Hamming Distance

If we want to correct 10 bits in a packet, we need to make the minimum hamming distance 21 bits

$$d_{min} = 2t + 1$$
$$d_{min} = 2(10) + 1$$
$$\underline{d_{min} = 21} \rightarrow \text{no. of bits (red) that you will send with dataword}$$

Another example:

If our data is 99 bits $\rightarrow$ 255 bits and we need to send 156 bits and it will fix only 23 bits

That is BCH

2nd Technique USING XOR

We can fix the error by using XOR operations as shown in figure

Using XOR

Another recommendation is to use the property of the exclusive OR operation as shown below.

$R = P_1 \oplus P_2 \oplus \dots \oplus P_i \oplus \dots \oplus P_N$

$P_i = P_1 \oplus P_2 \oplus \dots \oplus R \oplus \dots \oplus P_N$

$N = 4 \rightarrow 25\% \text{ extra data send.}$

$N \text{ chunks} \rightarrow N+1 \text{ chunks are sent.}$

$XOR \text{ (one out of 4 chunks)}$



Let we have Data $\rightarrow p_1$ to P_n

We have $P_i = P_1 \oplus P_2 \oplus \dots \oplus R \oplus \dots \oplus P_N$

Our overall packet will be divided into N chunks

Let if we have $N = 4$

Means we need to add 25% extra data in this technique of XOR we will be able to fix one out of 4 chunks.

Chunk Interleaving

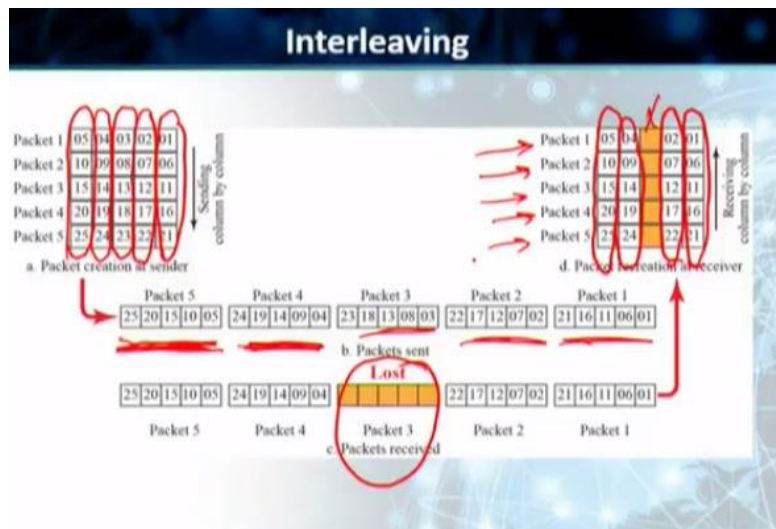
Another way to Achieve FEC in multimedia is to allow some amount of chunks in every packet that won't put effect and that could be rebuild at receive end so we can miss them.

We can't afford all chunks to be missing but yet we can afford one chunk be missing in each packet.

Let we have list of packets at sending end and we are sending it horizontally are sent

On a receiving end we lost 1 packet

And other packets will take their place and will keep space empty for the packet which is lost so whole packet isn't loosed only a chunk it loosed.



Combination of Hamming Distance and Interleaving

Another technique

Here in this technique

We first create n-bits packets that can correct T-bits error

Then we interleave m rows and send the bits by column by column

To correct Burst error we will multiply $M \times T$

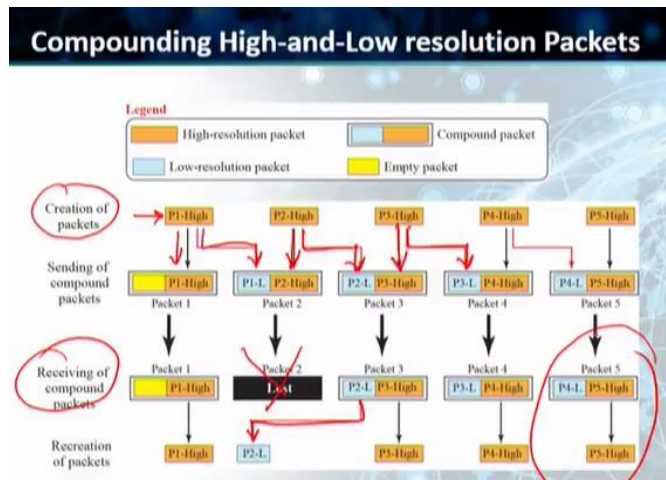
Compounding High- and Low-Resolution Packets:

In this Technique

We create a low resolution redundancy version of each packet and will send it with next packet.

Like 1 is send then low res of packet one will send with packet 2

As detailed shown in picture below



So on the receiving end if we lose any packet we can use low resolution packet to recreate high packet.

In this technique we can't fix last packet coz this is not low resolution packet send after that so expect this one we can fix all other packets.

Data Link Control (DLC) Services:

As data link is divided into two sub layers so 1 is DLC or Data link control other one is MAC

DLC is responsible for communication between two adjacent nodes whether the link is dedicated or broadcast

Functions DLC controls

Framing

Flow Control

Error Control

DLC Services:

DLC available for Broadcast link and point to point while MAC isn't

Framing:

As Data link send packets in the form of packets and all packets are distinguishable from another.

Our postal system practices a type of framing. (when letter are separated there is a separate address for each letter then its sent to its specified address).

Framing separates a message by adding sender address and the destination as well.

Frame Size:

Why not one Big Frame?

Why we do chunking?

We can create a bit frame but even if we had 1 bit error that small error will make us to discard whole frame.

So chunking is done to make less to less part to be discarded

Frames can be fixed in size:

Size of frame acts as a boundary/delimiter. I.e we got a 16 bits frame means all frames will be sent of 16 bits each

All receiver should see that it received 16 bits only not less not greater.

Variable Size:

Is most commonly used in Data Communication

How to Define Beginning and End of a Frame?

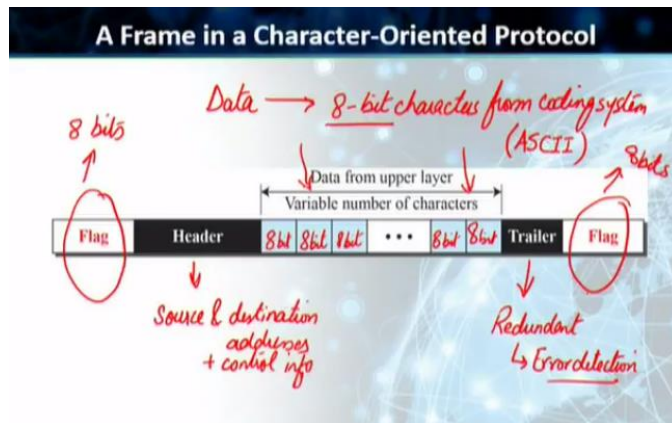
Variable Framing Techniques:

1 – Connection Oriented Framing: Also known as Byte Oriented Framing.

2 – Bit Oriented Framing:

Variable framing Techniques are famous in LANs

Frame in a Character Oriented Protocol:



The data in this are made up of only 8-bit character from any coding system (ASCII)

Here we got two flags one in the beginning and at the end as well. To tell receiver start and end of frame.

Then Header comes that consist of Source and Destination addresses, Control Info.

Then There is Trailer. That carries some error detection redundant bits that can perform correction as well.

Header and Trailer both are 8 bits each

Flags are protocol Dependent.

All variable numbers of characters are 8 bit each

Byte Stuffing in Connection-Oriented Framing:

Byte Stuffing means inserting a byte.

Initially connection oriented framing was used as data format text only

As we used flags in the start and ending and text is used to that time unused character in text we used to take out those characters and utilize those text in Flags it used in past times.

Now adays

Text is not only form is sent we are sending everything like pics, audio, videos etc...

Character used for flags can be part of data.

Inorder to avoid this confusing situation for reciver we do Byte Stuffing.

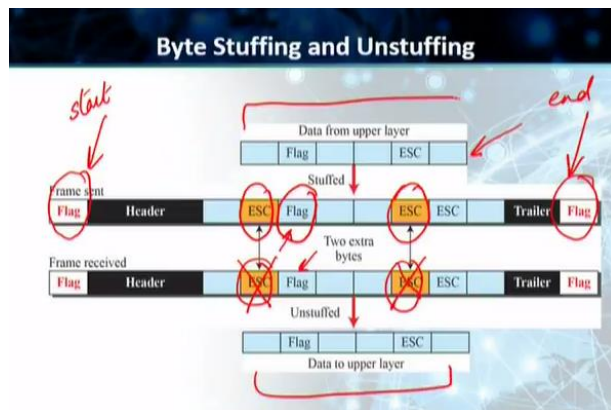
We add one or more escape character, as flag is being repated in main data when a flag is repated then we stuff a particular part of data. And we call it as escape character

There is issue with this. One or more escape characters followed by a byte with same pattern as a flag.

For 8 Bits Character

We stuff dat with a pre-defined Escape character (Byte) as a flag that prevent this problem from happening

Byte Stuffing and Unstuffing.



Byte stuffing is done at sender and unstuff at receiver.

There is two flag in the start and the end but when a flag comes within data we add a Escape byte/character that tells receiver that this flag isn't start or end of data its part of data.

That extra flag is removed at the receiving end and we get complete data.

Bit-Oriented Framing

Character oriented uses 8 bit or 1 byte only.

In Bit oriented Framing our data section of frames consist of a sequence of bits.

We need a boundry/delimiter to show where does the frame start and where does it end

A special 8 Bit pattern 01111110 to define the begging and end of a frame

Same issue as connection-oriented framing.

A Frame in Bit Oriented Protocol:

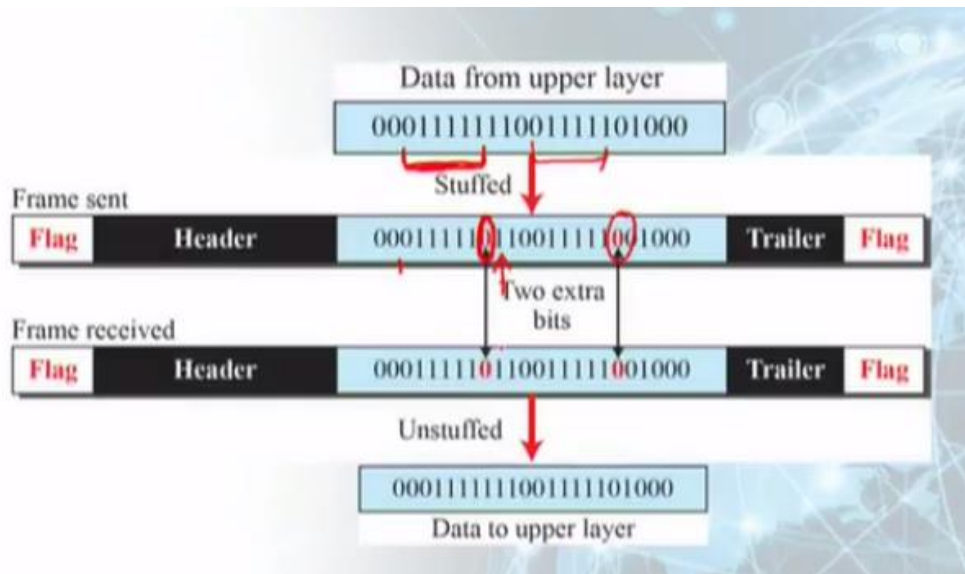
We got flag In front and last with pattern of 01111110 in the start and end.

If this flag pattern somehow comes within data we need to inform here as well that its not end of data.

That information can be done with the help of bit stuffing technique.

In this technique if 5 consecutive 1's comes after zero then we use bit stuffing technique

When 5 1's come after zero than we add 1 zero after 5 1s to show that its not a flag that zero will be removed at the receiving end.



Flow and Error Control:

these are sub layers of data link.

Flow Control:

It maintain the balance btw production and consumption rates

When frames are produced faster than they are consumed at the receiving data link layer the frames will be discarded

If we don't keep flow control then and sender will continuously producing data then receiver will keep on discarding frames away which would be usefull.

Flow control can be done by the use of small pockets that is called buffers.

Buffers are at both sending and receiving end.

Flow Control at the Data Link Layer.

Flow control is used to maintain btw producer and reciver to make sure what is produced it wouldn't have wastage.

Buffers to control flow.

Once buffer is full it will inform producer to slow down the production until remaining are utilized then it will inform producer to start producing again

Buffer will inform when it's full and when there are vacancies

Error Control:

Error control at Data link uses CRC

We know in cyclic we add redundant bits to make sure whether data is corrupt or not.

Scenario with CRC

- If data is good it moves forward, if it's corrupted it will be discarded
- If frame is corrupted it will be discarded and if it's good it will move forward and an acknowledgement is sent to the sender.

Connectionless and Connection Oriented Protocols

A Data link control protocol can be either connectionless or connection oriented.

There is no relationship btw frames in connectionless. All frames are independent.

In Connection-Oriented: Frames are numbered and sent in the order. It will do setup, transfer and connection teardown.

We will combine flow and error control:

For our process to tell sender whether the flow is completed and it's error free.

Data Link Layer Protocols:

Four protocols have been defined for the data-link layer to deal with flow and error control

- Simple Protocol
- Stop and Wait Protocol
- Go-Back-N Protocol (Rarely used nowadays)
- Selective-Repeat Protocol (Rarely used)

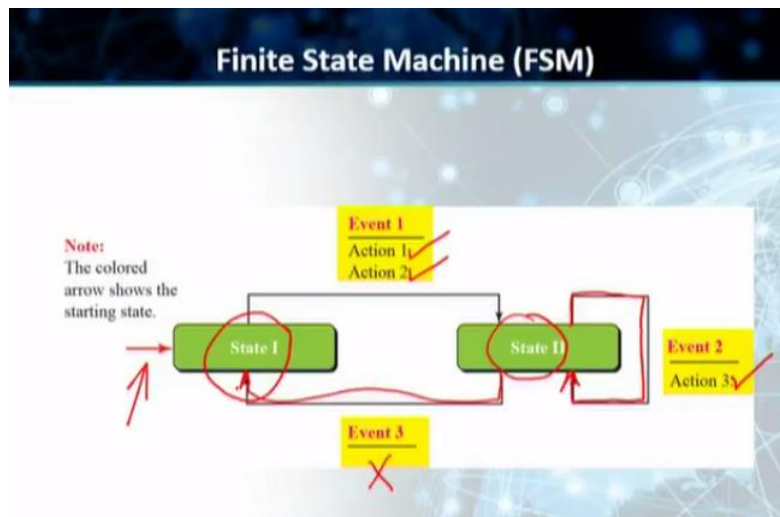
Finite State Machine FSM:

Machine with a finite number of states

Machine that will stay in one of the state until that specified event occurs. It will start with the state and will stay in the state until even happens

Each even has 2 reactions:

- List of Actions to be performed
- Determining the next State: for machine to move to next state.



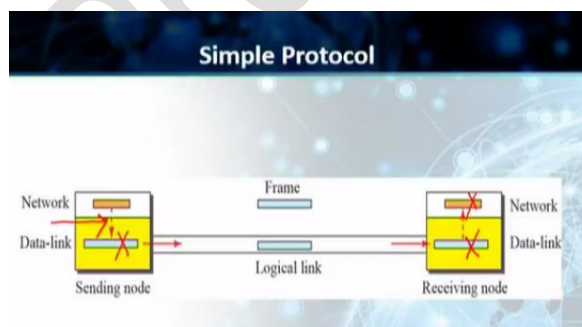
If event 1 occur so that event has 2 action that in state one it will perform those two actions and then it will move to state II and then action 3 will occur as there is no action in event 3 then it will move back to state 1.

Simple Protocol:

it doesn't not use any flow control and error control.

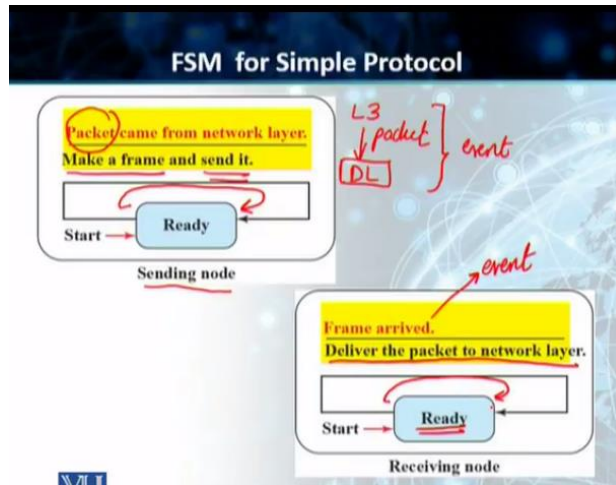
Receiver can immediately handle any frame it receives it will not get full

Receiver can't be overwhelmed with incoming frames.



From Network layer packets are converted into Frames and its sent to data link and that data link will convert back it to packets

Finite State Machine for Simple Protocol:



As here are there is a even of start state when packet is sent to data link layer then that event will perform another action of converting it to frames and once it done that layer will be available for process for next packet again.

Stop and Wait Protocol:

It uses both flow and error control.

In this protocol sender sends one frame at a time and waits for the acknowledgment before sending the next one.

To detect corrupted frames CRC code is used in it.

It is the most commonly used in Local Area Network

Here on the sending side we got Network layer and data link layer.

Same data link convert it into frames then it start timer and send frame to receiving end and that will check for the packets and will send acknowledgement to the sending end whether its okay. CRC are used with frames for errors detection.

Example:

Frame zero is send it got ACK

Frame 1 is sent it got ACK

Again Frame 0 is sent I again it ACK was corrupted and Frame zero sent again but at receiving end it was expected that Frame 1 is sent here. It

PiggyBacking:

Both simple and Stop-and-wait protocols are designed for unidirectional communication.

Data flows in one direction and ACK travels in the other direction in Simple and Stop and Wait protocols.

We will combine both of these signals.

In the first case A send data to B, so B ACK and send some data back to A again.

That make our communication more efficient rather than using unidirectional communication.

High Level Data Link Control (HDLC)

HDLC is a bit oriented protocol for communication which is based on Point to Point and multipoint links protocols.

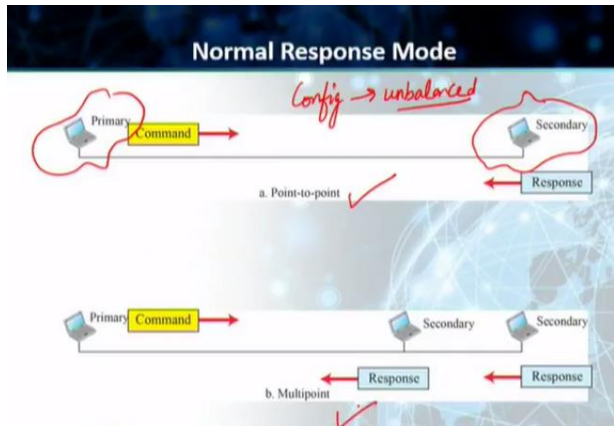
It implements Stop-and-Wait Protocol.

Most of the concepts defined in this protocol is the basis for other protocols such as PPP, Ethernet or Wireless LANs.

Transfer Modes in HDLC

1. Normal Response Mode (NRM)
2. Asynchronous Balanced Mode (ABM)

Normal Response Mode:



NRM in Point to Point

In it we say config is unbalanced.

Here we get 1 primary and 1 secondary station.

Primary station can send the command while secondary station can only receive command

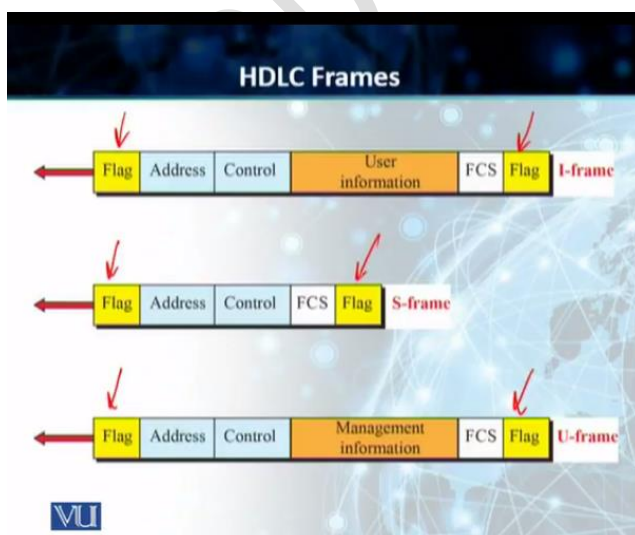
Asynchronous Balanced Mode:

It has 2 stations and each station can send and receive the messages.

Framing:

HDLC Defines three types of Frames

- Information Frames (I-Frames)
- Supervisory Frames (S-Frames)
- Unnumbered Frames (U-Frames)

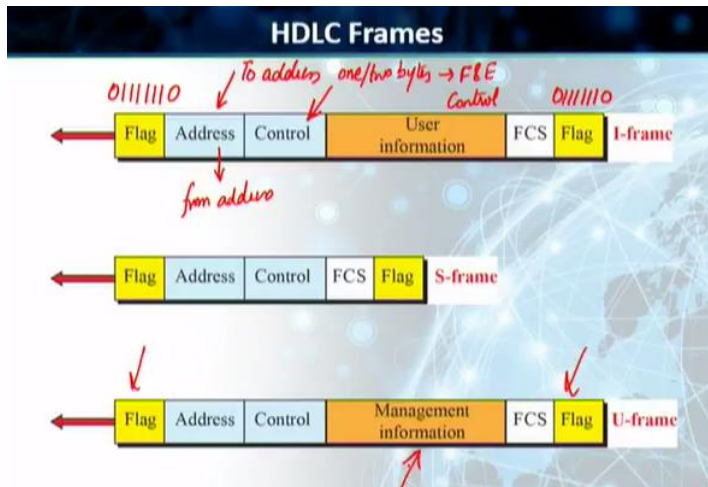


I Frame: it is used to carry user information and control the user data.

S Frames: are used to transport the control information.

U Frames: the information carried by this frame is the management information that is required to manage the links it self.

All frames do have flag fields in the start and end.



Flags are in the start as its bit oriented then it will have 01111110

2nd field consist of the address of secondary station. It will have from address as well.

3rd field: Control field is used for Flow and error control it has one or two bytes.

4th field: contains the user information.

4th : FCS Frames checks sequence

5th flag

Point to Point Protocol (PPP)

Most commonly used nowadays for point to point access.

Like users connect their computer with internet with PPP

To Control and manage the transfer of data, there is a need for a PPP at the data-link layer.

Services provided by PPP

The designers of PPP have included several services to make it suitable for a point-to-point protocol, but have ignored some traditional services to make it simple

Services Included	Services Not Included
Framing	Flow Control
Link Establishment and Data Exchange	Error Correction (PPP has CRC detection only)
Authentication	No Sequence Numbering
Multilink PPP Address configuration	Absence of sophisticated Addressing Mechanism
Network Address configuration	

PPP provides multiple Services

Framing:

Link Establishment and Data Exchange: how the two links are going to negotiate it will define in this services:

Authentication: Authentication of two devices who want to connect point to point.

Multilink PPP Address Configuration: that provides connection to multiple links

Network Address Configuration: it is useful for home user when you need a temporary network address.

Services Not Included:

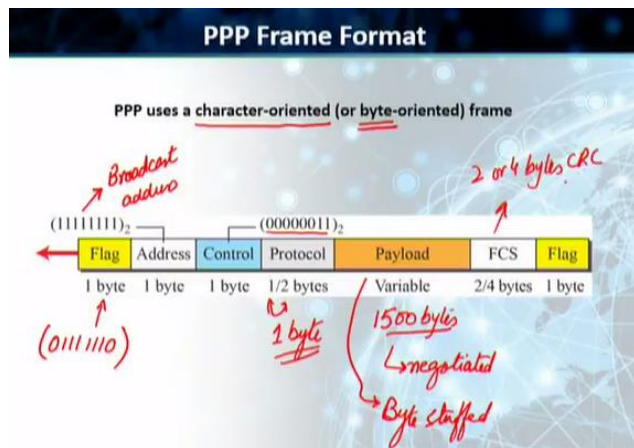
Flow Control.

Error Correction (PPP has CRC Detection Only)

No Sequence Numbering

Absence of Sophisticated addressing mechanism

Structure of PPP Frame Format



It is a character oriented protocol so it uses Bytes

We will have Flag field in the start and end

Flag field with all 1s will have broadcast frame

And control field will have 2 1s and remaining zeros it has constant value

Then we have 1/2 byte protocol field: that define what is carried in data field. Whether its user data or field data etc...

Payload: it carries user data and it can carry other information as well. It can carry 1500 bytes max.

Payload is Byte Stuffed. Mean adding 0 after every 5 similar number of flag

FCS: Frame check 2 or 4 bytes CRC for error detection.

Multiplexing in PPP

As PPP is a link layer protocol, it uses another set of protocols to establish the link. Authenticate and carry the network layer data.

Three set of protocols:

- Link Control Protocol (LCP)
- Two Authentication Protocols (APs)
- Several Network Control Protocols (NCPs)

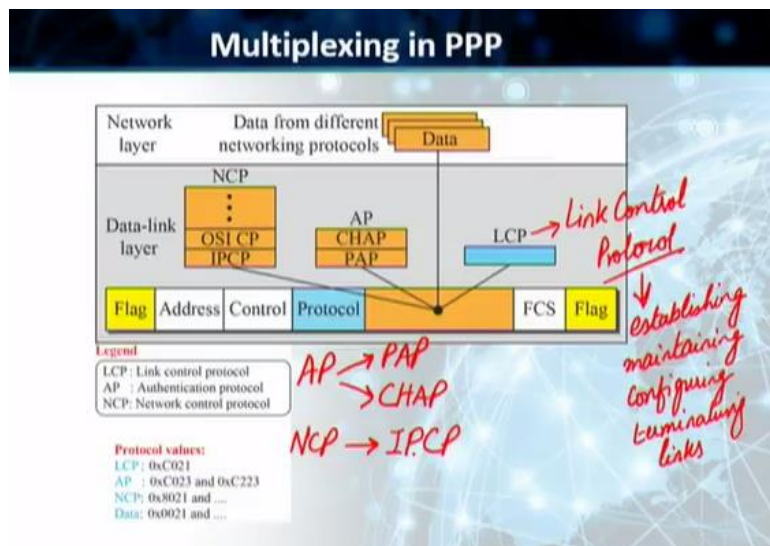
LCP is responsible for: Establishing, maintaining, configuring and terminating all PPP links.

AP: two protocol used PAP and CHAP

NCP:

OSI control protocol

IP Control protocol



LCP Packet Encapsulated in a Frame:

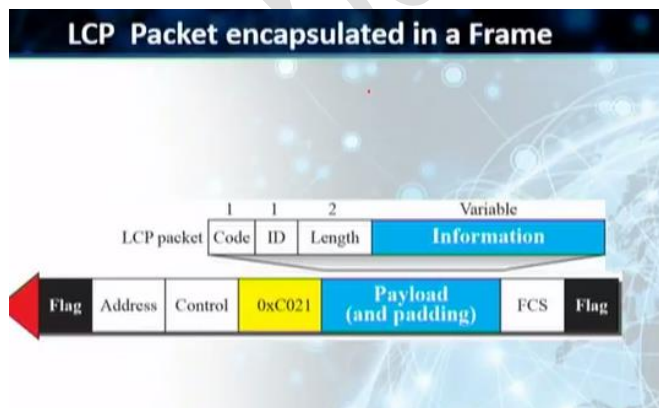
LCP packet consist of

Code Field: that contain different codes that we are going to use for configuration, termination...etc

ID field: that hold values that matches with the reply.

Length: that consist of length

Variable Information: consist of information for Payload part.



LCP Packets

LCP Packets		
Code	Packet Type	Description
0x01	Configure-request	Contains the list of proposed options and their values
0x02	Configure-ack	Accepts all options proposed
0x03	Configure-nak	Announces that some options are not acceptable
0x04	Configure-reject	Announces that some options are not recognized
0x05	Terminate-request	Request to shut down the line
0x06	Terminate-ack	Accept the shutdown request
0x07	Code-reject	Announces an unknown code
0x08	Protocol-reject	Announces an unknown protocol
0x09	Echo-request	A type of hello message to check if the other end is alive
0x0A	Echo-reply	The response to the echo-request message
0x0B	Discard-request	A request to discard the packet

First 4 used for link configuration

5 and 6 used for link termination

Last 5 packet types are used for link monitoring and debugging.

Authentication Protocols in PPP

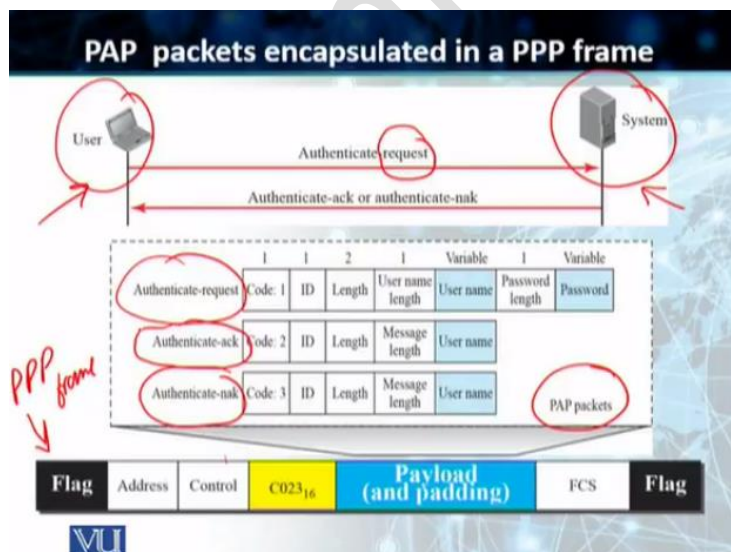
Two protocols are used:

Password Authentication Protocol (PAP)

Challenge Handshake Authentication protocol (CHAP)

PAP:

PAP Packets encapsulated in PPP Frame



Here we have PPP Frames in Flag

It's a simple protocol having only two ports only

User who want to access will send an authentication request by sending username and password.

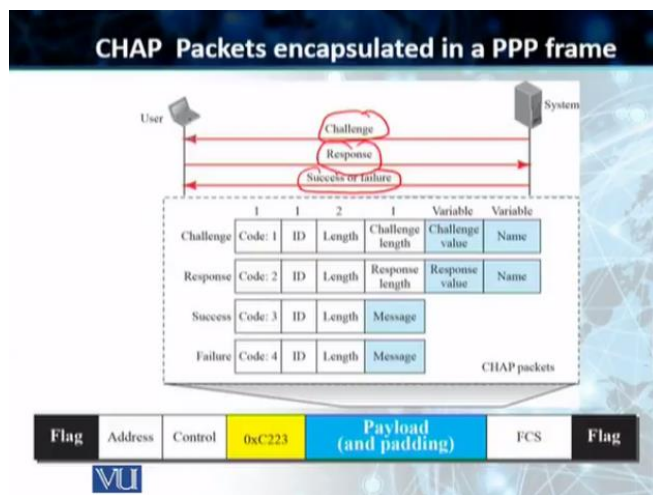
Then system will review username and password whether is right or wrong. So to allow or to reject decision will be taken according to it.

CHAP

In this protocol User receive challenge by system

Then user respond to it.

Then system replies success and failure.



Challenge: in which System sends challenge to user.

Internet protocol Control Protocol IPCP:

It the most commonly used protocol and most famous.

It consists of multiple protocols. i.e internet, OSI ...etc

It consists IPCP carries in Payload part.

CODE, ID and Length Field it contains network control information for internet protocol it used to configure IP package in internet.

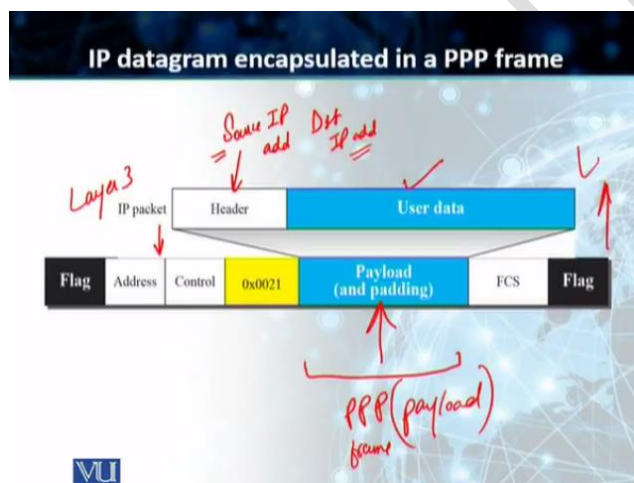
CODE Values used for IPCP Packets.

Code values for IPCP Packets	
Code	IPCP Packet
0x01	Configure-request ✓
0x02	Configure-ack
0x03	Configure-nak
0x04	Configure-reject
0x05	Terminate-request
0x06	Terminate-ack
0x07	Code-reject

IP Datagram encapsulated in PPP Frame:

that exist in Data link layer.

IP Packet: consist of Header that contains source IP Address, destination IP address and other control information along with user information it get embed in payload part of PPP Frame.



Media Access Control (MAC) Sub-Layer:

It is required in multipoint or broadcast link.

That manages multiple access protocol to coordinate access to the link.

It allows when there is multiple connection overthere

It prevent frames collision and utilize frames with turn by turn.

These sets of rules are defined by multiple protocols that are called multiple access protocols.

They are under the sub layer of data link which is media access control MAC layer.

Taxonomy(Categories) of Multiple Access Protocols:

There are 3 cateories which are further divided as well.

1: Random Access Protocol:

- ALOHA: Base multiple access protocol developed in 19s that form base for later protocols
- CSMA/CD: Carrier sense multiple access Collision detection
- CSMA/CA: Carrier sense multiple access Controlled Access

2: Controlled Access Protocols:

- Reservation
- =Polling
- Token Passing.

3: Channelization Protocols

- FDMA
- TDMA
- CDMA

Random Access Protocols:

In Random Every station is treated equal to other no station is superior to the other and none is assigned control over the other.

Each of those station will take turn to transfer the data.

Such set of protocols are used to tell station when to send media and when to not.

This decision depends on the state of medium (either idle or busy).

It is used to avoid collosion of frames. To prevent data being garbage.

There is another name of this protocol is **contention protocol**.

It doesn't have anytime whenever the data is required it checks station state and data is transferred as per rules.

In these stations kept on competing with each other.

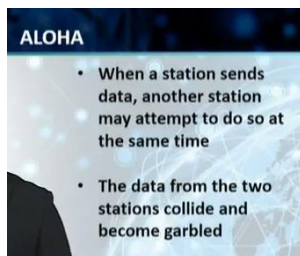
Types of Random Access Protocols:

1: ALOHA: Earliest random access method was developed in early 1970s that works on multiple access.

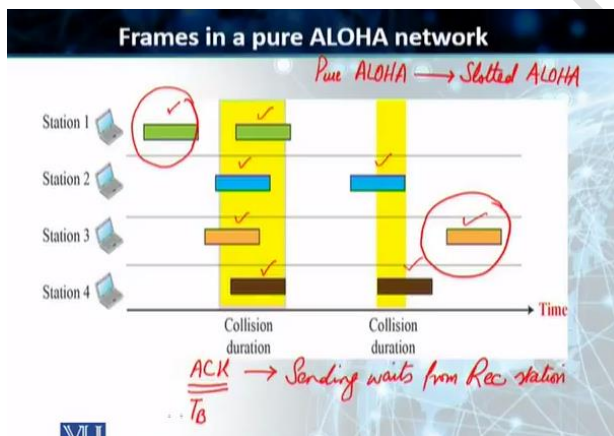
Further development was made on ALOHA and it is considered as the base of multiple access.

When multiple access are sending data on shared media then there are lots of chances for data collision.

It was designed for Wireless LAN, but it can be used on any shared medium.



Frames in a Pure ALOHA Network:



Just a consideration.

Here we have 4 station everyone is trying to access the main channel.

There is no collision avoidance technique is used in pure aloha.

In the end only 2 frames meant to each to destination remeaning had collision.

A method is used to prevent collusion is assigning a transmission time for everystation to send frames.

Procedure for Pure ALOHA Protocol:

Here we have

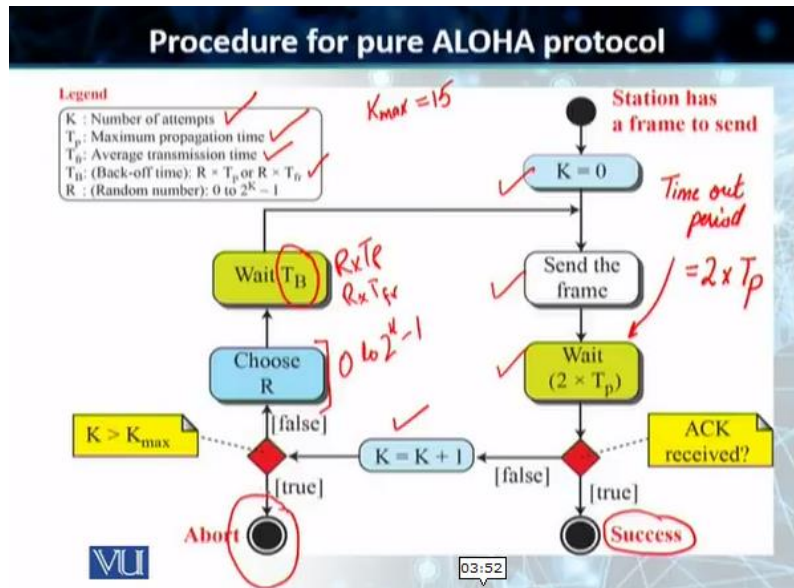
$K_{max} = 15$

Timeout period will be $(2 \times T_p)$ T_p = propagation time.

Initially we send the frames and then as per time rule we will have wait for a specified time.

If the ACK is received mean we are succeeded and if we fail it will be abort.

But if we fail then we Chose R and again wait for TB and process this loop again.



In this case K is greater than kmax

Vulnerable Time for pure ALOHA Protocol:

Vulnerable time defines the length of time at which propagation is possible.

Here at the start of station we have $t - T_{fr}$ at 2^{nd} we have t and at 3^{rd} we have $t + t_{fr}$

Slotted ALOHA

It is the 2^{nd} type of ALOHA

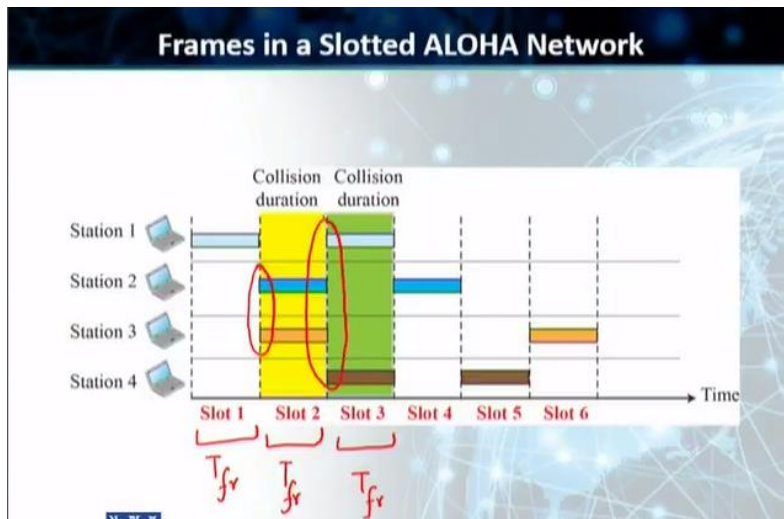
ALOHA is multiple access protocol that has no rules for access.

Slotted ALOHA was developed to improve efficiency of ALOHA.

In this we divide time in different time slots which is T_{fr} and all station will be able to send only in the beginning of the slot.

If a station misses the time slot then it have to wait for its turn of timeslot to come again.

Frames in Slotted ALOHA Network:



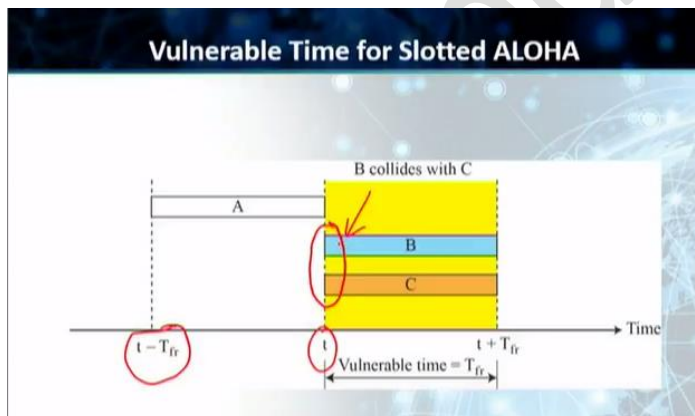
Here we have 4 Stations.

Stations will be allowed to send in the beginning of time slot.

If any station miss any slot it have to wait for next slot.

In this collusion could happed if the two station start at the same time.

Vulnerable time for Slotted ALOHA is solution for it.



What is B and C starts at the same time. Although we reduced difficulties but this if two starts at the same time that could create problem. But still our system is more efficient that Pure ALOHA.

Carrier Sense Multiple Access (CSMA)

2nd type of Random-Access protocol.

It was developed to improve performance over ALOHA.

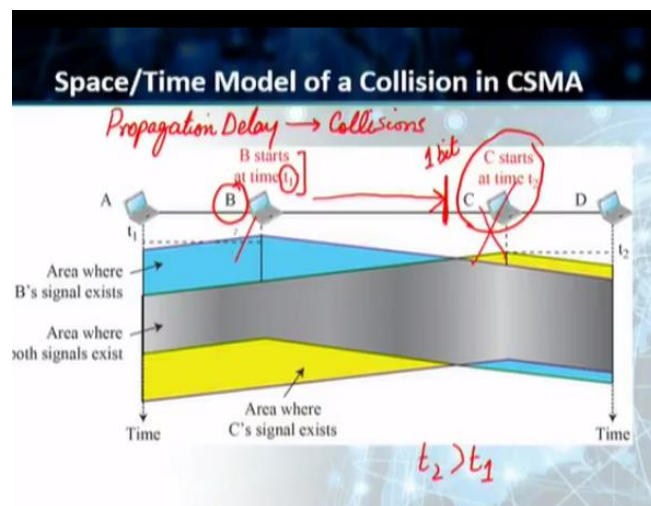
In this type of protocol every node has to listen/sense medium before sending data. If medium is busy data will not transfer if its ideal it will transfer.

It is used to minimize the chance of Collision

CSMA still can't overcome collision completely because of propagation delay.

Here more in Figure.

In CSMA it still get some time before reaching to other station there is a very small delay. That delay is propagation delay.



The delay to reach all other station is propagation delay.

At Time T_1 , B sense the medium and find its idle and start sending at T_1 b

And at Time T_2 C senses the medium and find it idle. B/c first bit data of B isn't arrived yet and during that delay C given idle to T_2 but the data from B already on way to the C and that time two signals collide and data wasted.

In CSMA Vulnerable time is Equal to propagation time.

If T_1 is sending data B,C,D all will sense the data any found it idle it will allow it.

Behavior of the Three Persistence Methods:

I-Persistent: after the frame find station Idle it sends frame immediately. It has heighest colloision rate.

I-Persistent is used in Eithernet.

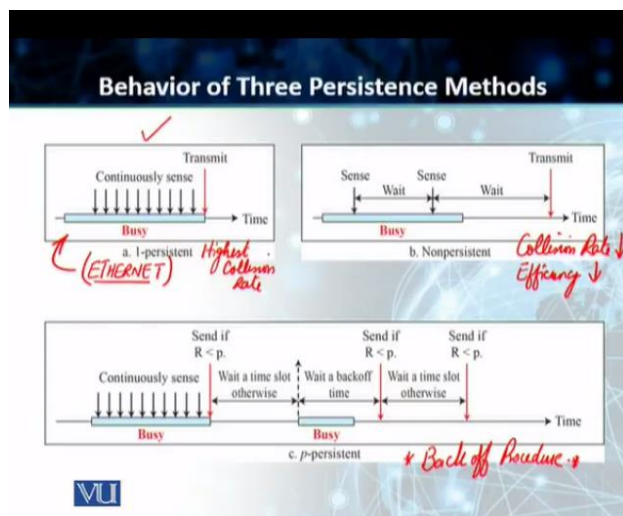
Non-Persistent: if channel has time slot it sense the line if the line isn't idle it wait for random amount of time until line get idle.

In this collision is reduced but efficiency is reduced a lot as well.

P-Persistent: it combines advantages of both I-persistent and Non-persistent.

If the line is idle it will produce if its not then it will consider it as collision happened and it will work as backoff procedure.

Backoff procedure makes it a better choice against others.



Carrier Sense Multiple Access / Collision Detection:

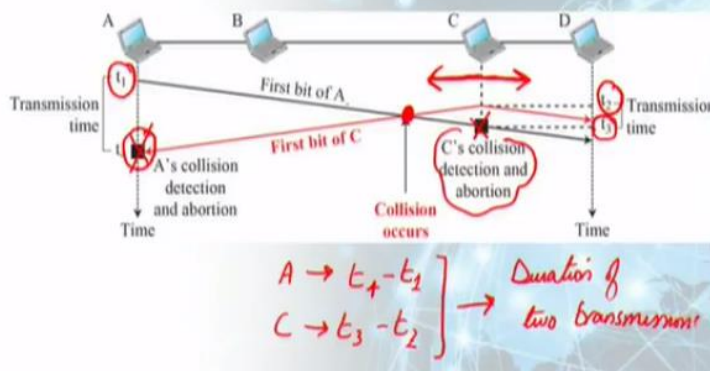
CSMA/CD: it argument CSMA protocol to handle the collision.

The station after sending frames it checks the medium if it sent successfully it there was a collision it will send data again.

Collision of the first Bit in CSMA/CD

Let take a example at t_1 data was being travelled to C but it wasn't reached and after a bit of time there was a collision occurred. And C detects this collision at T_2 and it aborted transmission. And station A aborted the transmission as well.

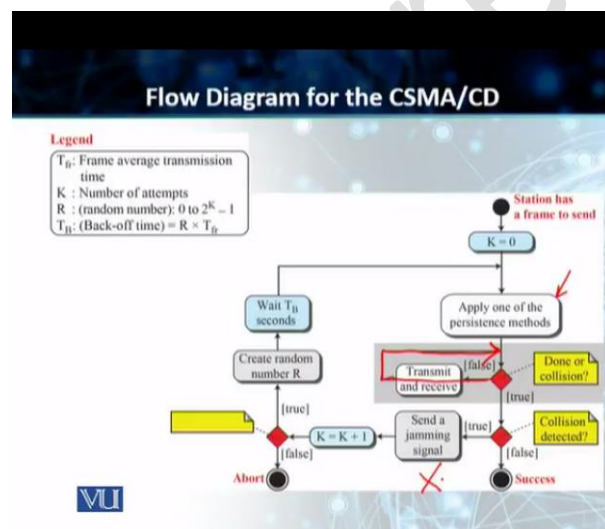
Collision of the First Bits in CSMA/CD



Flow Diagram:

Its similar to pure ALOHA but only 3 net method is added.

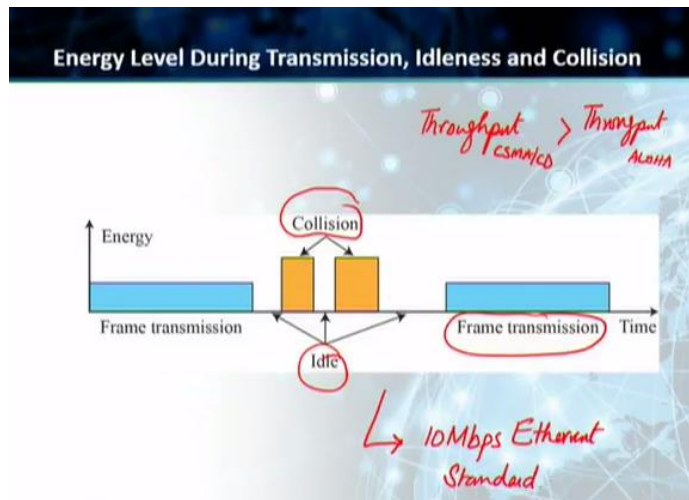
- 1 – is we apply a persistence methods.
- 2 – Transmission and Continous collision detection is checked here.
- 3: send a jamming signals. (to make all other station aware of collision).



Throughput of CSMA/CD is better than Throughput of ALOHA

Its also said 10Mbps ethernet

It's called traditional ethernet, and it was the first ethernet version that was used.



Carrier Sense Multiple Access/Collision Avoidance.

It was mainly invented for Wireless Networks.

It makes the system more efficient by using 3 following strategies:

- The Interframe Space: by the introduction concept of interframe
- The Contention window:
- Acknowledgements:

Flow diagram for CSMA/CA

At attempt one it checks for channel when it's free.

Then it looks for particular amount of time that is said interframe space and then it checks for medium whether it's idle or not.

If the medium is idle the station waits for the fixed amount of time which is equal to IFS(interframe space) before it sends.

IFS allows us to further avoid collision.

IFS allows different stations than already started transmission

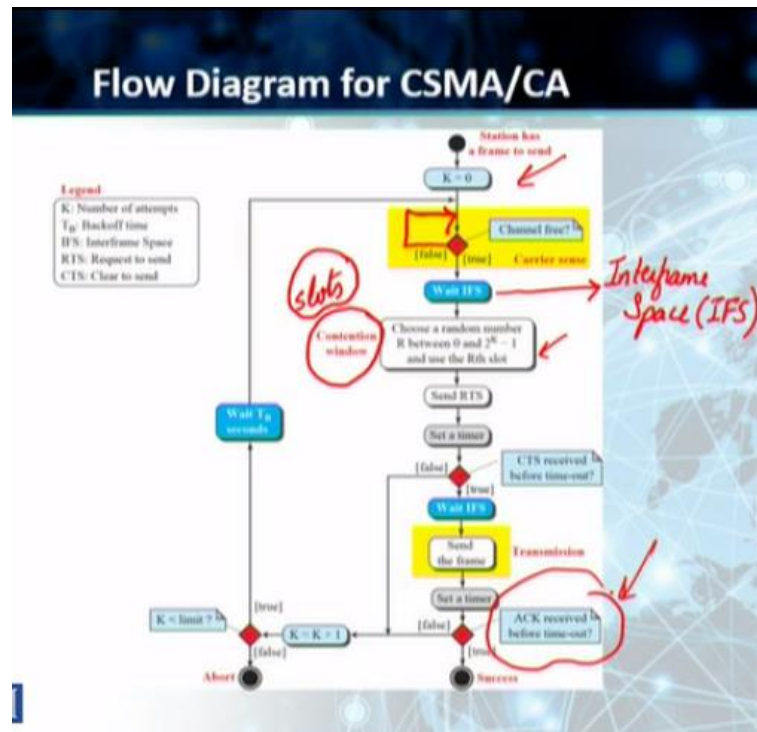
After IFS the system checks the particular medium then it starts transmission.

IFS is also used to check transmission priority to avoid collision

Contention window is used in CSMA it's just a amount of time which is divided into slots.

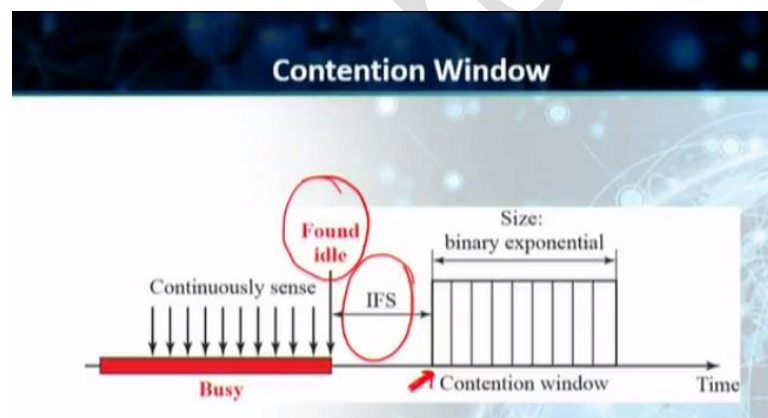
These slots in windows it changes according to binary backoff strategy.

If channel is busy it will double the slot for that station.



Interframe Space IFS: collisions are avoided by deferring transmission even if the channel is idle.

Contention Windows: in which we divide time into different slots, after system finds idle and wait for its slot...

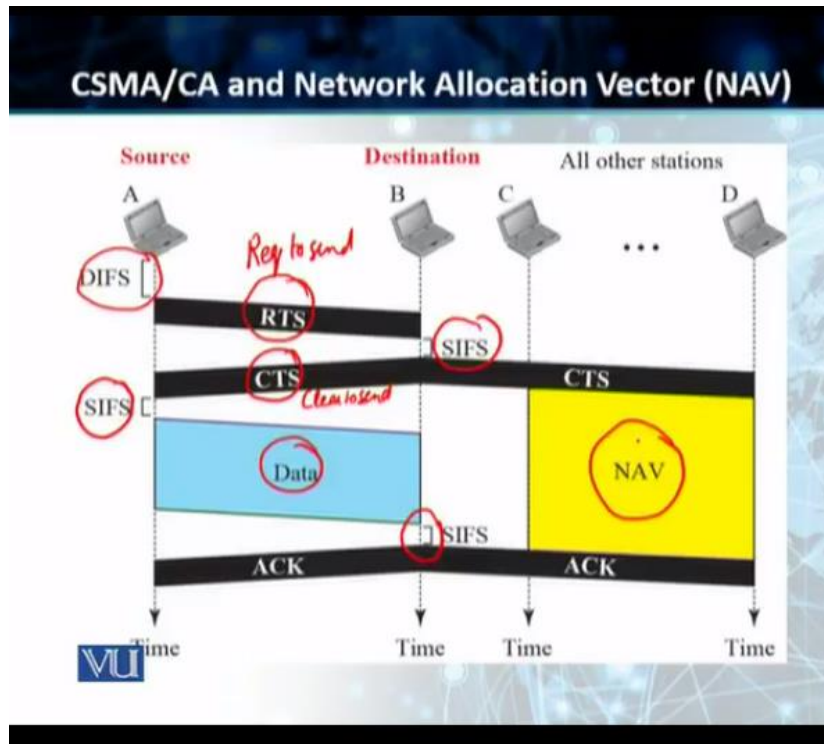


As its found idle it waits for IFS

Once both done and if the system found busy then we will implement the concept of contention window which is the binary exponential backoff. We will send to medium if its idle then we will send it at randomly chosen timeslot. If the system after this is busy then number of time slot will be double and then we transfer after this contention windows.

Even after the Contention window we do have collision.

Then Acknowledgement concept is used and data transmission only consider success only ACK received.



Control Frames are sent from A to B. then there is SIFS = Short interference Frame space

CTS = clear to send. Then there is SIFS again and it start sending the data once data is received then there is a short wait and then ACK is sent back to the source which sent the Data.

There is a new concept of network.

NAV it's a timer it get effect when RTS is sent and it shows the how much time must pass before checking the ACK.

Controlled Access:

Controlled Access protocols.

In this case stations need to concern with each other station that which station has right to send. Data can't be sent unless other stations are agree for its to send.

3 Methods:

- Reservation
- Polling
- Token Passing

Reservation:

In this station need to make reservation before sending the data before using the shared medium.

No of Station and N minislots.

The first frame is for Reservation of frames or stations then data is followed or processed according to this reservation.

Like in first reservation only 1,3, and 4 has got reservation.

Once its done then we make another reservation. Like in 2nd reservation 1 has only reservation.

Polling:

In case of polling we got a primary station and other stations are secondary station.

In polling we discuss we take permission from primary station before sending data. Even data is for 1 station yet we have to take permission from primary station.

Primary device controls the link while secondary devices follow its instructions.

Selection and Poll Functions in Polling Access Method:

Technique used here is Select and Poll to avoid collision.

It is used when primary has to send something to secondary. But in that case primary device will not know whether the target device is prepared to receive or not.

For this reason before sending the frame it created Selection **SEL** frame. To make sure to which system we have to send system.

ACK is used to get Acknowledgement. In which system will send ACK

Then data is sent.

This is Select method.

Poll:

Remaining things are same but in this primary polls first it will ask from A if its target system then A will give ACK other wise NAK (nothing to sent) then it send polls to B if get ACK it will process the data.

Drawback in Polling: if primary goes down whole system goes down.

Token Passing.

In this method Station is organized as a logical ring.

Every station there is a predecessor and successor.

Predecessor is the station which is logically before the station in the ring.

The Successor is the station which is after the in the ring.

Data transfer in Token ring is happened with the help of special packet is called token.

In order to transmit data any station should have possession of token. If station don't have it can't transmit the data.

If token is to successor then station needs to wait for its turn to receive the token.

Token Management is required to manage the possession time (how long station can hold token.)

It will manage Token Monitoring, priority, assignment etc...

Token is received from processor and then after station its forwarded to successor.

If system get offline then token gets trashed.

Logical Ring and Physical Topology in Token-Passing Method:

1. Physical Ring Topology

In this method predecessor will send to 2 and other successor will not get any info regarding this.

If this link btw 1 or 2 or system fails then whole system fails.

2. Dual Ring

It consist of dual ring 2nd ring is only used when main ring fails.

3. Bus ring Topology.

Its also called bus topology

It got a logical topology.

When the station sends the data it leaves the station then stations adds the address of successor after releasing the data. In this topology address is required.

4. Star ring.

There is a hub is used in this topology that is connected with all system. It works as connector system connected to it are connected by two wires.

If one system goes down yet others will be kept on working.

Channelization (Channel Partition)

It is the categories of multiple access protocol.

In this we divide available bandwidth of shared medium in time, frequency, and available code.

Frequency Division Multiple Access (FDMA):

We divide the bandwidth into different frequency band.

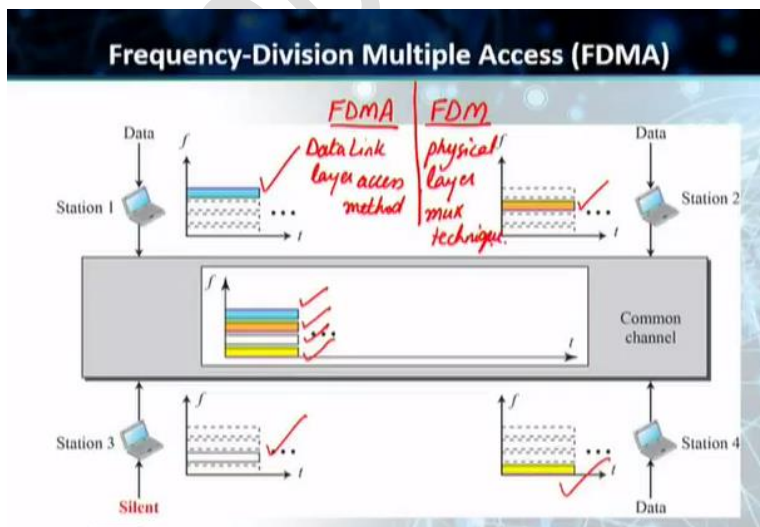
Each station is allocated a band to send its data.

Band is reserved for that specified station only. It belongs to the station all the time.

Each Station also uses a bandpass filter to confine the transmitter frequencies. So they don't overlap with each other

We can send the data at the same time as different bands are there allocated to different stations.

FDMA



Here there are band for all station.

Which they use to send their transmission to common channel.

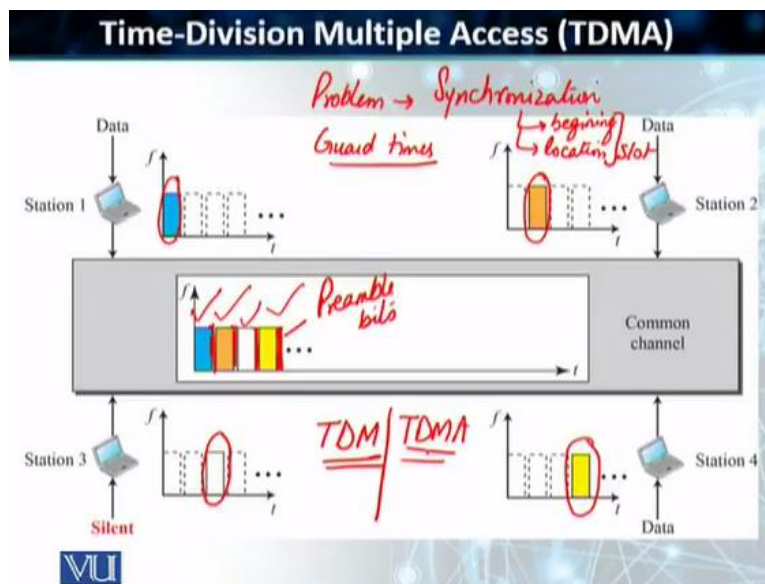
FDMA and FDM both are different.

In FDMA Data link layer tells each station to create a bandpass channel.

Time Division Multiple access TDMA

In this bandwidth is divided of the channel in time.

Each station is allocated is assigned with one single time slot and that channel use that specified time slot for sending the data.



Here in Example

Each station have different time slot.

Problem in TDMA:

Its Synchronization: you have to achieve Synchronization btw station.

Each station need to know beginning of its time slot and location of its time slot.

We get TDMA propagation delays.

Guard time: that is used to delay the propagation delays

At the beginning of the each slot there are some bits that will send the station when the time starts that bits are said to be preamble bits.

TDM and TDMA both are different.

TDM is physical layer technique it transfer data from slower to faster channel.

TDMA it's a access method.

CODE Division Multiple Access (CDMA)

Its more efficient than other 2 technique we discussed.

In this whole bandwidth is used for one particular changes.

It is different from TDMA.

All stations sends data at different time by using whole bandwidth.

In CDMA we are assigning code each station like $c_1, c_2, \dots$ etc

If we multiply these codes we will get 0 i.e $c_1 \times c_2 = 0$

If we multiply each code by it self then we will get number of station. In this case $c_1 \times c_1 = 4$

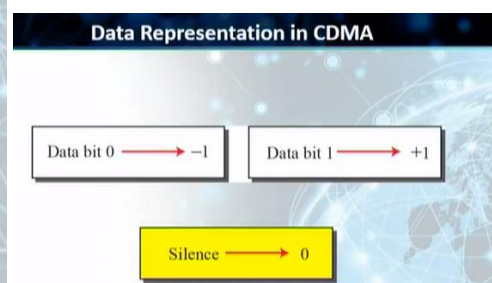
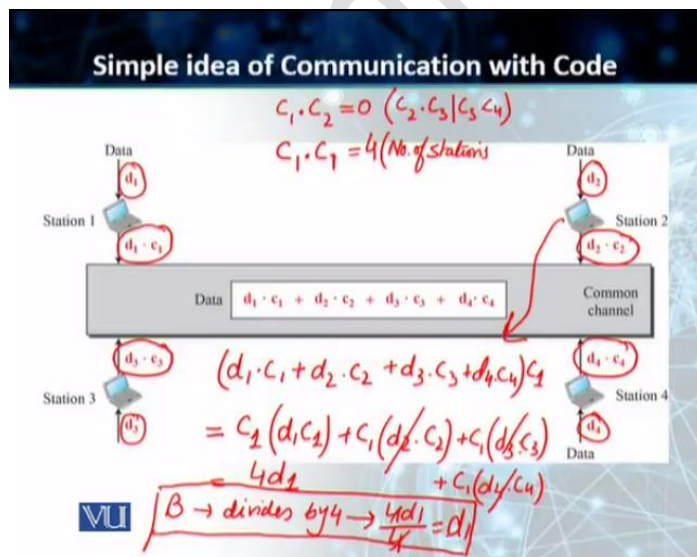
Each channel has different code and they will send that to the common channel.

Here if station 2 wants to know about data of Station 1 it will do

$(d_1 \cdot c_1 + d_2 \cdot c_2 + d_3 \cdot c_3 + d_4 \cdot c_4) \cdot c_1$

Further solved in photo.

All terms will be zero as when different term multiply it will be zero.



This is how data of any station is scanned.

Ethernet Protocol:

Its one of the famous LAN protocol.

In physical layer and Data link layer we don't have any specified protocol these layers can work with all layers it can be actually adapt.

There are some LANS which are not connected to WAN they are connected to internet or connected to eachother by wide area technique.

In Ethernet we will mostly talk about CSMA/CD

It released initially in 1970s

Ethernet is the only thing that survived data communication era.

It was released that way that it can adopt the upgration.

Initially it was desgied for 10 migs.

Totally be have that can carry 10 gigs.

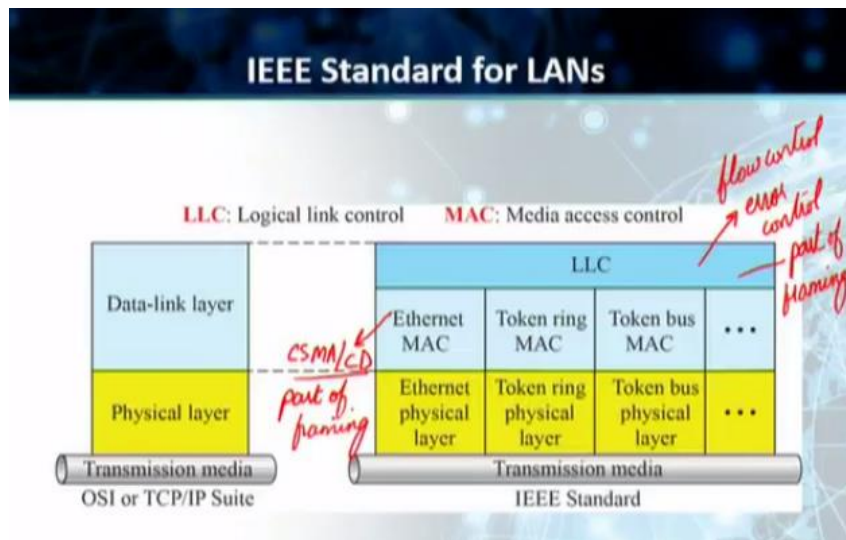
IEEE Project 802

In 1985, the computer society of IEEE stated this project.

It specify about the way different LANs technologies will interact with eachother on physical or data link layer.

Project 802 did not seek to replace any part of the OSI Model or TCP/IP protocol Suite

IEEE Standard for the LANs



This project divide data link layer into two layers 1 LLC and Nack

LLC = Flow control, error control, and part of framing.

MAC layer: specified access layer.

In case of Ethernet we will use CSMA/CD.

Ethernet Evolution:

The Ethernet LAN was developed in 1970s.

It has gone to four generations.

- Standard Ethernet (10 mbs)
- Fast Ethernet (100 Mbps)
- Gigabit Ethernet (1 Gbps)
- 10 Gigabit Ethernet (10 Gbps) mostly being used nowadays.

Standard Ethernet:

it was able to support data rate upto 10 mbps

most implementation of ethernet is now moved to either 1 gigs of 10 gigs

Yet some features are still sames that are not changed during the evalution.

Services of Ethernet:

Connectionless and Unreliable services that ethernet provides:

in this each frame is dependent.

Connectionless mean: frames are not connected they are independent.

Each frame can take its own particular route...

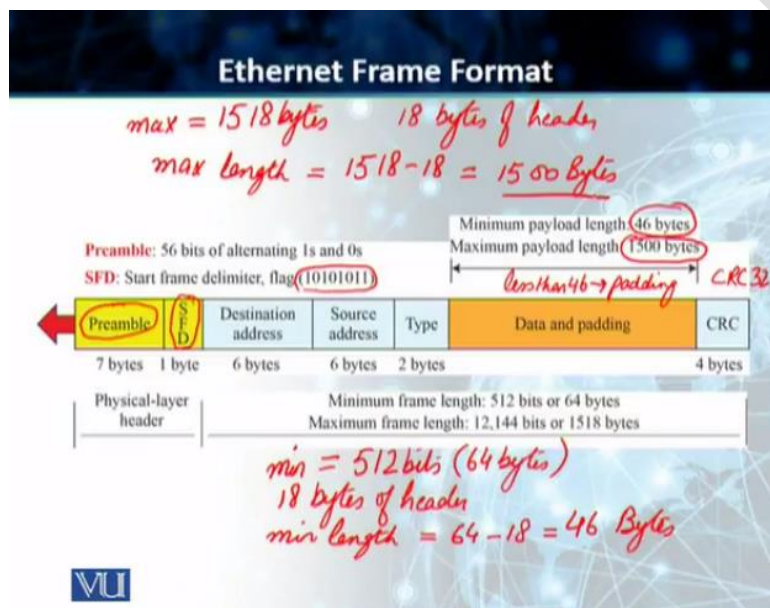
In this we don't have to set connection btw our source and receiver.

Unreliable features of ethernet:

Its unreliable like IP and UDP

There is no to know about that frame is corrupted so receiver will silently drop it.

Ethernet Frame Format.



It has 7 fields.

1st field is: preamble, it consists of 7 bytes its here to alert the receiver signals. That tells when the frames are coming.

2nd: SFD (Start frame Delimiter): this field beginning of the frame, its last chance to make or to show frames beginnings.

These both layers are depended on the Physical layer in the form of physical layer header

3rd: Destination header: 6 bytes (48 bits), signals are sent in the form of broadcast.

When data is sent then it will look to the destination address if its matches then it will proceed it other wise data will be discarded.

Then we have Source Address: which is 6 byte as well which got link address.

Type: this is meant to specify the upper layer protocol it has 2 bytes. If it has IP data and it will specify that. Same for all other types

Data and Padding: Minimum payload length 46 bytes:

Max payload length: 1500 bytes.

If the data is less than 46 bytes then we do a process named padding to add some extra bytes to it.

CRC: its is used for Error detection as discussed before.

Frame length: min size 512 bits (64 bytes)

Max (12.144 bits) 1518 bytes)

Min frame length = 64 bytes – 18 bytes = 46 bytes

Reason we got min length: CSMA/CD words better if we get specific shorter length.

Max length = 1518 bytes – 18 bytes of header = 1500 bytes.

Reason to specify max length as memory used to be very expensive before so max length was defined to reduced buffer length. It was defined to avoid station monopolize the links.

Addressing in Standard Ethernet

Each Station on Ethernet has its own interface Card (NIC)

Every NIC got a unique link layer / physical address.

In case of standard ethernet its 7 bytes and (48 Bits)

Its normally written in hexadecimal notations with a colon between the bytes.

Example:

Ethernet MAC Address

4A:30:10:21:10:1A

How the MAC Address is sent online

Address: 47:20:1B:2E:08:EE

Transmission of Address Bits

How the address 47:20:1B:2E:08:EE is sent out online.

The address is sent left to right, byte by byte; for each byte, it is sent right to left, bit by bit, as shown below:

Hexadecimal	47	20	1B	2E	08	EE
Binarys	01000111	00100000	00011011	00101110	00001000	11101110
Transmitted ←	11100010	00000100	11011000	01110100	00010000	01110111

LS → unicast
 MS → multicast
 → broadcast

How we transmit bit it doesn't mean this is how we transfer bit. It is sent by the following methods.

The Address is sent left to right, byte by byte; for each byte.

It is sent right to left, bit by bit

Here we have 47 we will convert it binary and then we transmit least significant bit first and most at last.

Like here 1 is sent first and 0 is sent at last.

Least significant bit will tell receiver whether its unicast, multicast, or broadcast

If least significant is 0 it will be unicast.

If its 1 then it will be multicast.

Source address will be always unicast.

Destination address can be unicast, multicast, or broadcast.

Example: 13.2

Example 13.2

Define the type of the following destination addresses:

- a. 4A:30:10:21:10:1A ✓ 4A A → 1010 (0 → LS)
- b. 47:20:1B:2E:08:EE ✓ 47 7 → 0111 (1 → LS)
- c. FF:FF:FF:FF:FF:FF

To find the type of the address, we need to look at the second hexadecimal digit from the left. If it is even, the address is unicast. If it is odd, the address is multicast. If all digits are Fs, the address is broadcast. Therefore, we have the following:

- a. This is a unicast address because A in binary is 1010 (even).
- b. This is a multicast address because 7 in binary is 0111 (odd).
- c. This is a broadcast address because all digits are Fs in hexadecimal.

With 3 different destination addresses:

In first case:

- a) 4A:30:10:21:10:1A

It is a unicast address as A in binary is 1010 as 0 is least significant so its unicast.

- b) : 47:20:1B:2E:08:EE: 47 = 7 = 0111, 1 is least significant so its multicast.

- C) It consist of all F means all 1s

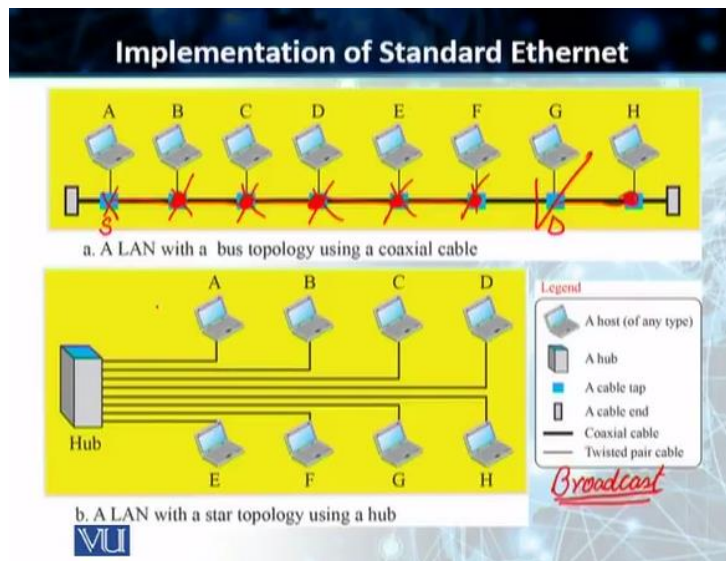
Implementation of Standard Ethernet:

How unicast, multicast, and broadcast are different from each other.

All the frames are by nature are broadcast.

If all are by nature broadcast then how will we identify whether its unicast of multicast.

It is identified when the we sent any data from one system to another



Like here if A send unicast data it will pass from all stations i.e B,C,D

And all will see if its unicast then it will discard it and G will see and accept this particular frames.

In Multicast.

In multicast only particular group will look at the station and only destination one will receive it.

In broadcast all system will see the data and accept it.

Access Method in Standard Ethernet:

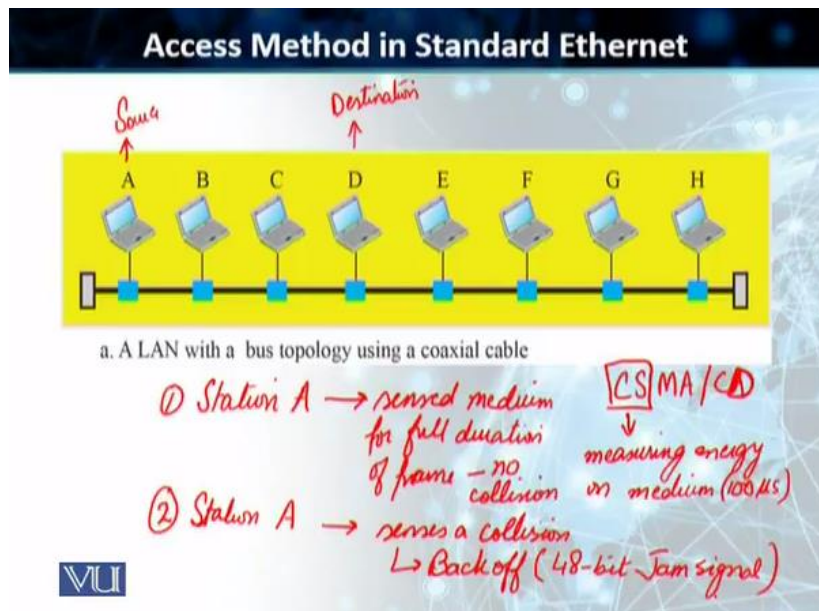
Everything is sent on Ethernet is in the form of Broadcast.

Before sending data we need to employ MAC address.

Standard Ethernet uses CSMA/CD with 1 Persistent method.

In case of CSMA it has to sense the medium before sending the data.

Access method using CSMA/CD with 1 persistent method:



Let A is source and D is destination

Station A will sense the medium before sending anything. It is done atleast for 100 microseconds.

If there is any energy sense it will sense that there is transmission is being occur and it will wait for it to be completed.

Once medium is found idle. And Station A already started the transmission that time medium sensing wasn't stopped over there.

b/c station A sent the medium for the full duration of the frame and it found out there is no collision

It will sense station is idle after the transfer is completed.

It also senses the collision

If collision will happen then both station will backoff and stop sending signals. And that time station A will send 48 bits jamming signals to alert all other stations.

After every collision the backoff period will increase.

Efficiency of Standard Ethernet.

It is the ratio of the time used by a station to send data to the time the medium is occupied by the station.

Formula to calculate efficiency

$$\text{Efficiency} = 1/(1 + 4\alpha)$$

Example:

To find Efficiency:

Example

In the Standard Ethernet with the transmission rate of 10 Mbps, we assume that the length of the medium is 2500 m and the size of the frame is 512 bits. The propagation speed of a signal in a cable is normally 2×10^8 m/s.

Handwritten calculations:

$$\text{Propagation Delay} = \frac{2500 \text{ m}}{2 \times 10^8 \text{ m/s}} = 12.5 \mu\text{sec}$$
$$\text{Transmission Delay} = \frac{512}{10^7} = 51.2 \mu\text{sec}$$
$$a = \frac{\text{Prop. Delay}}{\text{Trans Delay}} = \frac{12.5}{51.2} = 0.24 \rightarrow 0.24 \text{ of a frame occupies medium}$$
$$\xi = \frac{1}{(1 + 64 \times a)} = 39\% \rightarrow \text{moderate only 61\% of time}$$

Boxed notes: IDEAL, $a=0$, $\xi=1$



Here we have transmission rate is 10 mbps

Length of medium is 2500 m

Frame = 512 bit

Propagation speed of signal in cable is normally 2×10^8 m/s

We need to find propagation delay

Propagation delay = $2500 \text{ m} / 2 \times 10^8 \text{ m/s}$

Transmission delay = $512 / 10^7 = 51.5$ micro second.

Now finding a = propagation delay/ transmission delay

Further solved in following image.

And then putting values we will get efficiency

Only .24 of a frame occupies cable medium.

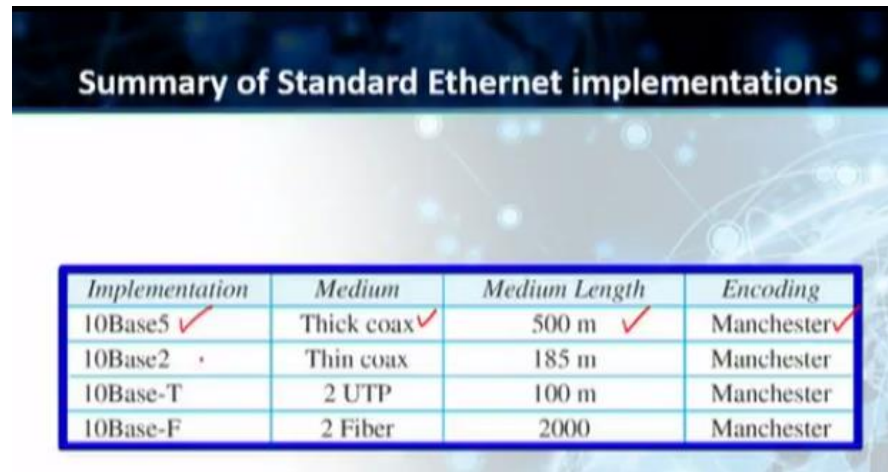
39% of efficiency is moderate efficiency.

In case of idle case our a should be equal to zero.

The lower the value of a means higher the efficiency is.

Implementation of Standard Ethernet:

There are several but few are used



The image shows a table titled "Summary of Standard Ethernet implementations" with a blue header and a light blue background. The table has four columns: Implementation, Medium, Medium Length, and Encoding. It lists four Ethernet standards: 10Base5, 10Base2, 10Base-T, and 10Base-F. Each row includes checkmarks in the Implementation, Medium, and Medium Length columns. A large, faint watermark "Prepared by amza" is visible across the table.

Implementation	Medium	Medium Length	Encoding
10Base5 ✓	Thick coax ✓	500 m ✓	Manchester ✓
10Base2	Thin coax	185 m	Manchester
10Base-T	2 UTP	100 m	Manchester
10Base-F	2 Fiber	2000	Manchester

1st is 10Base5

Complete details are mentioned in image

For 10Base2 we use thinner version of Coax cable

Encoding in Stander Ethernet

All standard ethernet use encoding technique Manchester Encoder and then we convert it to the digital signals.

Then Manchester decocder receive it and after decoding it forward the data.

10Base5 Implementation:

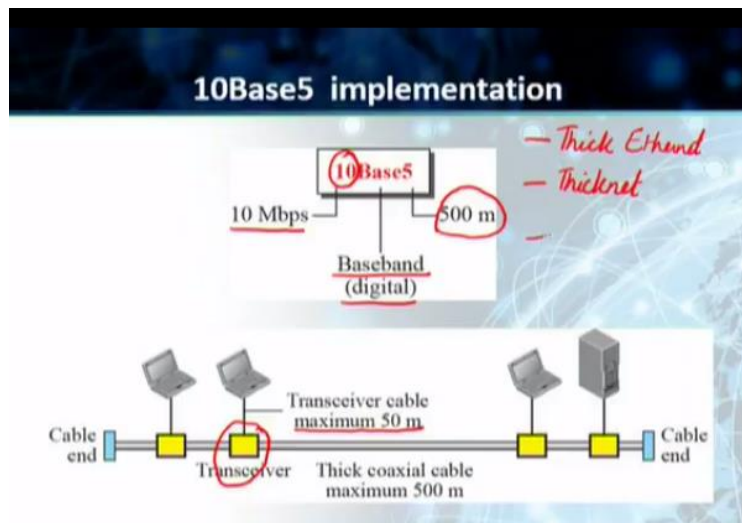
Uses thick coaxal cable

It works on bus topology

10 in this base represent data rate.

Base represent Baseband signals

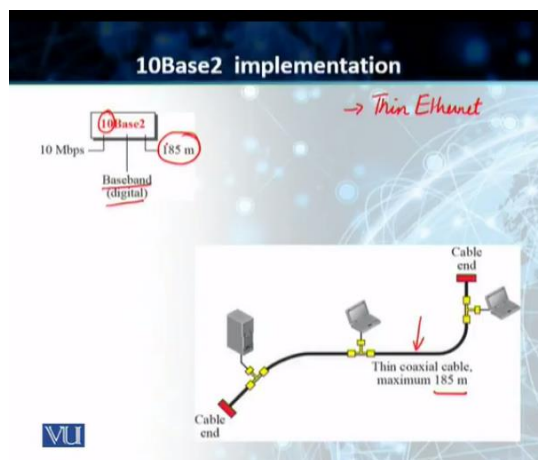
5 represent length of cable (completely shown in figure.)



If more than 500 m is needed then we will use repeaters.

10Base2

Here 2 means about 200m but actually it is 185 its near about 200

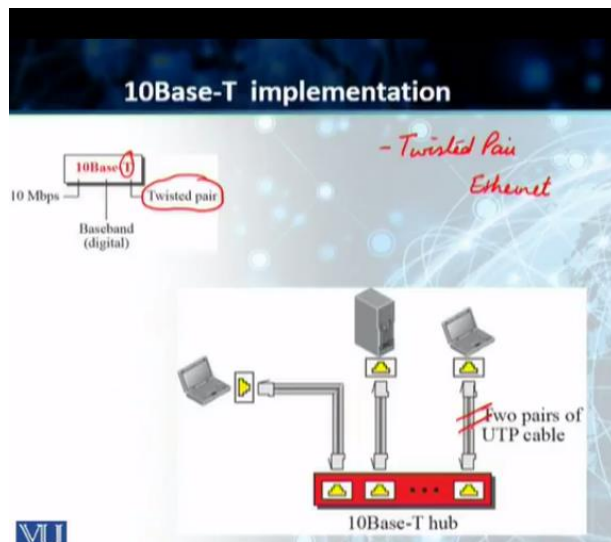


10Base – T

T represent we use two twisted pairs

It uses physical star topology it has hub and stations are connected to that HUB

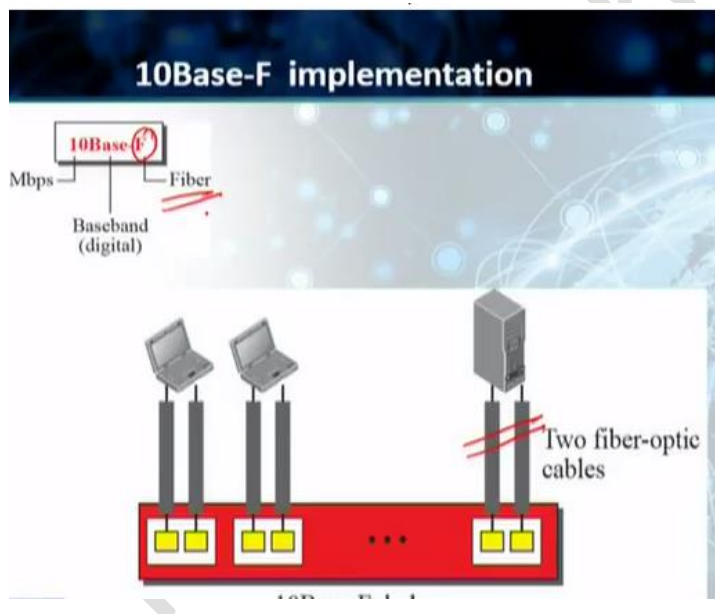
In this collision is only happen in hub not in cables.



10Base-F

F for Fiber

In this system is connected to hub by using two fiber cables.



Changes in Standard Ethernet

Changes that we have made with evolution in Standard Ethernet.
3 different development made in it.

- Bridge Ethernet
- Switched Ethernet

- Full-Duplex Ethernet

Bridge Ethernet.

If we got max bandwidth is 10 mbps

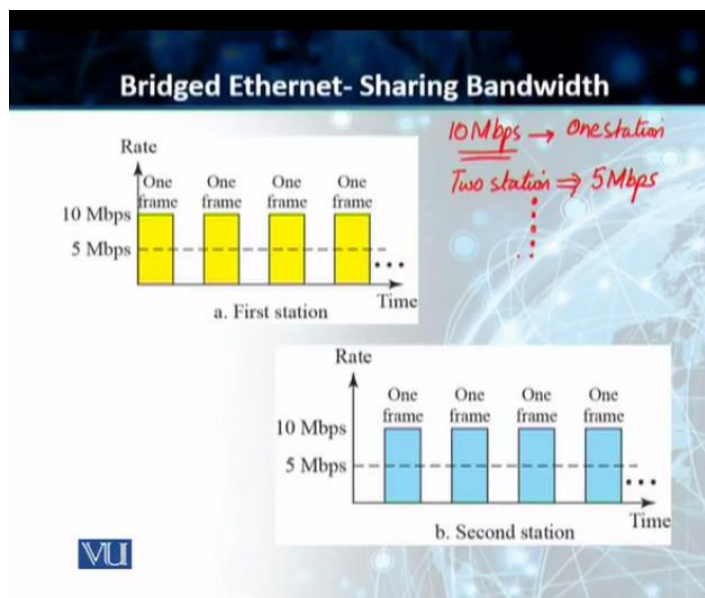
And there is only one station that is using shared medium to send the data. That time whole of the bandwidth of medium available for station.

Then we get two station connected.

That time efficiency of 10 mbps will be divided.

But when one station is sending the data other stations will remain quite.

If the four frames are sending data then that time four frames will divide the efficiency



Solution to that is bridge

Bridge Divide network frame into 2 or more networks.

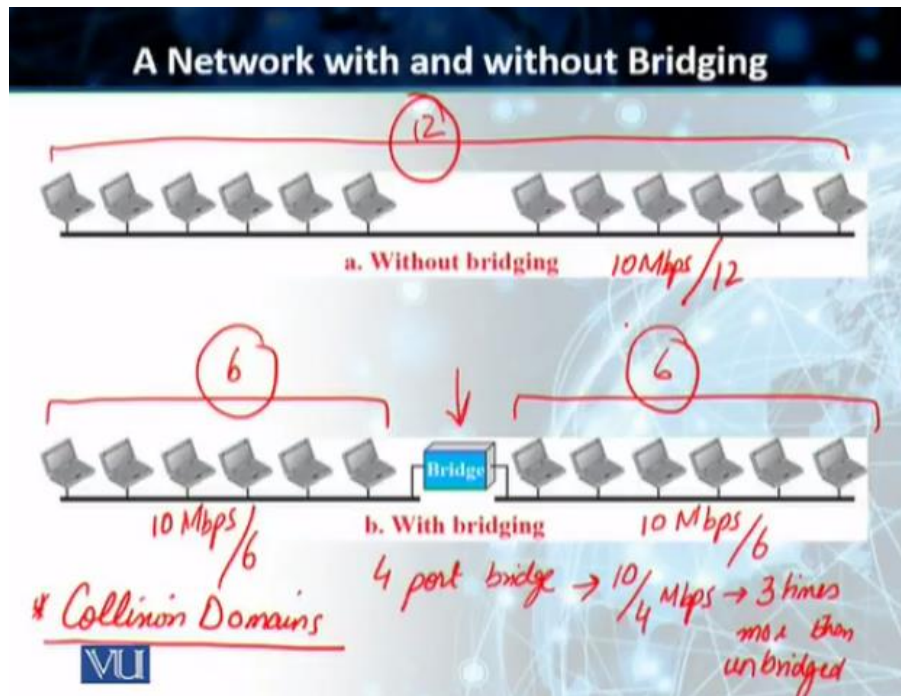
Like in image we have divided network into two networks 6 station each

And we can divide 10 mbps after bridge for 6 and 6 after not 12 at a time.

It is used to improve efficiency

Increasing bridge with increase power /efficiency

Bridge helps in separation of collision domains.



Switched Ethernet:

Another change made in Standard Ethernet.

Bandwidth capacity was increased by using Bridge.

In this we use a switch and domains are connected to that switch.

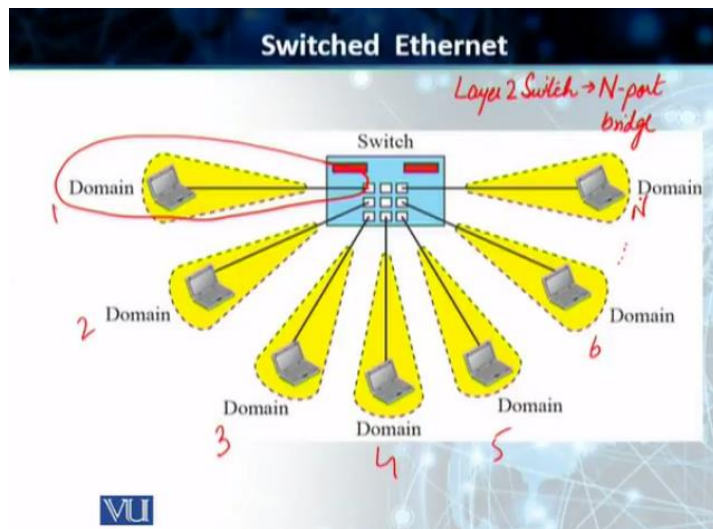
Layer 2 Switch $\rightarrow$ N Port bridge

Bandwidth is shared only btw domain and switch.

Domains would be dependent on the number of ports on switch.

We got N port switch that is layer 2 switch.

It was one of the major change that was made in bridge ethernet and it is reason for fast data we are using now adays.



Full – Duplex Switched Ethernet

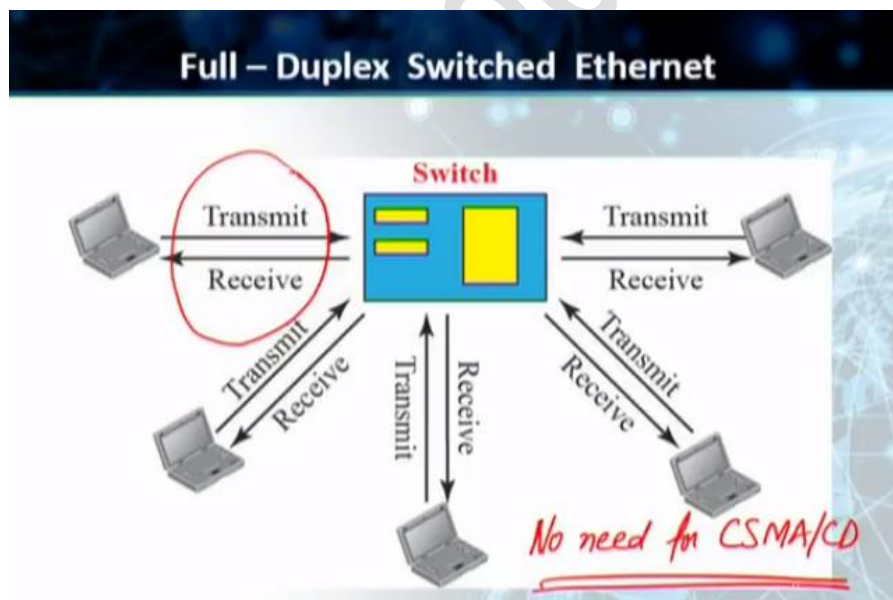
In previous communication was half duplex.

Switched Ethernet was advanced to full duplex switched ethernet.

It was able to made two way communication by making links.

Another advantage that there was no need of CSMA / CD

Each link could transfer data without any hesitation of collision.



Fast Ethernet:

It is the second generation of Ethernet

In 1990s a great progress was made from 10mbps to 100 mbps.

To make it compatible with the standard ethernet the MAC sublayer was left unchanged.

There was certain element that was used to upgrade the transmission speed.

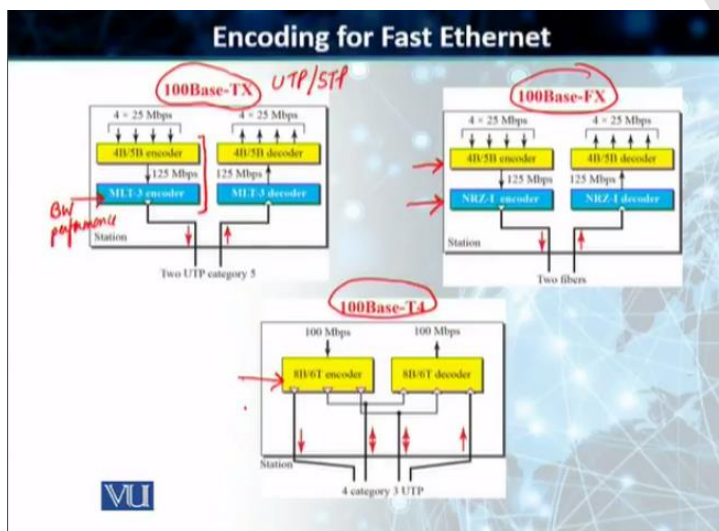
Goals was set for Fast Ethernet.

- Upgrade data rate to 100 mbps
- Make it compatible with standard Ethernet
- Keep it same 48 bit address.
- Keep Same Frame Format.

There was several changes made to handle 100 mbps

Changes was made at the physical layer.

Changes Made:



1) Encoding for Fast Ethernet.

1st implementation 100Base-TX

It used two cables of twisted pair cable it could be UTP/STP

ML T3 encodder was used in this.

Its combination made of 4B, 4B encoder.

2) 100Base- FX

Used with fiber optics.

As it could handle large data so NRZ was used

There is long sequence of 0s and 1s that's why we are using 4B/5B

3) 100Base T4

For this case we used category 3 of UTP

Encoding used in this case was 8B/6B to support 100Mbps speed.

Implementation of Fast Ethernet implementations				
Implementation	Medium	Medium Length	Wires	Encoding
100Base-TX ✓	STP ✓	100 m	2	4B5B + MLT-3
100Base-FX	Fiber	185 m	2	4B5B + NRZ-I
100Base-T4	UTP	100 m	4	Two 8B/6T

Gigabit Ethernet:

It was able to support quite higher data rate. To make it this able IEEE Standard of 802.3z Gigabit bit.

For Gigabit there was certain goals

Goals:

- Upgrade the data rate to 1 Gbps
- Make it compatible with standard or fast ethernet
- Use same 48 bit address
- Use the same frame format
- Keep same minimum and maximum frame lengths.

MAC Sub-Layer.

MAC sub layer was a issue for Gigabit

This evaluation was to keep MAC sublayer untouched.

To upgrade 10x speed changed was required to made in MAC

Gigabit Ethernet has two distinctive approaches for medium access:

- Half Duplex
- Full Duplex.

Physical Layer in Gigabit Ethernet required lots of changes it became more complicated than standard and fast ethernet.

Encoding in Gigabit Ethernet.

It is either 2 wire or 4 wire network.

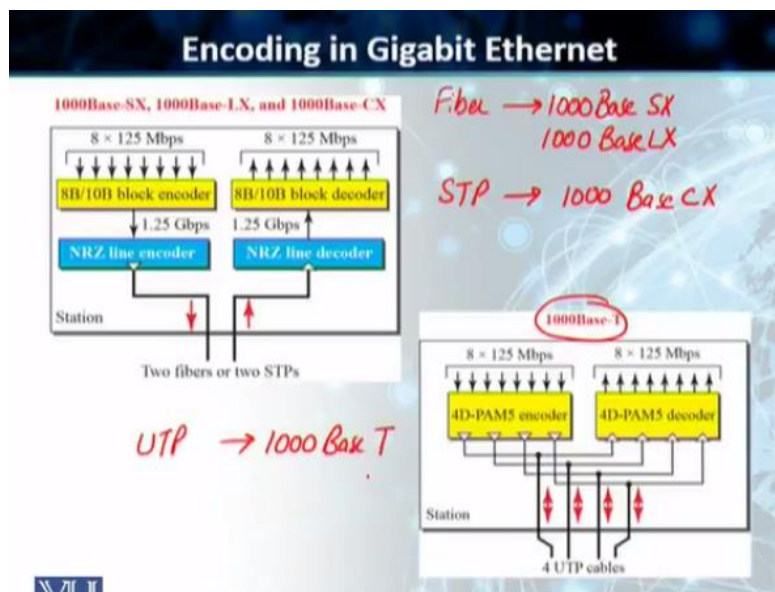
Fiber is used for 1000mbps so it will be → 1000BaseSX

STP → 1000 BaseCX

There we have 4 wire version.

It was designed for people who already has twisted pair cables.

It uses UTP → 1000BaseT



Summary of Gigabit Ethernet Implementations				
Implementation	Medium	Medium Length	Wires	Encoding
1000Base-SX ✓	Fiber S-W ✓	550 m ✓	2 ✓	8B/10B + NRZ ✓
1000Base-LX ✓	Fiber L-W ✓	5000 m	2	8B/10B + NRZ ✓
1000Base-CX	STP	25 m	2	8B/10B + NRZ
1000Base-T4	UTP	100 m	4	4D-PAM5

SW = short ways

LW = Long way fiber

10 Gigabit Ethernet.

Fourth Generation.

Being used mostly nowadays.

It being used in LANS and Metropolitan networks.

It only operates in Full – Duplex Mode. Means no contention for CSMA/CD

Four Implementation is most common given below

Implementation	Medium	Medium Length	Number of wires	Encoding
10GBase-SR	Fiber 850 nm *	300 m	2	64B66B
10GBase-LR	Fiber 1310 nm	10 Km	2	64B66B
10GBase-EW	Fiber 1350 nm	40 Km	2	SONET
10GBase-X4	Fiber 1310 nm	300 m to 10 Km	2	8B10B

Other Wired Networks:

Incase of Wired LAN commonly used is ethernet

Two types:

Access Networks: Networks that connect small LAN to an ISP. i.e 1 single user is using internet in his computer.

Wired Network: is the type of network that is being used to transfer data over long distances.

Telephone Network:

Started in late 1800s

Early POTS: Plain old telephone system was used it was analog using analog signals to transmit voice.

In later they started carrying data as well.

Now a days telephone can carry analog and data as we..

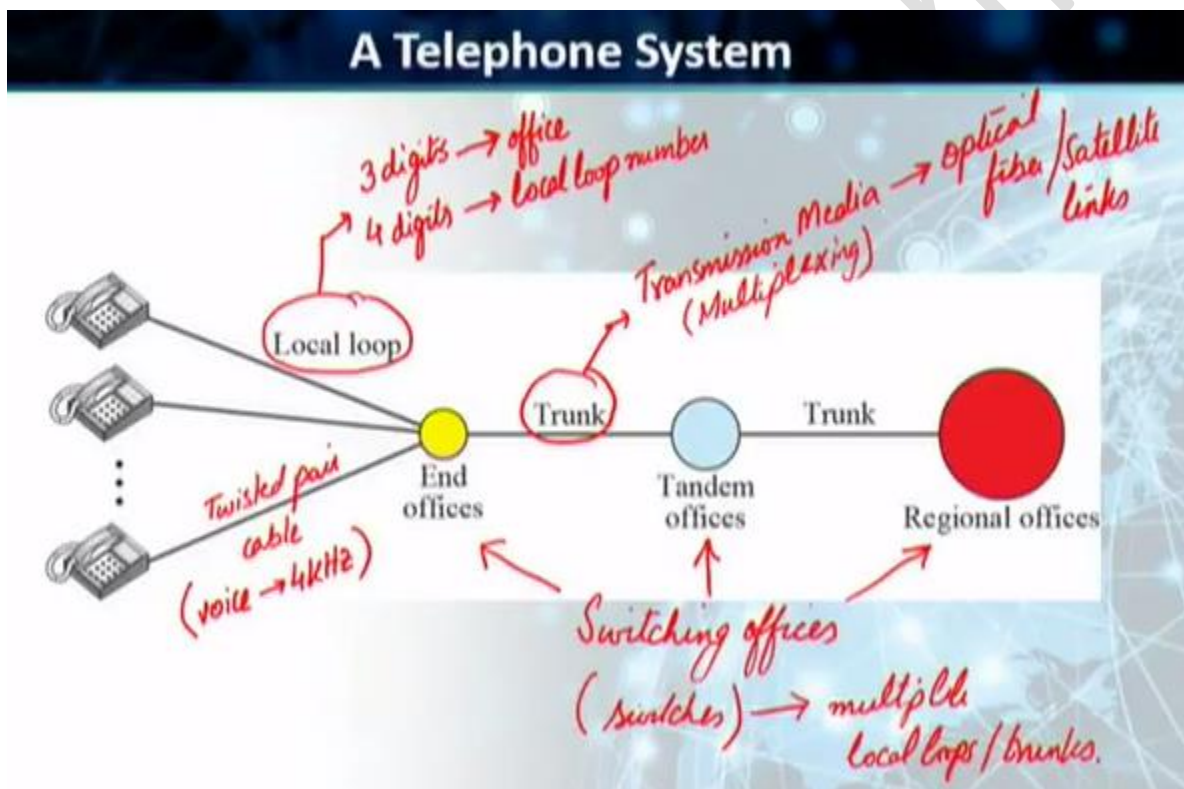
Component of Telephone System:

- Local Loops
- Trunks
- Switching Offices:

Types of Switching Offices:

- End Offices
- Tandem Offices
- Regional Offices

The Component of A Telephone system:



We have local loops its just twisted pair cable that connect subscribed to the end offices its called local loop

When its used for voice I has bandwidth of 4KHz

First 3 digits of these local loops identifier they represent the particular offices.

Next 4 digits define exact local loop numbers

Trunks: it is just a transfer medium it can handle multiple connect 100 to 1000

Switching Offices:

These are switches shown in figure that connect multiple local loops and trunks

Local-Access Transport Areas (LATAs)

Lata can be small small or large metropolitan area

Small state can have 1 lata, large can have several.

LATA boundary could overlap with the state boundary, like a Lata part of another state could be part of same our another mean 2 or more.

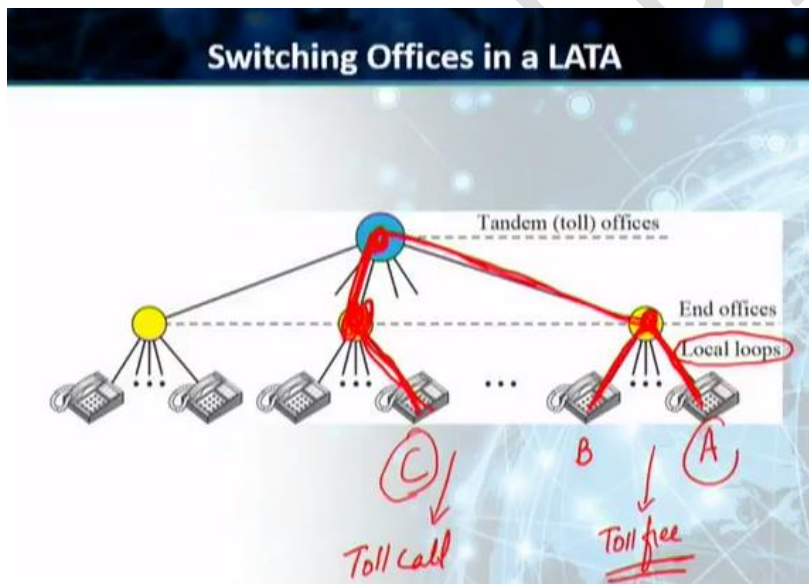
Services that is offered vby companies that operate btw LATAs are called Intra-Lata Services:

Companies that provide services within the LATA or underLATA are said to be Local Exchange Carrier (LEC)

Telephone companies that provide Services between Two LATAs are Called Interexchange carriers (IXCs):

Switching Offices In LATA

IT HAS two components

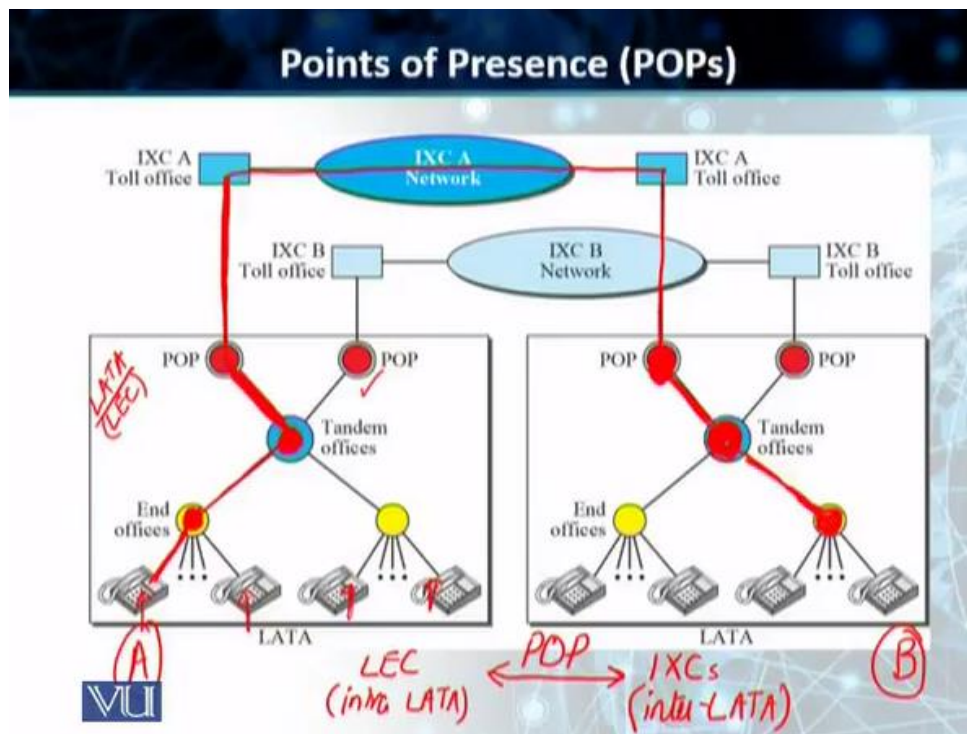


- 1) Local Loops: that connect devices to End Offices:
- 2) Tandem Offices: uses to do communication when its btw two different end Offices

If the call is within the same end office then Tandem office is not involved and call is said to be Toll Free.

When call goes through tandem office Toll fee is charged.

Points of Presence (POPs):



Local Exchange carriers LEC interact with IXC's by using a particular switching office that is called POP

LEC has to provide each of the subscriber data to POPs:

Let suppose A wants to call B which is in different LATA: A will contact end office that will connect tandem office that will select pop as per inter carrier we want to use as per figure we have two different carrier. And that will forward to pop and same process continues.

LEC are used to interact with IXCs and then IXCs interact with another IXCs that will connect with LEC ...

Signaling:

Initially signaling network was all circuit link. Circuit switched network with dedicated link was used for voice calls. But all of these we were doing manually

There was a central office with an operator connected.

Like one subscriber used to call operator and then that operator connects 1 subscriber to other.

And for disconnection one of the parties again contacts operator for disconnection.

It's in-band communication.

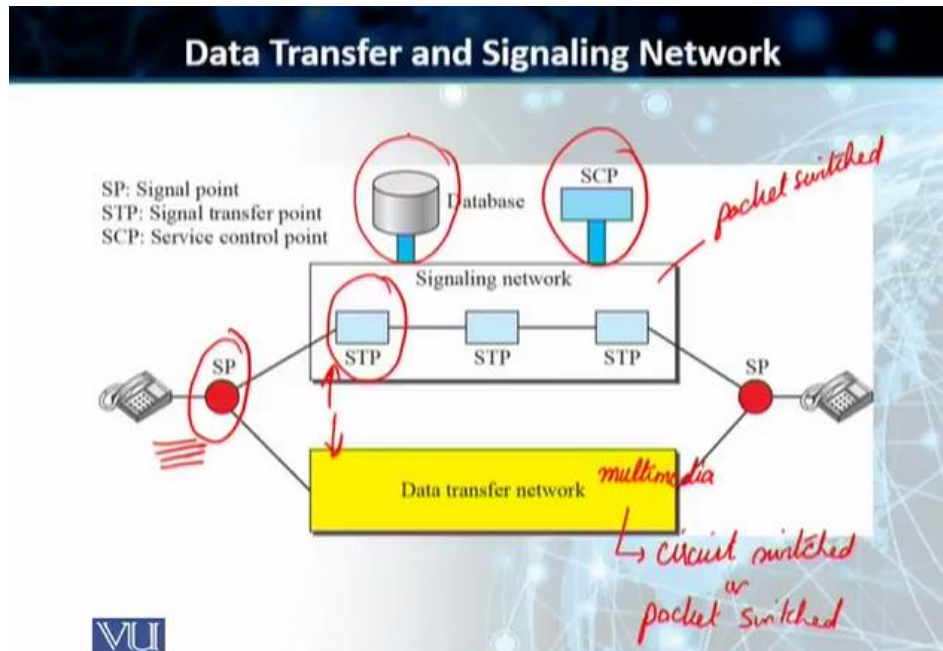
It was manually but now it's automatic.

Rotary telephones were invented that sent a digital signals defining each digit in a multidigit telephone number.

When telephone networks evolved into a complex network the functionality of the signaling system increased.

To make it control busy tone, ring tone, etc...

Data Transfer and Signaling Network:



In this most part is circuit switched network or packet switched.

Consist of mostly same type of protocol.

Data Transfer network carries multimedia.

Signaling Network Carries Packet switched.

In this case subscriber are connected to SP signaling point.

Signaling point connects to the STP (Signals transfer point): it is connected to the SCP that controls whole the operation of Signaling network.

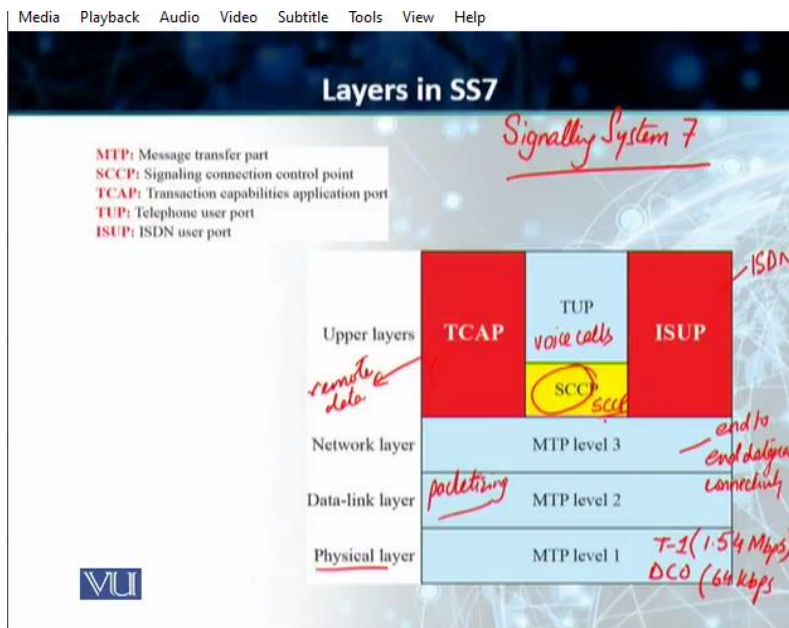
Other networks can be included in this connection like here database is included.

Signaling System 7 (SS7)

SS& is normally being used now adays in our telephone networks.

Layers in SS7

There are various layers in it.



1 MTP layer 1: it has physical layer specs. It got T-1 (1.54 Mbps)_DCO (64 kbps)

2 MTP Layer 2: based on Data link layer. That is used for packetizing.

3 MTP Layer: based on Network layer. Used to end to end datagram or connectivity.

TUP: used to setup voice call.

TCAP: Transition capability application protocol. That provides remote calls.

ISUP: It replaces telephone port to provide services ISDN we required.

SCC: Signaling system Control point that I used to control the whole signals system.

Services By Telephone Companies.

Two Types:

- 1) Analog Services
 - a. Analog Switched Services
 - b. Analog Leased Servers: leasing a analog line to create a point to point connection btw systems.
- 2) Digital Services
 - a. Switched /56 Services: it can provide data rate upto 56 per second
 - b. Digital Data Service: equal to Analog lease service 65 kbps.

Digital Subscriber Line DSL

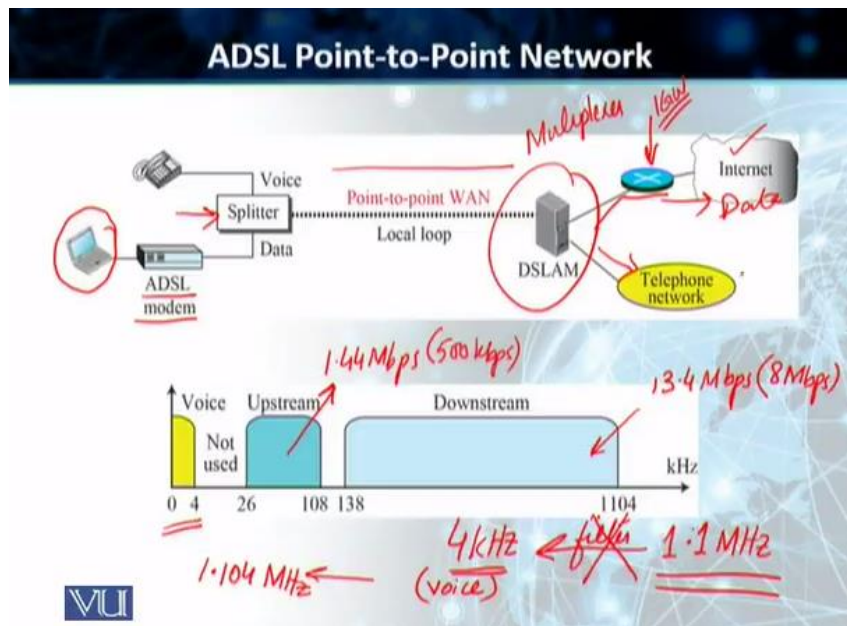
After Traditional Dial up services telephone companies developed another Technology which is DSL to provide Higher Speed access to the internet.

Old wasn't enough as 65 kbps was too slow so that's why DSL was developed.

It supports High Speed Digital Communication over twisted pair line.

There is multiple variant of dsl (ADSL, VDSL, HDSL, and SDSL)

ADSL Point to Point Network:



It use existing line just like local loop

ADSL achieve high data rate with same line.

Our telephone line can carry 1.1MHz

In case of dial up model we used to have filters that reduce our bandwidth from 1.1MHz to 4KHz.

These filters are now removed

Now adays we can use exact amount of bandwidth 1.104MHz

0 to 4 KHz used for Voice.

26 to 107 used for Upstream

138 to 1104 used for Downstream.

ADSL allows to use Phone line and internet as same time.

In case of DSL it can achieve 1.44Mbps but normally we get 500 Kbps Upstream (upload)

For Downstream 13.4 Mbps are possible but we only get 8Mbps with DSL technology

Splitter is used to split voice and data channel

Then simple Point to point line or Local loop is connected to DSLAM (Telephone Network) (connecting Multiple Local Loops). It connects telephone network for voice and Internet Gateway to Internet

Cable TV Network:

Initially created to provide TV connection to remote subscriber i.e. living in mountains or remote areas.

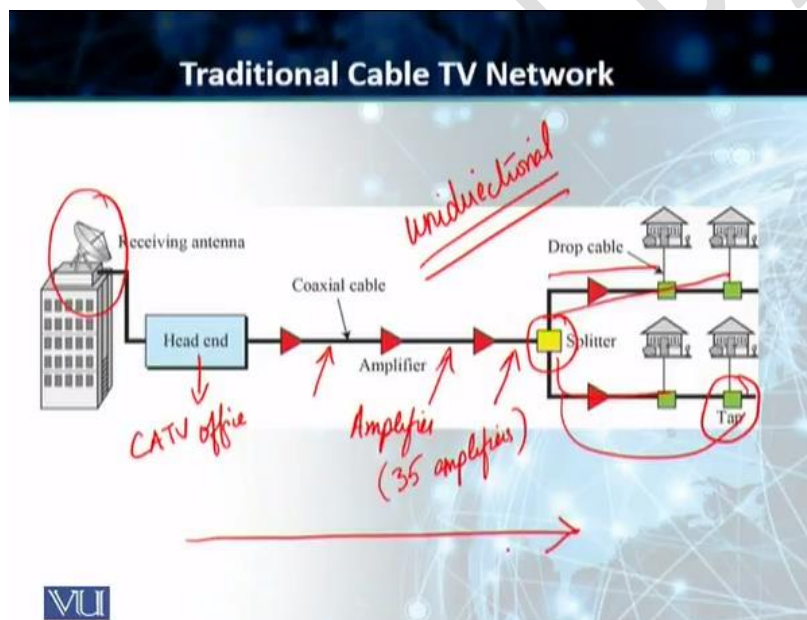
Later Cable TV networks enabled access to remote broadcasting station via microwave connections

Cable TV are nowadays could also be used to carry the data.

Traditional Cable Networks:

Cable networks were being used to broadcast video signals in 1940s

It was called community antenna television (CATV) b/c an antenna is used at the top of a tower or building to receive the TV signals.



Here we can see the antenna on the building that is our cable office which receives the signals.

Then there is a cable TV office named head end

It is receiving signals from cable office and using coaxial cable

There is amplifier is used to renew the signals

We have 35 amplifiers btw headend and subscribers.

Then Splitters are used to split Connection then TAP is used to give connection to each home.

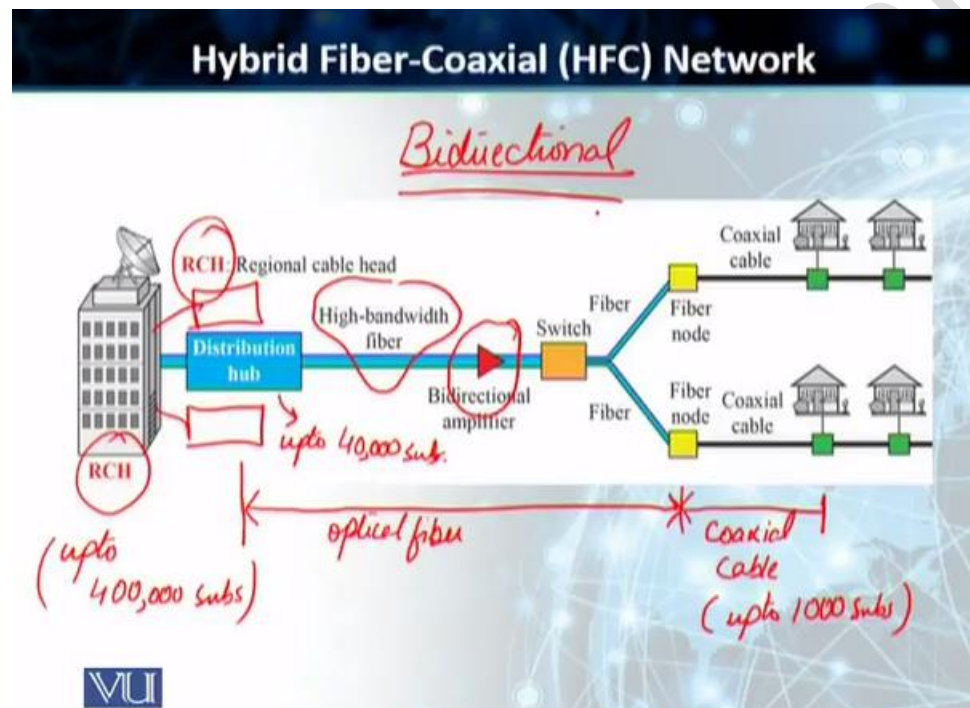
There is attenuation problem in signals.

CATV is only unidirectional.

Hybrid Fiber Coaxial HFC Network:

It is the 2nd generation of Cable network.

It uses combination of Fiber-optic and coaxial cable.



Here we are using High Bandwidth fiber cable

There are Distribution HUB and Fiber Node.

From fiber node to subscriber there is coaxial cable is used.

RCH (Regional Cable Head): 1 RCH can deal with upto 400k subscribes.

1 Distribution HUB can have upto 40k subs

Fiber cable send each signal to each coaxial cables.

For Fiber cable we don't need many amplifier

Reason for moving Coaxial to fiber is: to make communication Bidirectional.

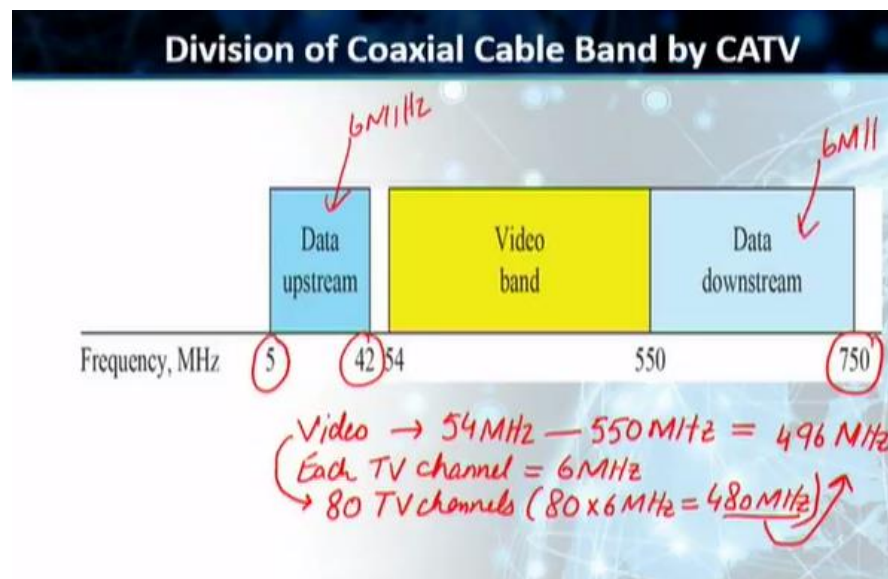
Cable TV for Data Transfer:

Some Cable companies are competing with telecom companies for high speed data transfer.

DSL technology provide quite high data rate connection to subscribed as twisted pair used that isn't that much efficient it that it is limited in carrying speed that make DSL less efficient

Solution: we can use cable TV network to carry data.

Division of Coaxial Cable Band by CATV



It divide band in 3 sub band

We have video band from 54 to 550 MHz = 490 MHz

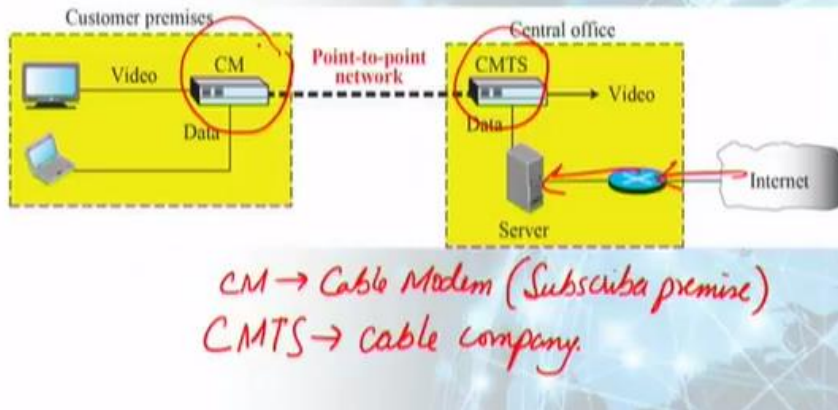
Each of TV channel takes 6MHz so we can total carry 80 TV channels with this band

Upstream to 5 to 46 . also divided into 6 MHz channel

Data downstream from 550 to 750; also divided into 6 MHz channel

Cable Model Transmission System (CMTS):

Cable Modem Transmission System (CMTS)



We must have CM (cable modem) at subscriber end to receive data

Then CMTS which should be at cable company end.

CMTS receive signals from internet through servers then send to CMTS and that is forwarded to CM.

It also receive signals from subscriber as well that is passes to internet...

Cable company would also have to ISP to provide data to users.

CM separates Video and Data as well.

Synchronous Optical Network (SONET):

It is a Wide area network transport technology to carry loads from WAN.

Fiber optics cable are used in SONET.

Fiber optics is capable to carry high data rate.

SONET uses Synchronous Digital Hierarchy SDH and ITU-T standard is used for it.

Architecture of SONET system consist of

- Signals
- Devices
- Connections

Signals:

- **Synchronous Transport Signals STS:** by ASMI Standard
- Optical Carriers OCs by ASMI Standard
- Synchronous Transport Module STM by ITU-T Standard

SONET Devices

- STS MUX/DEMUS
- Regenerators
- Add-Drop Multiplexer and Terminals

Connections:

- Section
- Line
- Path

SONET SIGNALS:

Different Type of Sonet signals used.

STS	OC	Rate (Mbps)	STM
STS-1	OC-1	51.840	STM-1
STS-3	OC-3	155.520	STM-3
STS-9	OC-9	466.560	STM-9
STS-12	OC-12	622.080	STM-12
STS-18	OC-18	933.120	STM-18
STS-24	OC-24	1244.160	STM-24
STS-36	OC-36	1866.230	STM-36
STS-48	OC-48	2488.320	STM-48
STS-96	OC-96	4976.640	STM-96
STS-192	OC-192	9953.280	STM-192

Each STS signals support specified signals rate.

STS and OC is ANSI SONET Standard:

Then we have STM standard that should data rate it support

STS-1 and OC1 supports 51.840 mbps it is somewhere equivalent to DS3 as we studied that gives 51.840 mbps combining header etc...

We can multiply to get STS standard

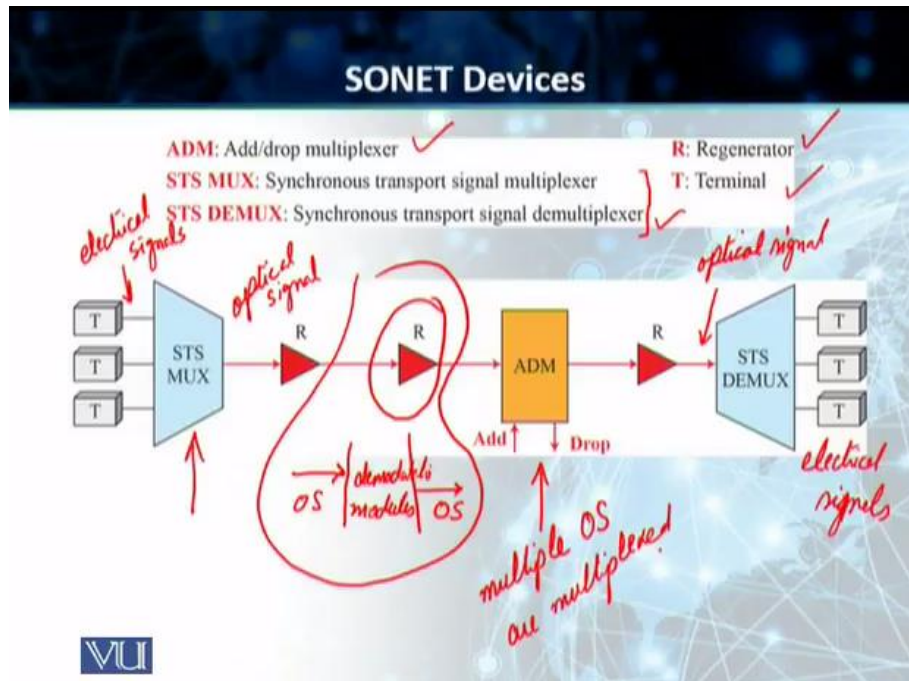
Like if we multiply STS1 x 3 we will get STS-3 Value and same we can do till STS-192 as well.

In ITU Standard which is minimum of it and it has minimum 155.520

ITU Standard is used in Europe and ANSI standard is used in USA.

SONET Devices:

ADM: Add/Drop Multiplexer.



Here we have STS MUX (multiplexer): it takes multiple electrical signals from end user terminal.

These signals are multiplexed by STS Mux that creates Optical signals that travels into Optical Line.

Demux receive these signals and demux it into electrical signals.

Regenerator (R): regenerate signals. The process of modulation and demodulation happen at R

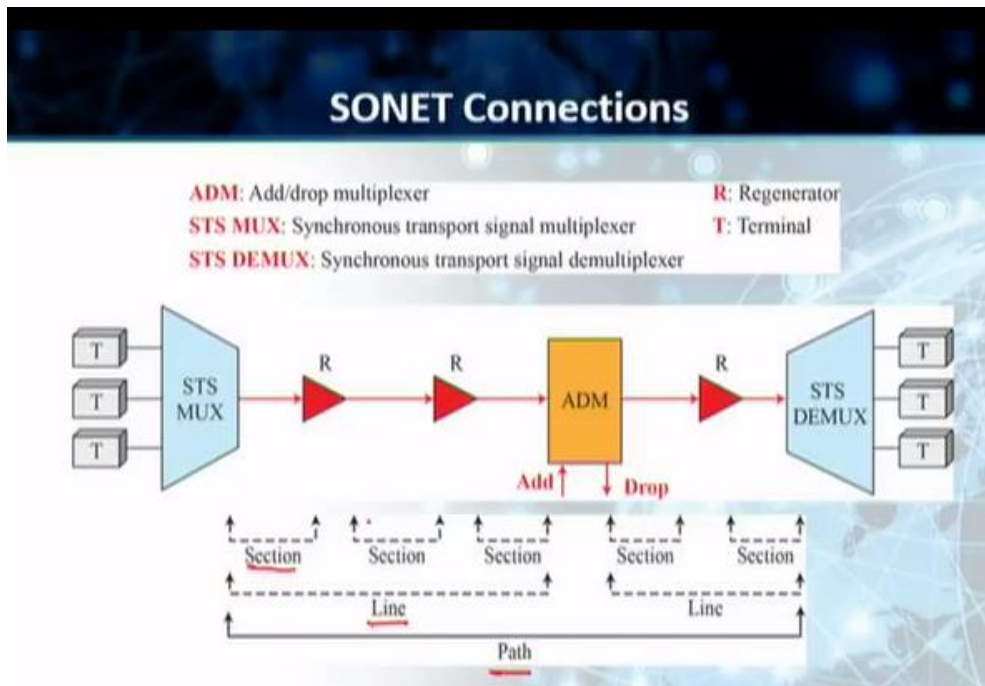
First it receive signals. Then it demodulate it to get electrical signals and then it module these electrical signals to get Optical signals back.

Then we have ADM (Add drop multiplexer): Multiple Optical Signals are multiplexed here so they could transfer over high bandwidth Optical Fiber Cable.

Regenerators are used in line.

In the last we get Terminals

SONET Connections:



1: Section: that connect two neighbour devices whatever device it when STS MUX to R, R to R anything.

Then we have LINE that connects Various section like here in pic we have STS To R, R to R and R to ADM

Path connect all lines from STS MUX to STS DEMUX or source to destination

SONET Layers:

4 Different layers as per sonet standard

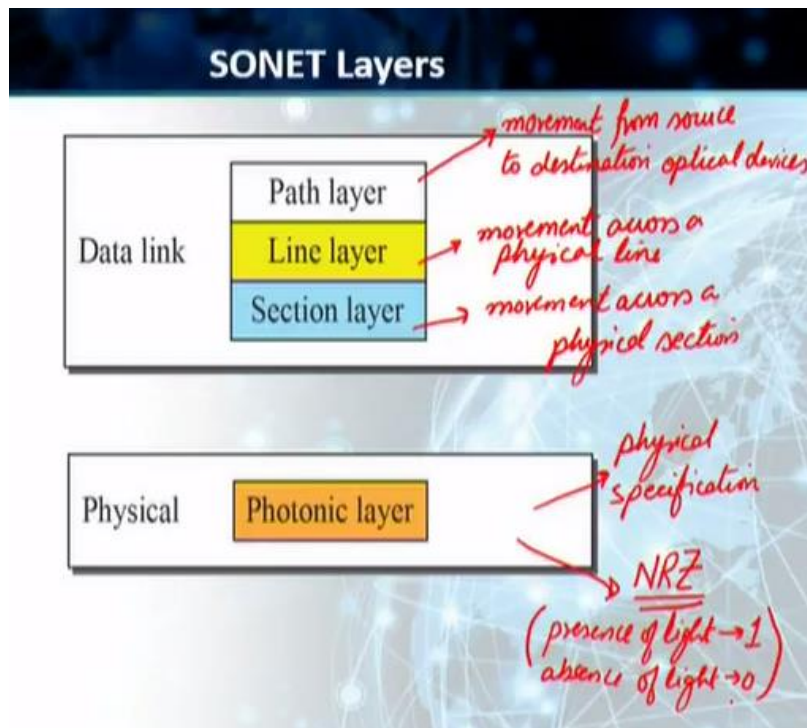
- Path Layer
- Line Layer
- Section Layer
- Photonic Layer (equal to physical layer).

At data link layer we got 3 layers.

Path Layer: following same concept of connection or movement from source to destination.

Liner Layer: is responsible for the movement of signals across physical line. Line layer overhead is added at this layer.

Section Layer: it is responsible for the movement of signals across physical section.



Physical layer consist of 1 layer

Photonic Layer: it specified physical specification that we want to use.

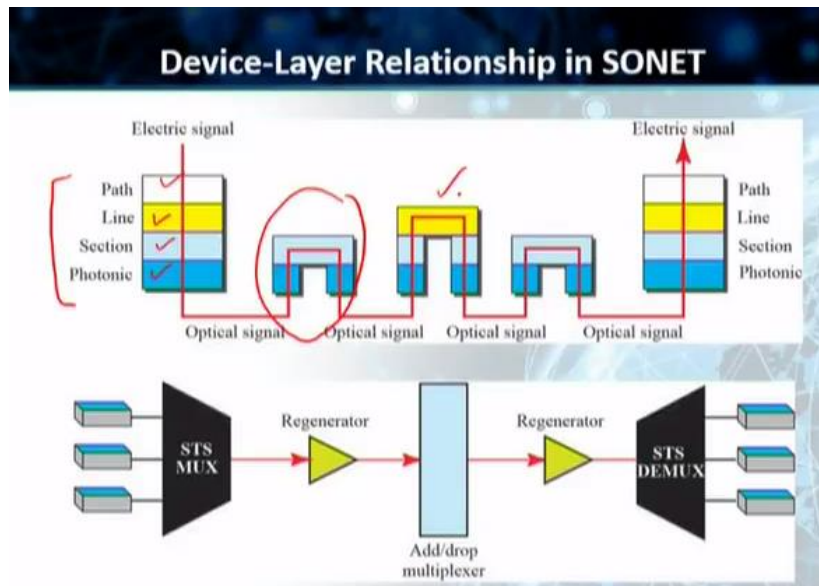
It uses NRZ coddling.

As SONET uses Light as its fiber optic

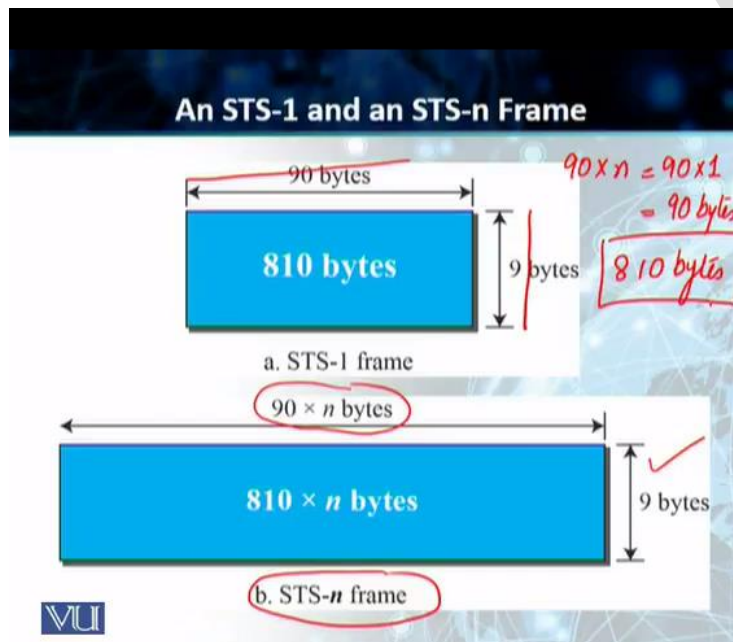
Then Presence of Light = 1

Absence of Light in line = 0

If we map these layers into devices so it will be as follows shown in figure



SONET Frames:



Each of STS-n Signals composed of 8000 frames.

It consist of two dimensional matrix of bytes that has 9 rows and $90 \times n$ columns.

In case of STS-1 will have 9 rows with 90 columns (819 Bytes)

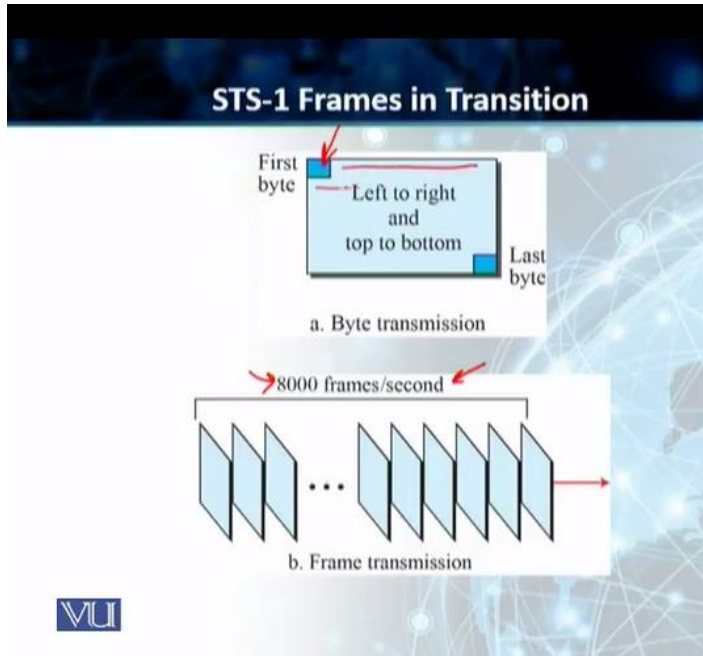
For STS-3 is 9 rows with 270 columns (2430 Bytes)

Here we multiple rows x columns to get bytes

As shown in figure.

I.e $90 \times 90 = 810$ bytes

Transmission:



STS signals are transmitted on 8000 frames per second

Example:

To find data rate of STS 1 Signals.

$N = 1$

As each STS 1 Signal send 8k frames per second

1 frame consist of 9 rows 90 columns.

Further solved here

Example

Find the data rate of an STS-1 signal

$$\begin{aligned} & 8000 \text{ frames per second} \\ & 9 \text{ bytes} \times (1 \times 90) \text{ bytes} \\ & \text{Data Rate} = 8000 \times 9(1 \times 90) \\ & = \underline{\underline{51.84 \text{ Mbps}}} \end{aligned}$$

STS-3: we have Data rate 8000

Here $n = 1$

So we will have

$$8000 \times 9 (3 \times 90)$$

$$= 155.52 \text{ Mbps}$$

Example

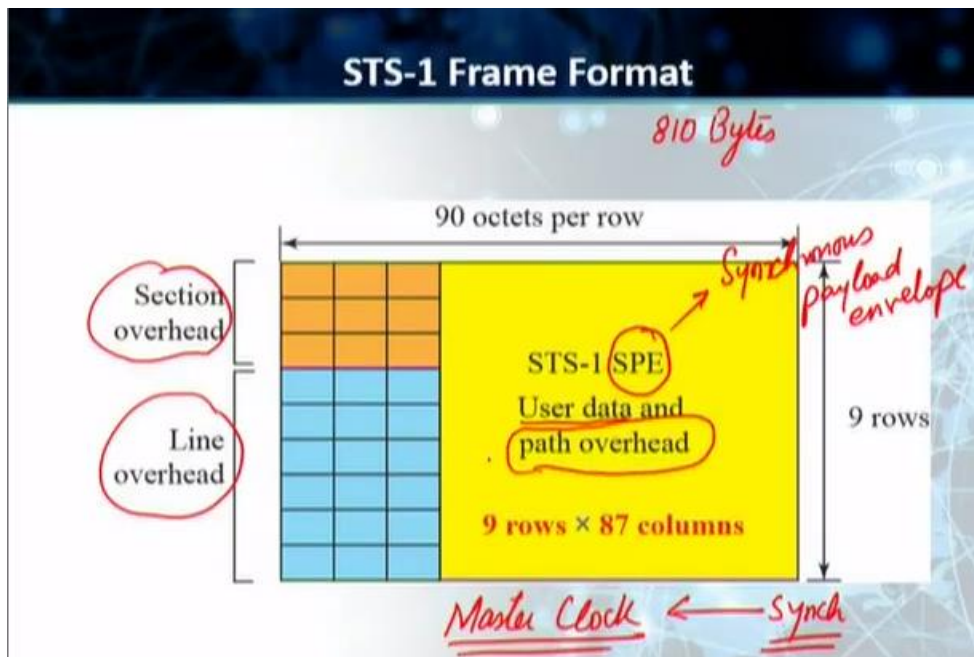
Find the data rate of an STS-3 signal

$$n=3$$

$$\begin{aligned} \text{Data Rate} &= 8000 \times 9 (3 \times 90) \\ &= \underline{\underline{155.52 \text{ Mbps}}} \end{aligned}$$

$$\begin{aligned} & \text{STS-1 Data Rate} \times 3 \\ &= \text{STS-3 Data Rate} \end{aligned}$$

STS-1 Frame Format:



It has 90 column and 9 rows

Section over head at first 3 rows.

Following 6 are line overhead

Remaining are STS-1 Synchronous Payload envelope

User data and path resides into this particular frame as well.

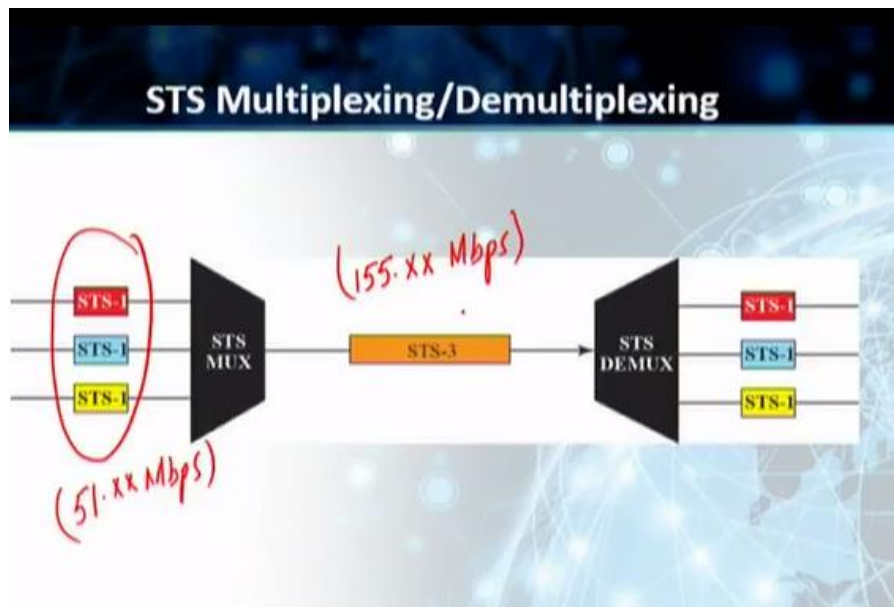
STS Multiplexing

SONET uses STS multiplexing.

Frames of lower rate can be synchronously time-division multiplexed into high rate frames.

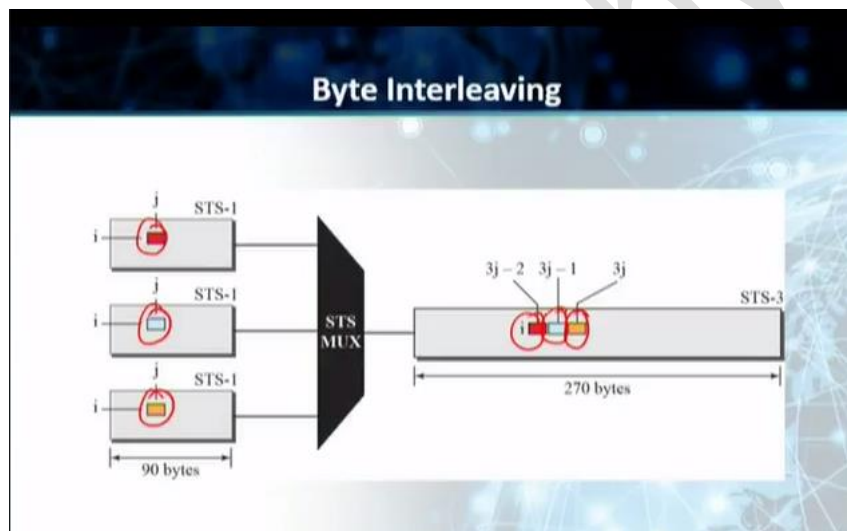
Example STS 1 signals can combined x 3 to get STS 3 signals.

STS Multiplexing/ Demultiplexing



As we 3 STS 1 combine them with STS MUX will get us STS 3 and then DEMUX it for 3 STS again

Byte Interleaving:



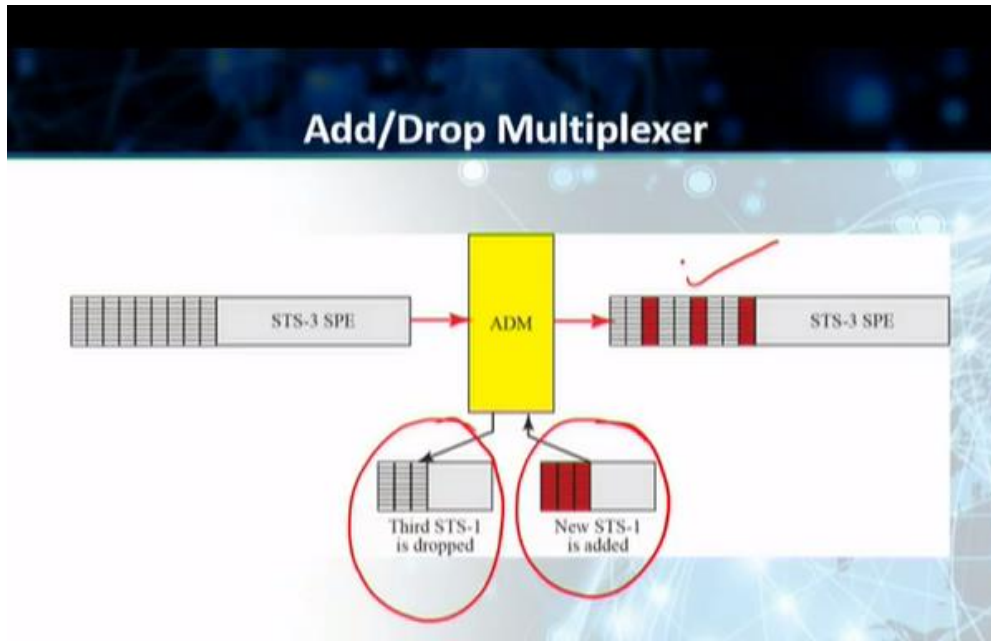
It used to get SONET frame.

Let suppose we have 3 STS-1 Signals these are multiplexed into STS-3

Each has 3 Bytes from each STS-1 Signals

Every byte of STS-1 will keep its row position. Column position changes when we do byte interleaving.

Add/Drop Multiplexer:



for several STS1 signals into STS-n Signals it is done by path layer in sonnet less.

There are several mux used let 2 being used

In between these muxers we have ADM, It replaces 1 signals without.

In ADM it actually operates on the Line Layer not path layer. It act as a switch.

Like we have 2 STS signals and we dropped from 1 and added from other

Like there is source multiple from others as we have STS in red and there is STS-3 gray so we dropped several STS-3 into STS-3 and Added other STS-1 with the help of ADM

SONET Network:

It can be used for High Speed Data carrying network. If there is high load / speed it required we can use SONET over there i.e ATM or IP

SONET can be divided into 3 types

- Linear Networks
- Ring Networks
- Mesh Networks

Linear Networks:

Further Types:

- Point to Point

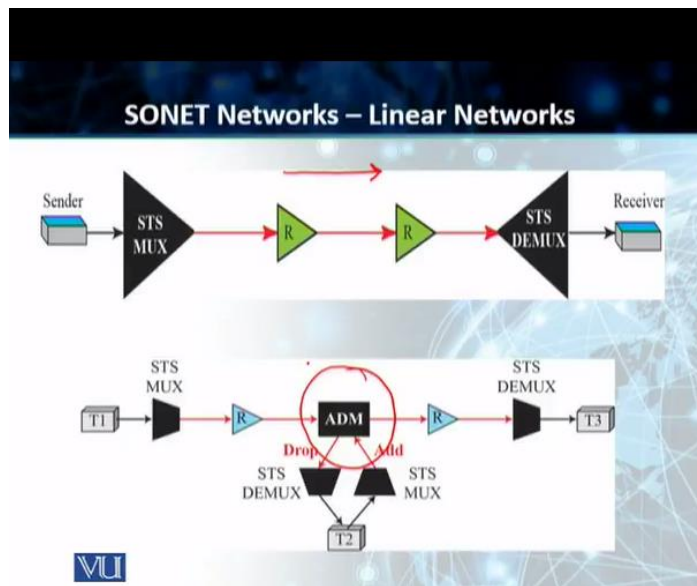
- Multipoint

Ring Networks:

- UPSR Unidirectional Path Switching Ring:
- BLSR: Biadirectional Line Switching Ring:

MESH

Linear Network



Point to Point.

Simple with STS Mux and Demux and several regenerators.

It is unidirectional

Multipoint

In case of multipoint we use ADM

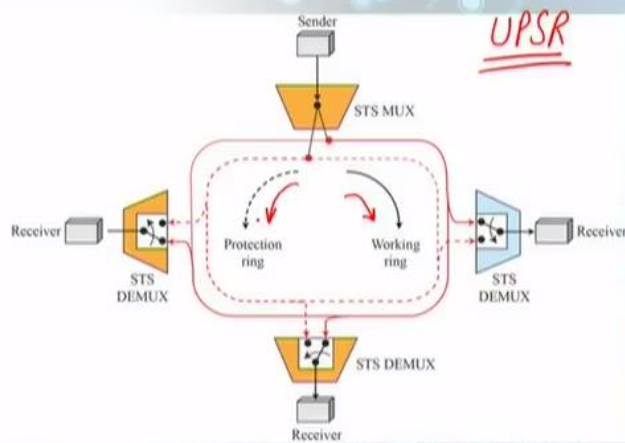
ADM is used when we have more than one signals need to be added up.

It can be bidirectional as well

Ring Network:

UPSR:

SONET Networks – Ring Networks



In this we got two rings 1 Working ring in clockwise direction

2nd protection ring: in anticlockwise direction

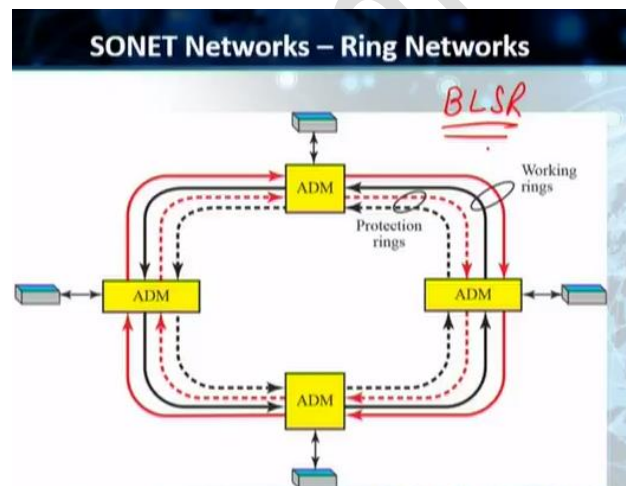
Both rings carry same type of data.

Each node receives same data from two different lines.

So node will select the best option. It has fast recovery rate.

But half bandwidth is wasted as two rings are doing the job of one.

BLSR:



Here we have bidirectional communication.

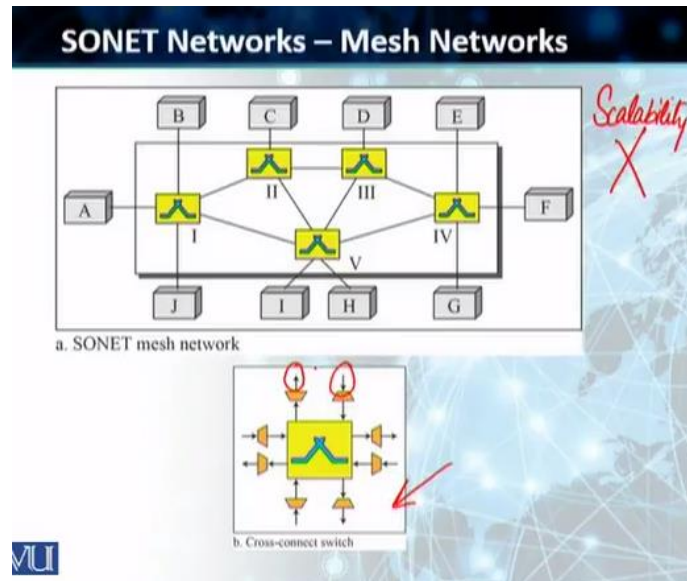
We need two for working line and two rings for Protection line.

It used 4 links. Failure rate is reduced while efficiency is reduced as well.

Problem in Ring Networks was Scalability.

We don't need to modify ring in case of adding any device but we need upgrade multiplexer as well

MESH Networks:



It was introduced coz of Scalability in Ring networks

It has a switch that connects multiple point.

It is bidirectional it do both operation of MUX and DEMUX for each device.

ATM Asynchronous Transfer MODE

It is a switched wide area network based on a cell relay protocol.

The combination of ATM and SONET will allow us to get extremely high speed.

Problems associated with existing systems:

ATM uses fix frame networks which is Cell not different frame sizes

Problems:

- Frame Networks (size): Most of the technology has varying frame size that can disable multiple links. MUX will pickup large frame if frames are varying and it will ignore other small frames.
- Mixed Network Traffic: We have different type of traffic like audio, video, etc... different type of data type need to be treated differently

Solution for these:

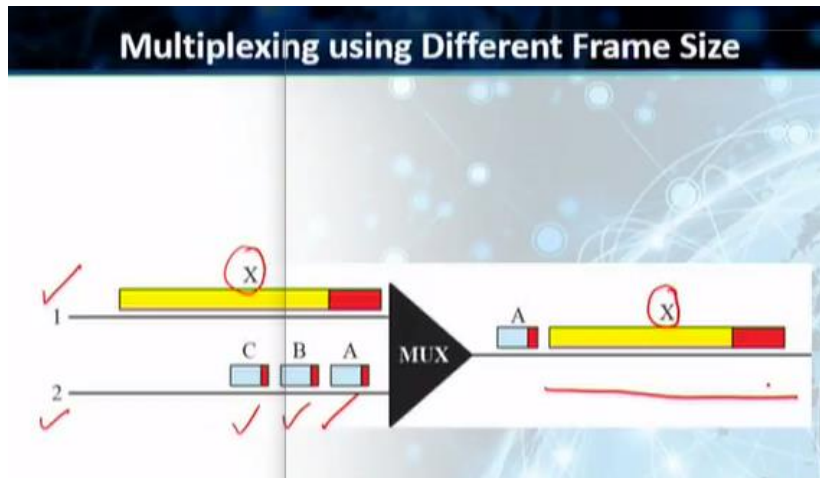
Cell based system introduced in the solution of these problems

ATM also introduced along with Cell

In this we use a fix slot and cell goes into these slots

We take a data and create fixed size slots and cells goes in these slots.

Multiplexing using Different Frame Size:

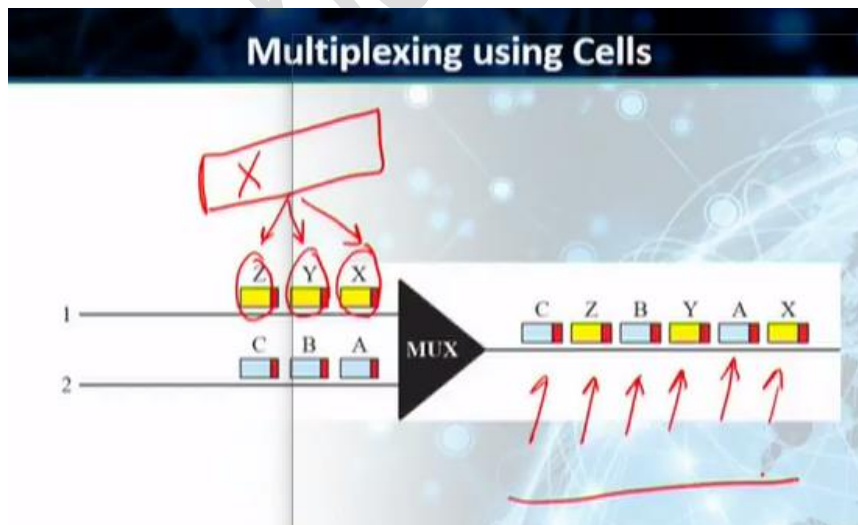


Let we have two lines connected to MUX

In 1 we have large frame which is x and in 2 we have multiple similar size

And if large come first then mux will give it priority in old system.

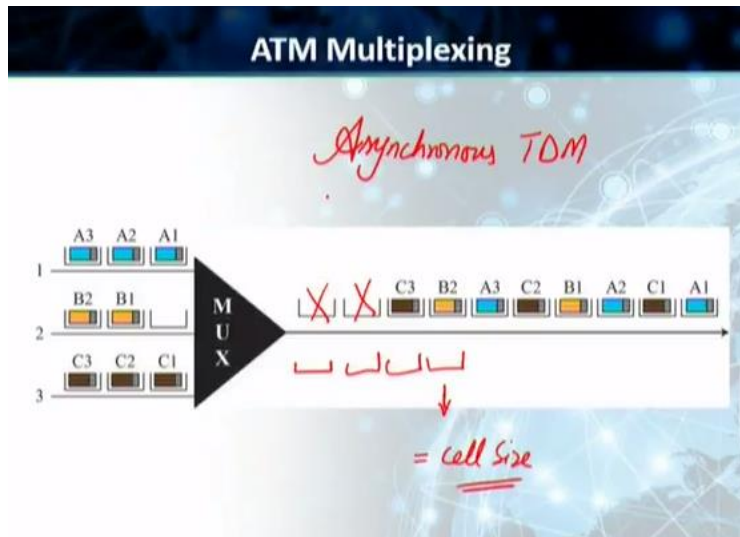
Multiplexing Using Cells:



In this data into multiple different cells.

In this MUX will select according to their turn.

ATM Multiplexing:



It used Asynchronous Time division multiplexing.

We have multiple slots.

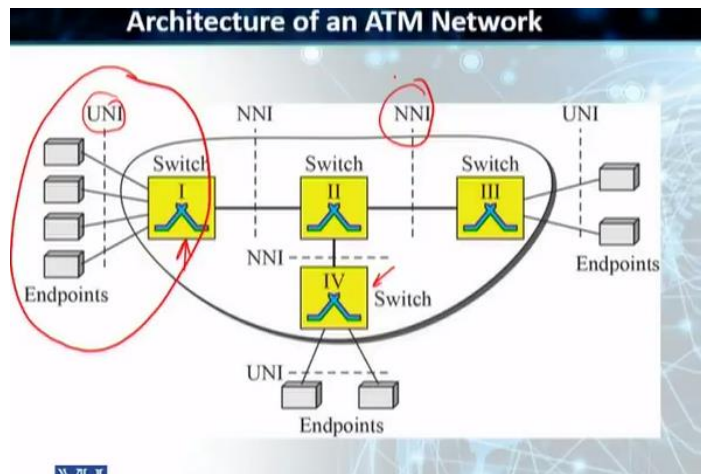
Each uses similar cell size. Slots = cells size

Architecture of ATM

Its is cell switched network.

User access device in it are called endpoints which are connected through UNI User to network interface to the switches inside the networks.

The switches are connected through Network to Network interfaces NNIs

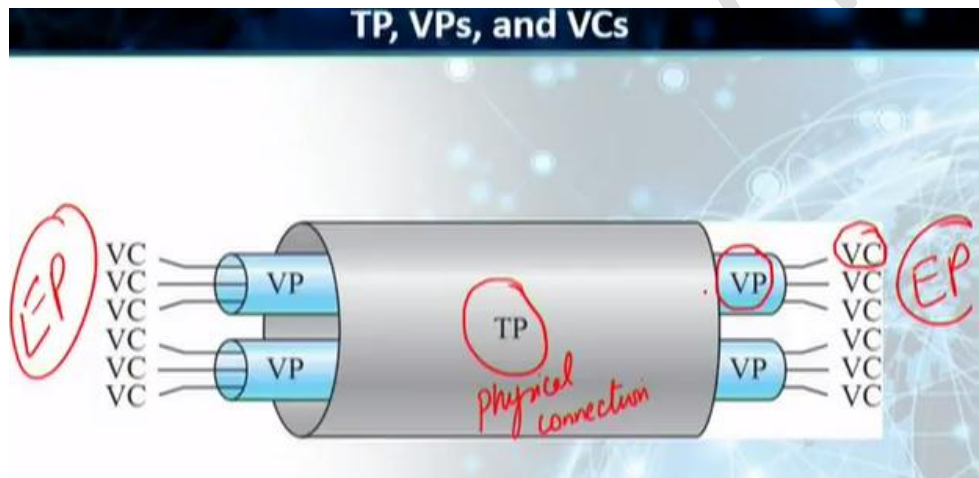


Here in this we have switches in middle

There is a switch that is connected to UNI

And multiple switches are connected to each other

Connection btw two points is NNI

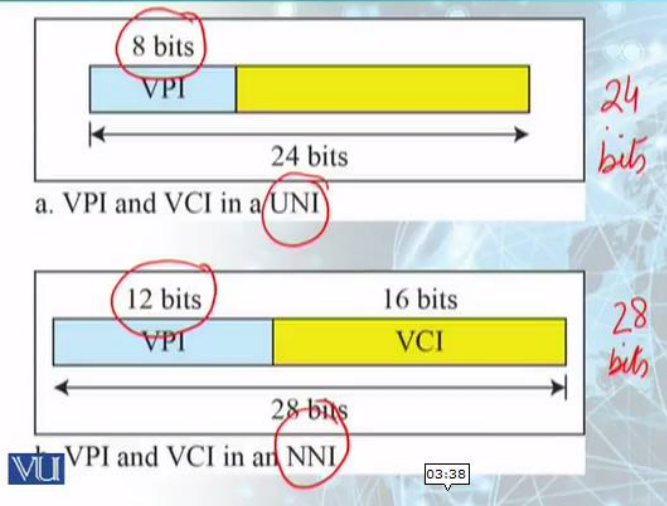


Here we have a TP that has several virtual path that provide set of connection btw two switches.

There are virtual circuit into VP.

Virtual Connection Identifiers in UNIs and NNIs

Virtual connection identifiers in UNIs & NNIs



We use VPI = virtual path identifier

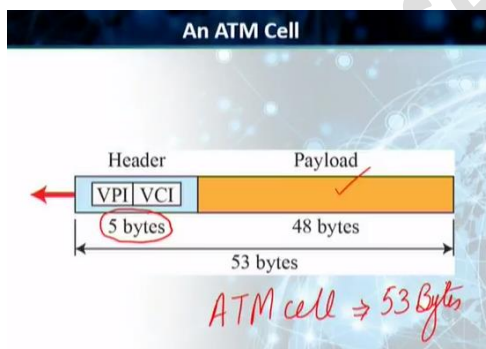
VCI = virtual circuit identifier.

VPI defines specified virtual path VP

VCI defines VC.

VPI is same for all Virtual connection into a VP

AN ATM CELL:

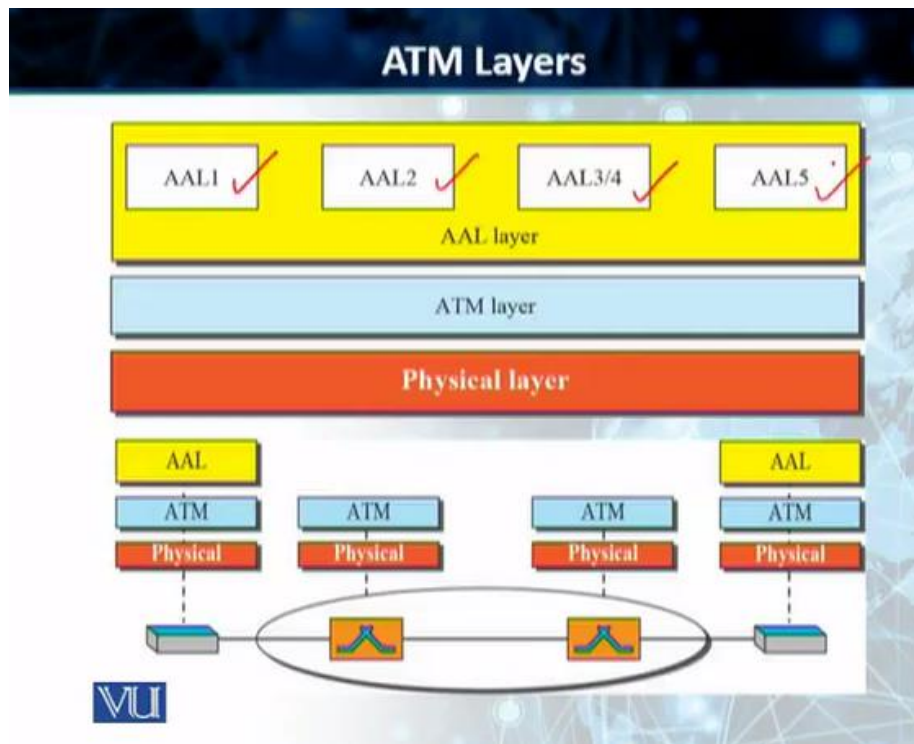


Here we have an atm cell

It has 5 bytes dedicated for header which is composed of VPI and CPI

Then we have Payload of 48 bytes

ATM Layers:



We got physical layer then ATM layer and then AAL = Application adaptation layer

Physical layer deals with the medium.

We have ATM layer.

AAL Layer is further divided into 4 different layers each one performs different function.

Wireless

Is the one of the fastest growing technologies.

We want to connect system without wire.

It can be found everywhere in home, colleges, offices...etc

WIFI is most common Wireless network type.

Wireless Architectural Comparison:

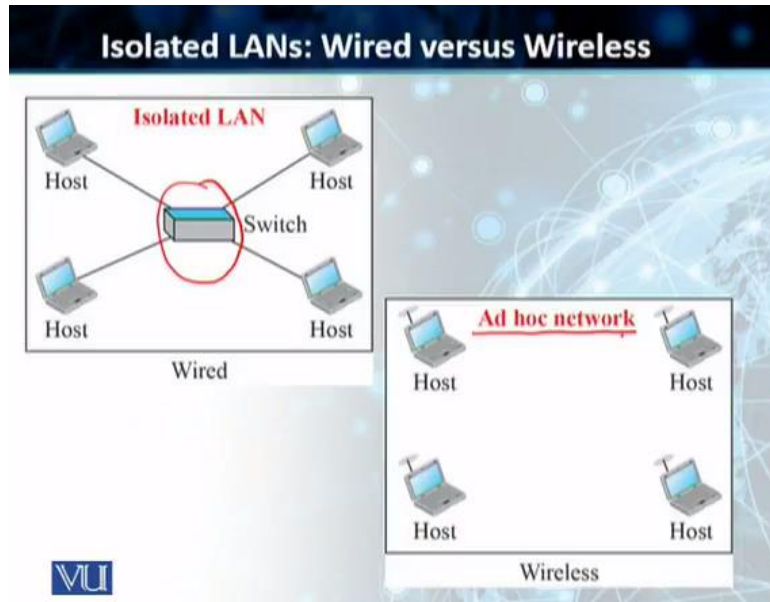
Comparing it with Wired LAN

Medium in wired LAN we required cables but in case of wireless medium is air.

Wireless work as broadcast.

Hosts: in case of wired were different end user machines which were connect to a central switch using cables moving in wired was tough coz we need to change direction of switch and wires etc... In wireless our host is free to move.

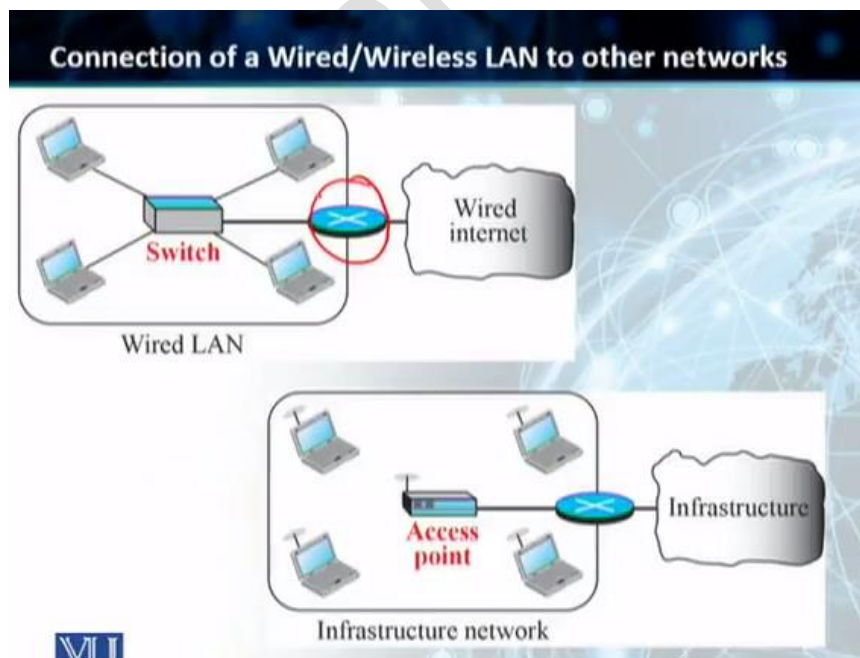
Isolated LANs:



In Wired hosts are connected to switch using wire.

While Wireless has nothing as switch

Connection of Wired/Wireless LAN to Others Networks.



Wireless use a centralized device which is access point which is connected to the wired connection but it provides that wired connection in the form of wifi

Wireless LAN exist on first two layers of TCP/IP protocol

To convert wire to wireless all we need to do is replace switch with access point.

And our host needs to have WIFI card to receive wireless signals.

Characteristics of Wireless LAN:

WIFI = Wireless Fidelity

Several Characteristics are related to wireless LANs not available for Wired LAN

- **Attenuation:** electromagnetic Signals: they get weaker as they travel. They are disperse in the air as well. When strength is not enough then attenuation happens.
- **Interference:** This happens when we using Air as communication medium there are chances of interference from all other signals available in the air.
- **Multipath Propagation:** We receive signals from our device but we receive multiple signals of the same signals because of continuous reflection which room, building etc.. it comes after reflecting that create noise and that is multipath propagation.
- **Error:** Due to all these, there must be error detection. We need to make sure our signals are strong enough so Noise would not effect it.

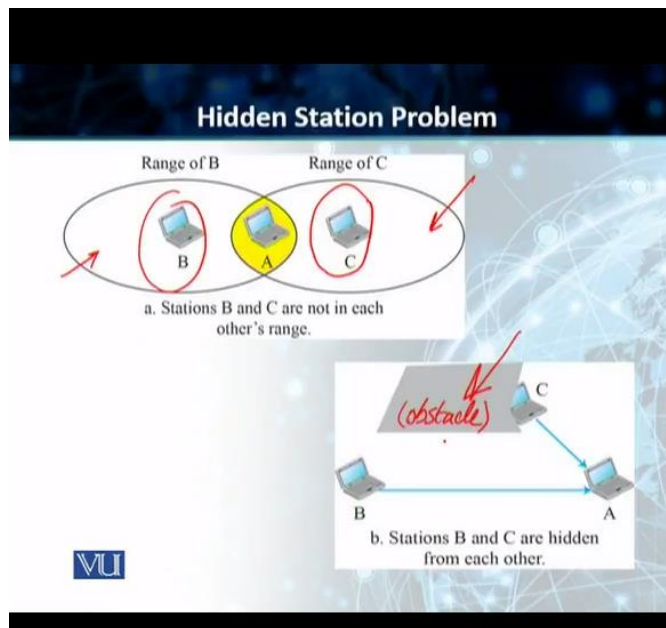
Access Control:

Most important issue in Wireless LAN is how a wireless host can get access to shared medium (Air).

Access control technology we used in wired was CSMA/CD

But we can't use it for Wireless medium as Wireless hosts don't have power to send and receive at the same time. b/c wireless hosts are mostly battery operator and we need more power.

Hidden Station Problem:



Here as Station B and C aren't in each other range. In that case they will not be able to send so CSMA/CD will fail.

Here in 2nd Scenario: C and B are hidden from each other while A can access both as there is obstacle in btw C and B that obstacle is blocking signals that causes the problem.

Distance:

Distance can be large btw two wireless source.

Due to large distance there are chances us to not receive signals.

IEEE 802.11 Project:

Wireless Standard that come from IEEE

IEEE has defined the specifications for the wireless LAN, that specs are called IEEE 802.11, that covers physical and data-link layers.

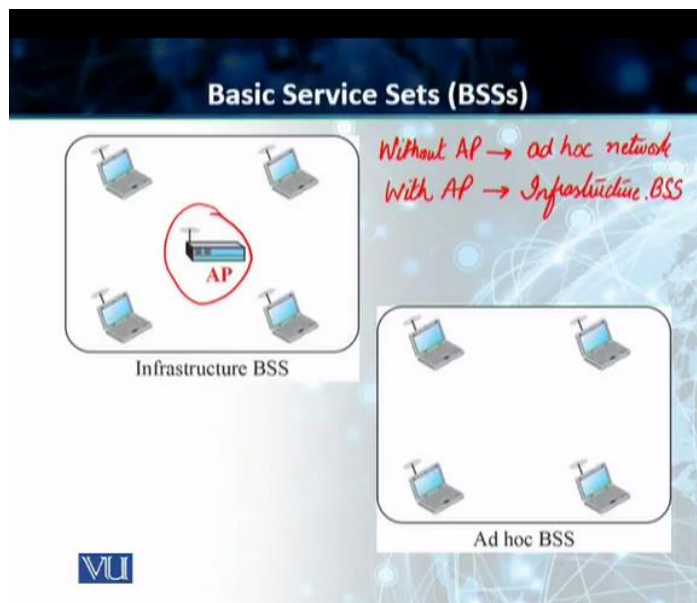
Sometime it also called Wireless Ethernet

Arhitecture of IEEE 802.11

Two services offers

- The Basic service Set (BSS)
- The Extended Services Set (ESS)

Basic Service Set (BSSs)



Infrastructure BSS: an Access point is used btw.

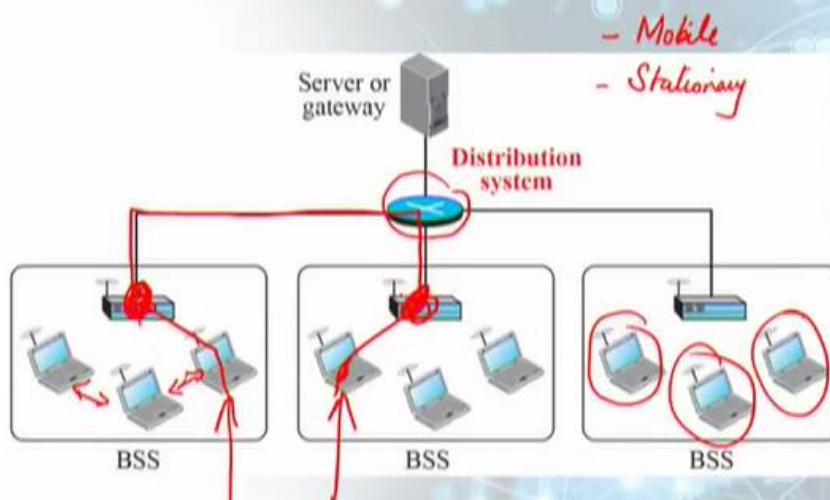
Host can communicate with it but the Host can send data to external BSS for this Access point is required.

Without AP we call it Ad hoc BSS Stations can be formed without need of AP.

Without AP wireless node still can communicate with each other that is said to be: Ad hoc BSS

Extended Service Set (ESS)

Extended Service Set (ESS)



It is made up of two or more BSS

All these BSS are connected to a Distribution system that can be wired or wireless.

It uses two type of station

1st one is mobile other one is stationary

Normal Station like our laptops are Mobile

Our AP are Stationary.

When these BSS are connected all stations connected to a distribution system can access without AP as well.

Types of Stations Defined by IEEE

These types are defined on the base of their mobility

- No-Transition Mobility: consist of all stationary station or station that moves with BSS
- BSS-Transition Mobility: that can actually move within BSS
- ESS-Transition Mobility: these are stations that can move from one ESS to another ESS

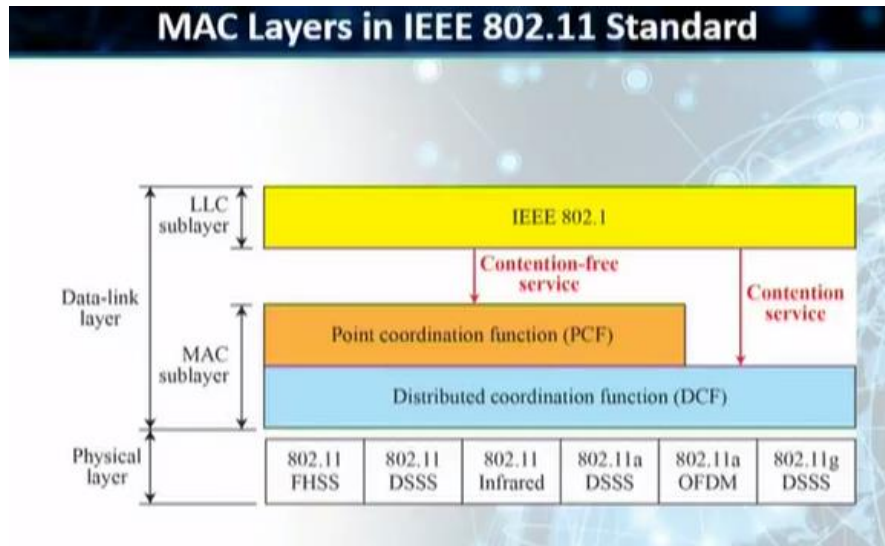
MAC Sub-Layer:

IEEE 802.11 Defines two MAC Sub-layers:

- The distributed coordination function (DCF)

- The point Coordination function (PCF)

MAC Layers in IEEE 802.11 Standard.



DCF is contention service. Which shows station those which are using this service to take ownership medium only then they can transfer

PCF: Its contention free. We will use it for station that has higher priority.

We got LLC sublayer as part of Data-link layer and physical layer.

CSMA/CA and NAV

CSMA/ CA Collision Avoidance.

As we discussed that has RTS, CTS and that do ACK as well.

In case of NAV

In this case if 1 station acquire access to the medium how the collision avoidance is going to be implemented. How system will tell station to when.

When Station send RTS frame it also include the duration of the time it needs to occupy the channel

Once this sent all other station will create a timer and all other station will not be able to send until the timer ends

Each station will check timer before sending.

Problem occurs when our Control signals get corrupted.

If our RTS get loss in this case the RTS is received by station B but not by C and further.

As the C and D are not in the range of a they will not receive RTS

As IEEE 802.11 defines even if CTS is received without RTS then Clear to send data is sent to all other station on the channel.

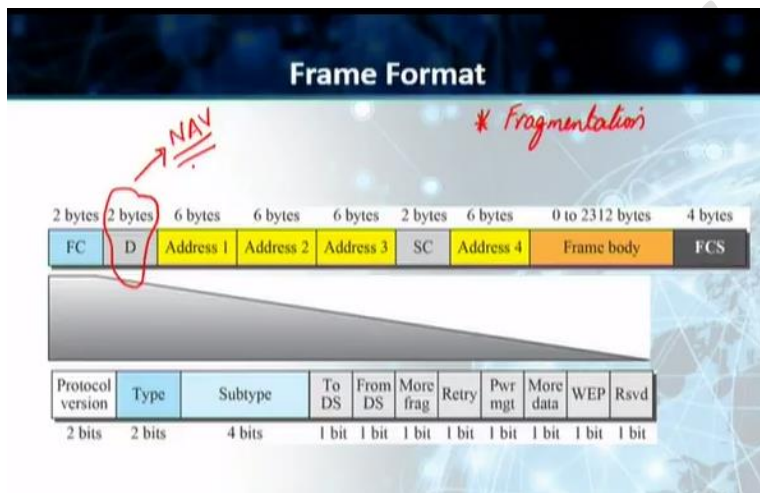
MAC Sub-Layer:

The Point Coordination Function PCF:

Its optional it is used in extremely time sensitive cases.

It is better technique than as consider than time.

Frame Format:



In Wireless environment

Fragmentation: is protocol in which we divide large frames into multiple smaller frames.

MAC Layer Frame consist of 9 fields

FC Field: it define type of Frame and control information we need to send

Subfields in FC Field

Subfields in FC field

Field	Explanation
Version	Current version is 0
Type	Type of information: management (00), control (01), or data (10)
Subtype	Subtype of each type (see Table 6.2)
To DS	Defined later
From DS	Defined later
More flag	When set to 1, means more fragments
Retry	When set to 1, means retransmitted frame
Pwr mgt	When set to 1, means station is in power management mode
More data	When set to 1, means station has more data to send
WEP	Wired equivalent privacy (encryption implemented)
Rsvd	Reserved

D Field: that defines the duration of transmission. NAV is set here

Then we have Four Address Fields:

In all these fields where we add all the addresses.

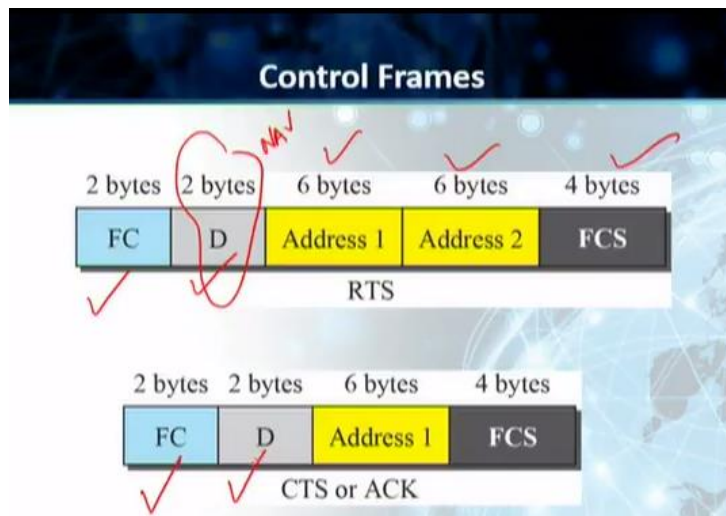
Then we have Main Frame Body: btw 0 to 2312 it contain information of Type and Subtype

FCS: Frame check sequence to check the frames.

Frames Types:

- Management Frames: Used for Initial communication
- Control Frames: used for request to send RTS, CTS clear to send and ACK
- Data Frames

Control Frame:



For RTS We got FC and D as discussed then we have Addresses and FCS

CTS and ACK all are same with 1 address

Sub types

Values of Subfields in Control Frames

<i>Subtype</i>	<i>Meaning</i>
1011	Request to send (RTS)
1100	Clear to send (CTS)
1101	Acknowledgment (ACK)

Physical Layer:

All physical Implementations comes under IEEE except the infrared which operates under the industrial, scientific, and medical ISM band, that defines 3 unlicensed bands in 3 ranges.

- 902 – 928 MHz
- 2.400 – 4.835 GHz
- 5.527 – 5.853 GHz

Specifications

IEEE	Technique	Band	Modulation	Rate (Mbps)
802.11	FHSS ✓	2,400–4,835 GHz	FSK ✓	1 and 2 ✓
	DSSS ✓	2,400–4,835 GHz	PSK ✓	1 and 2 ✓
	None ✓	Infrared	PPM ✓	1 and 2 ✓
802.11a	OFDM	5.725–5,850 GHz	PSK or QAM	6 to 54
802.11b	DSSS	2,400–4,835 GHz	PSK	5.5 and 11
802.11g	OFDM	2,400–4,835 GHz	Different	22 and 54
802.11n	OFDM	5.725–5,850 GHz	Different	600

Most current is OFDM which is capable of receiving 600 Mbps

Bluetooth:

It is wireless LAN Technology designed to connect devices of different functions when they are at a short distance from each other

It is an Ad hoc network. That doesn't required access point.

The devices find each other and make a network called piconet

Bluetooth Technology is the implementation of a protocol defined by IEEE 802.15 Standard.

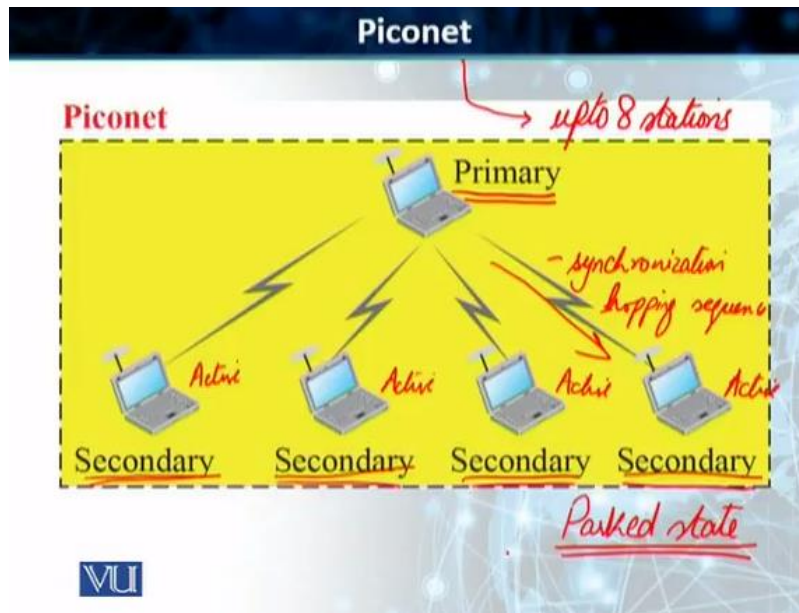
It was started by archerson company.

The Standard defines a Wireless personal Area network (PAN) operable in limited areas.

Bluetooth types:

- Piconet
- Scatternet

Piconet:



In this we can have upto 8 stations (devices)

In this case one of the device will be primary and other will be upto 7 secondary devices.

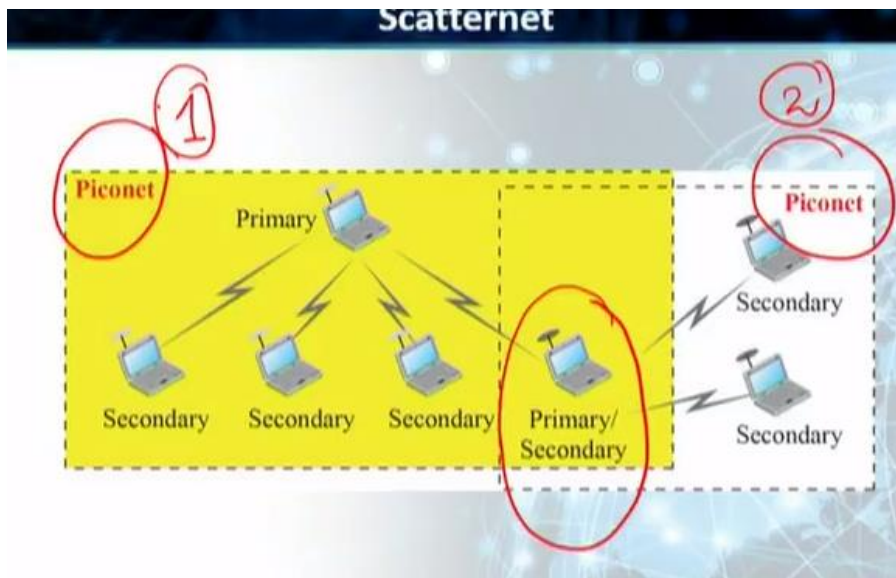
All the secondary get every type of data i.e time required. Hopping, sync from primary.

One to one communication or multi communication can be done it.

We can have more secondary but these will be parked now active only 7 could be active at a time

To make active parked device we have to remove it from active and make it active but will we have to make its space in active device.

Scatternet:



Multiple Piconet can be combined to make a network connection.

In this 1 station can be primary and secondary as well.

Bluetooth Devices:

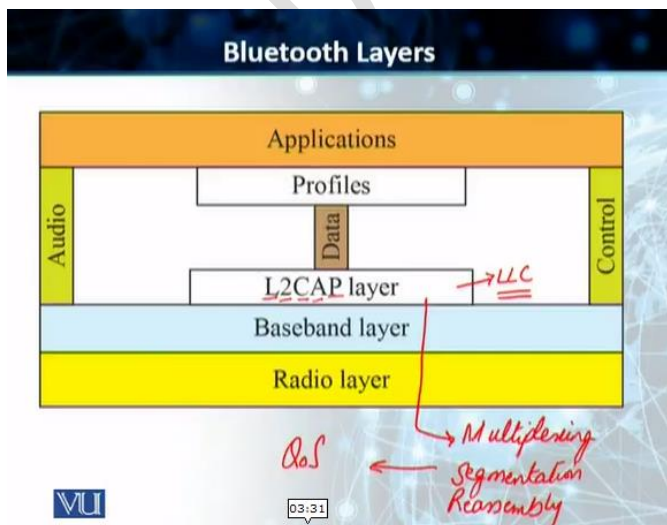
Every Bluetooth device got a built in short range transmitter.

The current Data rate is 1 Mbps with 2.4 Ghz bandwidth

This mean that there is possibility of interference btw the IEEE 802.11b wireless LANs and Bluetooth LANs as they are working in same LANs

Bluetooth's layer don't match with the internet layer we discussed already.

Bluetooth Layers:



Radio layer: used for transmission

Baseband Layer: as its required in all wireless devices

And other stuff control, audio, application

We have

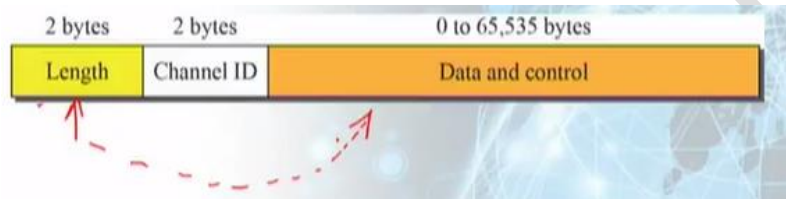
L2CAP Layer: it Is equal to LLC in LAN networks. It has similar function as LLC, Multiplexing is done here, segmentation, reassembly done here.

QoS is taken care at L2CAP as well

L2CAP Data Packet Format:

Length: that defines Data length then we have channel ID

And we have our data and Control with 0 to 65,535 bytes.



Single Secondary Communication:

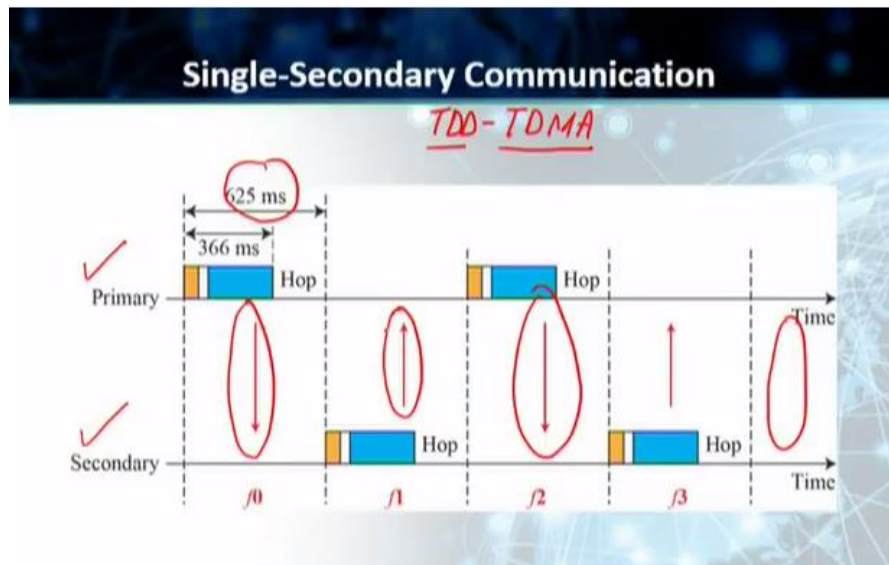
It uses TDD – TDMA for the communication.

In this case both devices can send and receive.

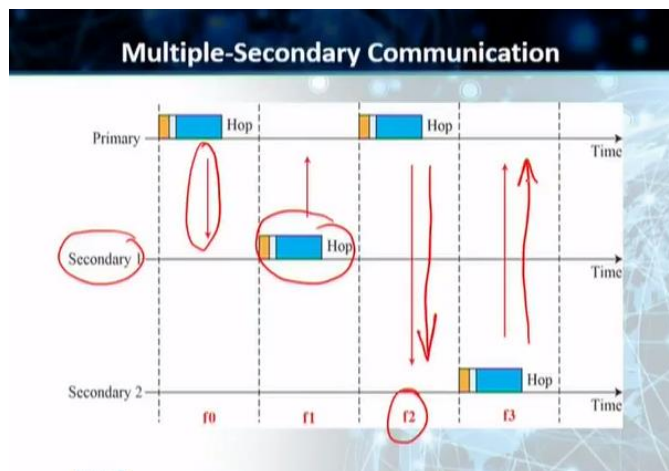
In this primary can send and receive as well while secondary as well but not at the same time.

Here we have 625 ms time slot In the hop used to send data to secondary.

Then we have secondary is also sending data but in different slot



Multiple Secondary Communication



In this we have Primary Secondary 1 and Secondary 2

In this 1 station can make communication at a time.

In this primary will communicate with secondary 1 and it will send message back then primary will contact secondary 2 again.

Connecting Devices:

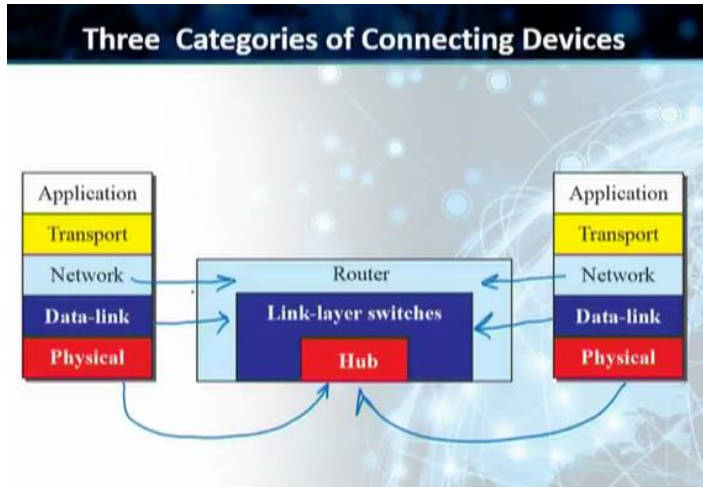
Normally host and network don't work in isolation they required a connecting device to make network so many host could connect together.

Connecting devices also connect networks to make internet.

Connecting devices can work in different layers of the internet model.

Types of connecting Devices:

- 1) Hubs: it works on Physical layer.
- 2) Link-Layer Switches: it's a layer two device work on Data link layer
- 3) Routers: it's a network layer



Hub: device that operates on the physical layer.

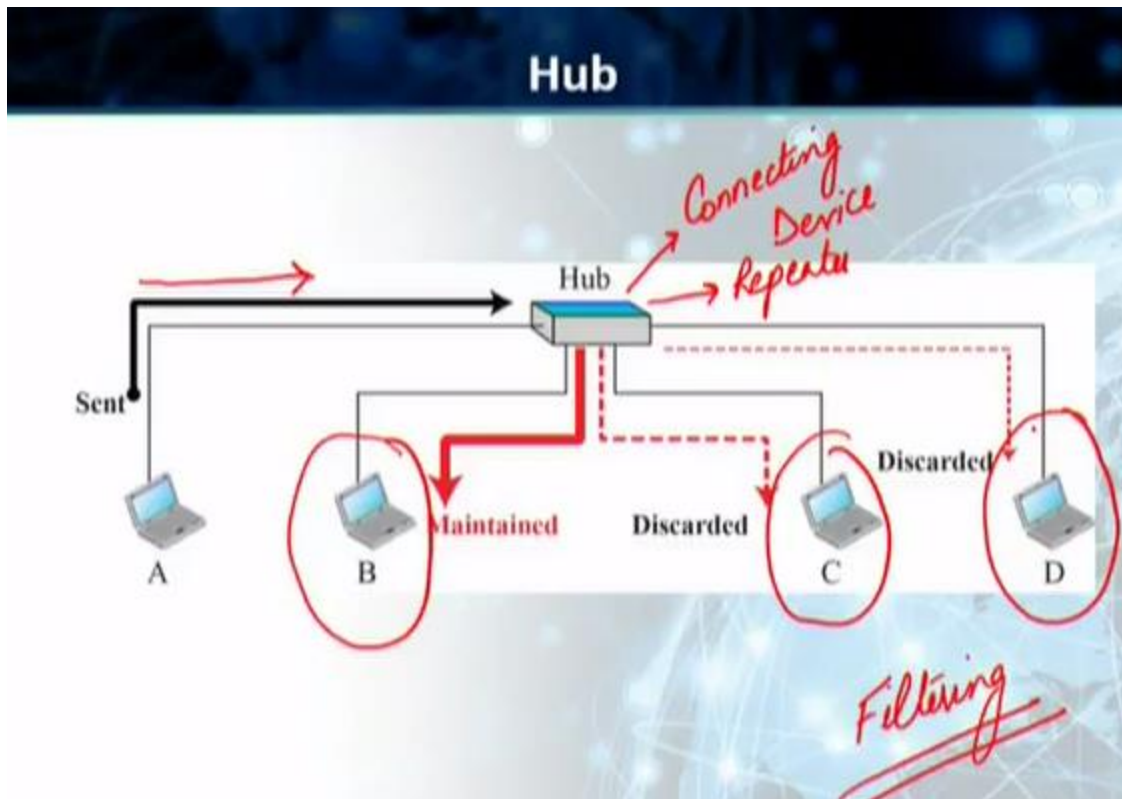
It consist of signals that carry information with a network. It can travel within a fixed distance before attenuation impacts the data.

A Hub (Repeater) receives a signal, before it become too weak or corrupted regenerates it.

This happens when station is wired to the hub.

After regeneration of data hub resend data to output.

Hub can't detect noise, if hub receive noise it will regenerate signals and noise as well.



Here in this example

Hub is connected to 4 different stations using star technology.

Hub is not only acting as connecting device but being repeater as well.

Hub can't perform filtering operation. It can filter noise etc.

Hub only regenerates the data. Whether its data or noise.

In this data pass through all device all will check header file if data belongs to them then they will keep it otherwise will discard it

Link-Layer Switches:

A link layer operates in both physical and data-link layers.

As it's a physical layer device, it regenerates the signals that it receives.

As its link layer device as well, switch can check the MAC address (source and Destination) contained in a frame.

Switch Versus Hub:

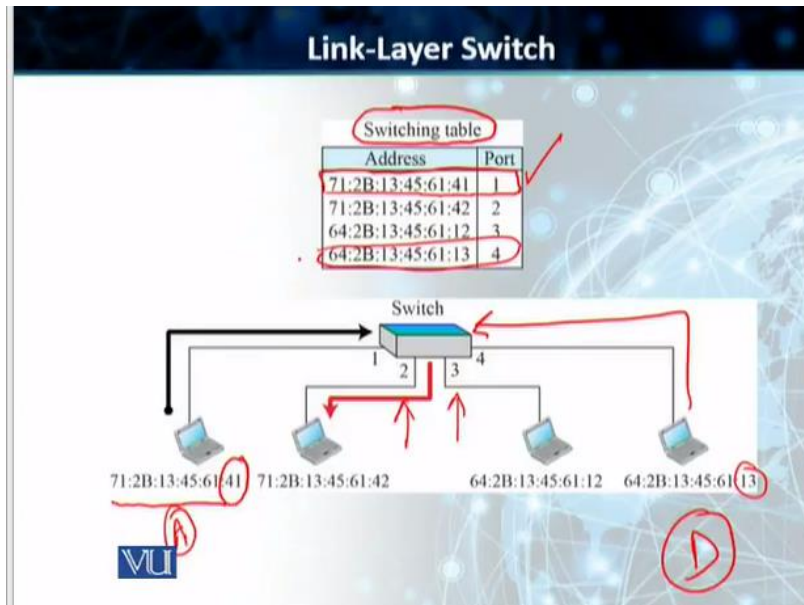
Switch has filtering capability.

Switch can check the destination address of frame and decide on which device it need to be sent.

Possibility of collision happening is approx. to zero so switch doesn't need to sense the carrier or medium

Switch can connect heterogenous devices.

Switch is an intelligent device.



Switch we are using is building an address table according to it.

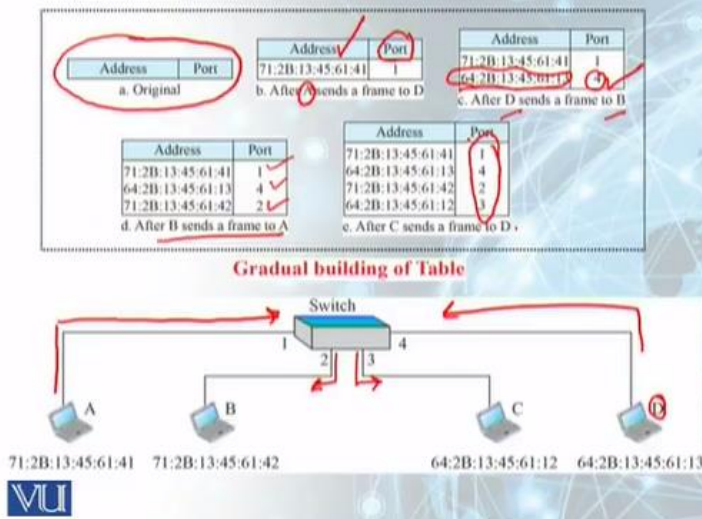
i.e. 1st device sends data to switch so as switch doesn't know whom to send then switch will send it all ports b/c in new sense switching table is empty and it will get address of all ports and save it for future.

If it got address of one device of 4 devices and 4th device sends data which is not for the 1st whose data is saved so switch will still send connection to other 2 ports and will save address of devices.

When switch got data of all ports then it will send data to specified address only.

Learning of switch

Learning Switch



After A sends frame to D it will save address of A and like this as communication will be kept on going it will get address all of the stations.

Issue with this process: if a station kept quite then switch will not be get address of that station but its quite rare

Loop Problem in Switch:

In modern data transfer redundant switches create loop in the system

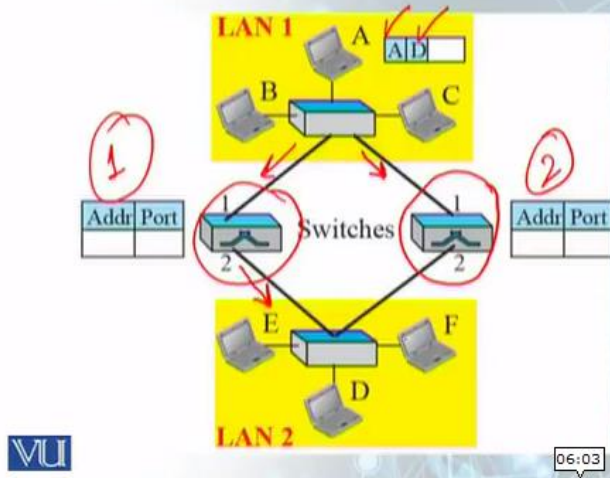
This issue creates when two or more broadcasting LANs are connected by more than one switch

We add redundancy by connecting more than one LANs

For connecting more than one LANs we need more switch and using more switch redundancy create.

Loop Problem in a Learning Switch (Part a)

a. Station A sends a frame to station D



Here as per figure.

Two LANs are connected to 2 switches.

These two switches connect to get redundancy if 1 switch fails so other one kept communication one.

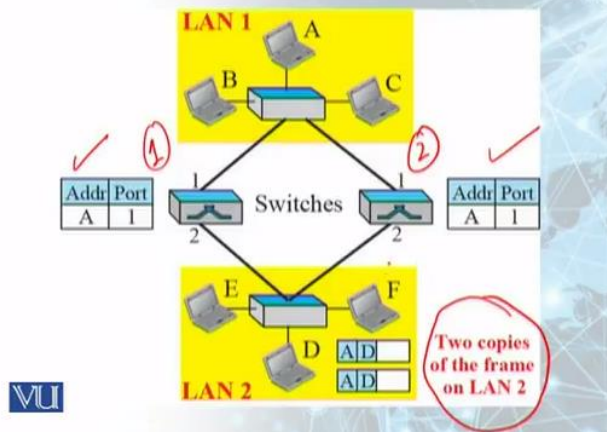
First switch got Address of A and D.

As switching table is empty and it will send to all ports.

PART B

Loop Problem in a Learning Switch (Part b)

b. Both switches forward the frame



After creating entry in Switching table.

As it doesn't have destination address of A so it will send address to all ports again that's why loop creates of learning period of switch.

Spanning Tree Algorithm:

In switching we discussed loop problem as switches in their learning period.

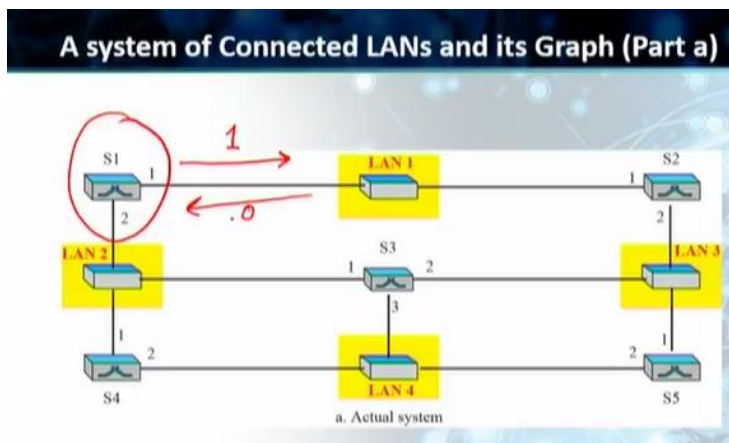
So solution to that problem is spanning tree algorithm.

Spanning tree is a graph in which there is no loop.

In a switched LAN, creating a topology in which each LAN can be reached from any other LAN by using one path only. As there is one path mean no Loop

We accomplish this mean no loop by assigning cost to each LAN link.

A System of Connected LANs and Its Graph (Part a)



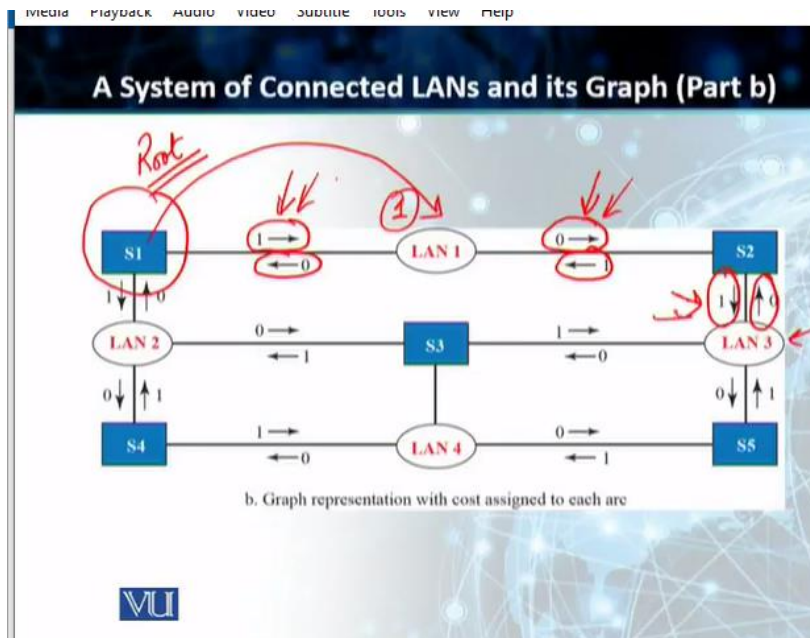
It has 4 LANs

In this case first we allocate cost metric to the each link. Cost will be upto system administrator what it sets and based on other available factors of line.

We need to have minimum hop to reach from 1 system to all other system.

Hop kind is 1 from switch to LAN and 0 in reverse condition.

Part B Represented in graph by spanning tree



We allocated cost metric of 1 and zero

Point in calculate topology to find shortest path btw root switch and all other destination

Let S1 is Root Switch.

Let shortest path from S1 to LAN 1 is 1

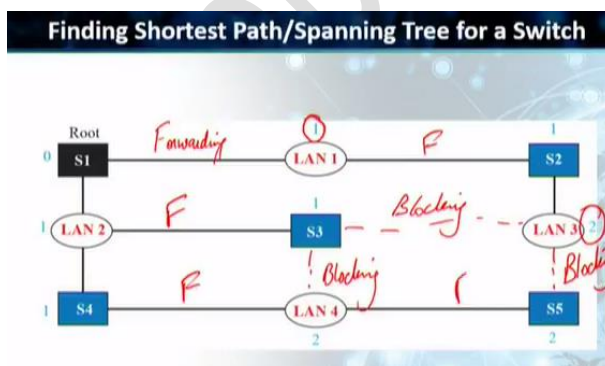
So Shortest path from S1 to S2 will be $1 + 0 = 1$

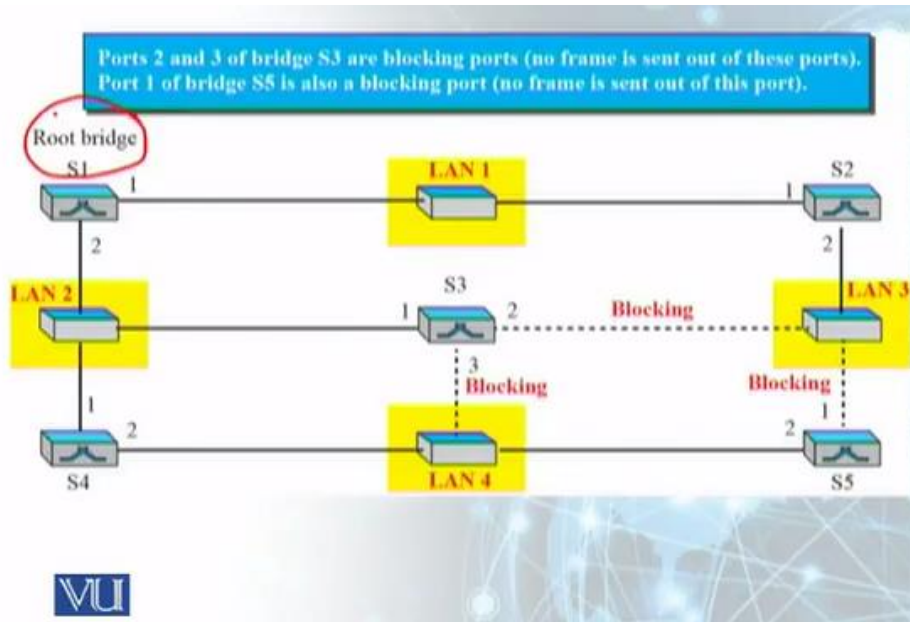
And from S1 to LAN 3 will be $1 + 0 + 1 = 2$

This topology is different from physical topology coz it has blocking and forwarding link here as well.

Links which are eliminated from switch are blocking link.

Links which are being used are forwarding link.





Routers:

It is another 3rd connecting device.

Router is layer 3 mean network layer device.

It operates on all 3 layers.

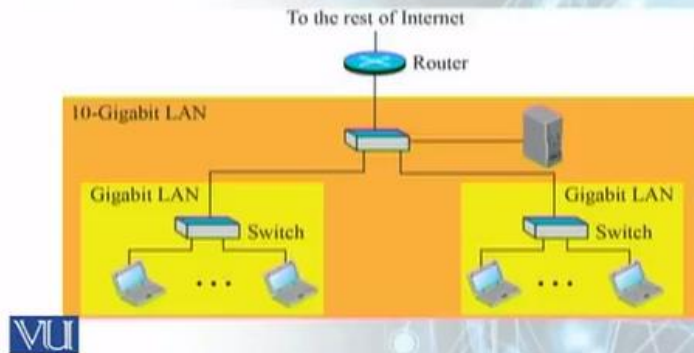
It operates in the physical, data link and network layers.

Router Vs. Switch:

Router vs. Switch

Three differences between a router and a repeater or a switch:

1. A router has a physical and logical (IP) address for each of its interfaces.
2. A router acts only on those packets in which the link-layer destination address matches the address of the interface at which the packet arrives.
3. A router changes the link-layer address of the packet (both source and destination) when it forwards the packet.



Here Switch are connected to Server and then that server is connected to Router so that's why its called layer 3 device as station, switches, servers are based on all layers are connected to router mean it uses all 3 layers.

Virtual LANS (VLANs):

LANS that exist in software not by physical wiring.

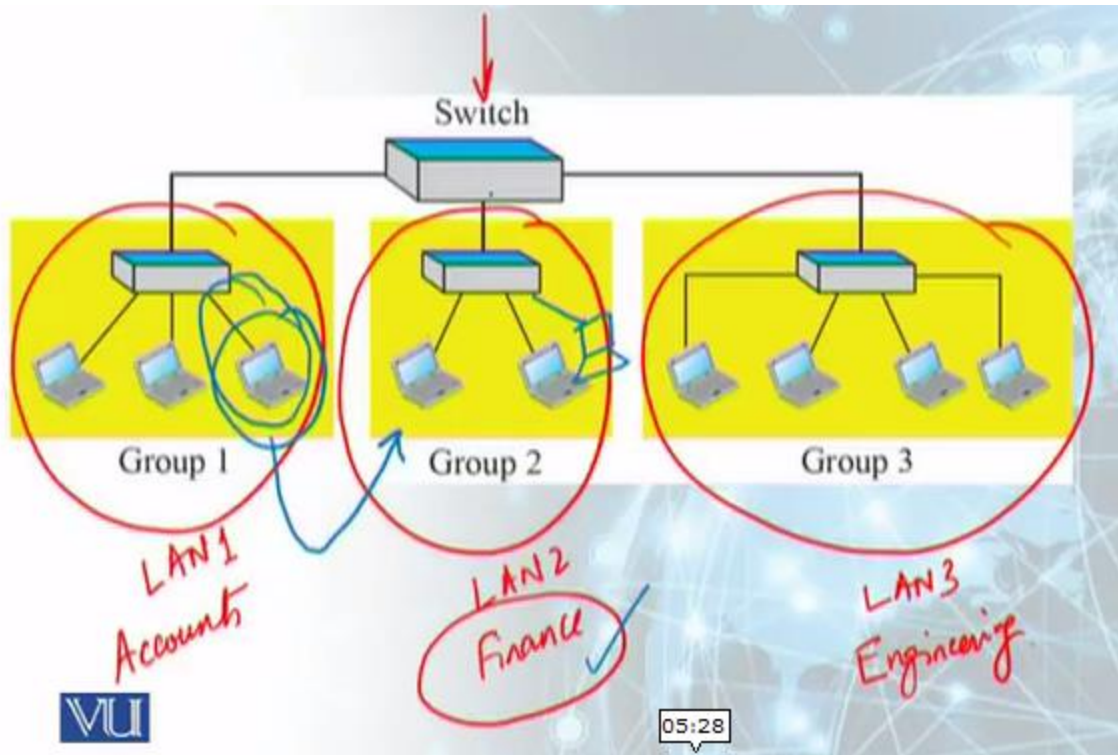
A station is considered to be part of LAN if it geographically connected to that LAN. i.e connected by piece of wire to any LAN.

VLAN provides virtual connected btw to LANS.

Here as in figure we have 3 groups connected to using switch to another switch

3 groups are LANS (LAN1, 2 & 3)

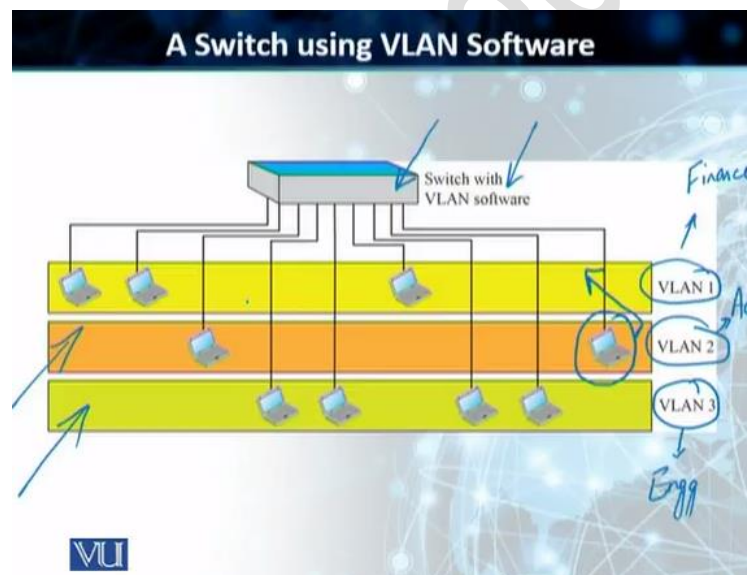
If any station want to be connected in Group 2 that time we have to physically reconnect by doing whole wiring.



But in

such problems VLAN helps us.

Here for this problem we use a particular software which is deployed on the switch by with all stations are connected. And if any station's position to be changed we can simply doing in software and after changing it will be able to access other group easily.

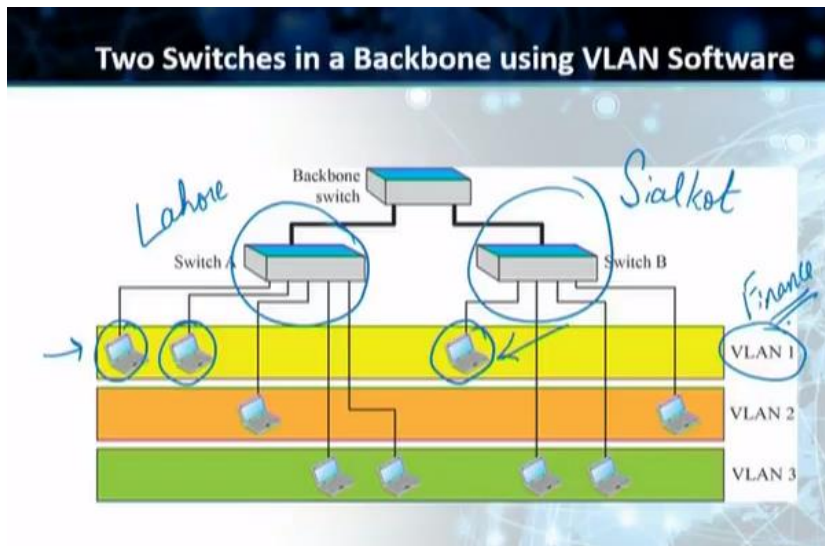


We can connect multiple switches with VLAN as well.

In which some user could be part of switch A and Switch B.

For this case we must have a backbone switch connected both switches.

So we still can give access by just doing VLAN changes.



Membership of VLAN:

What characteristics can be used to group a station in VLAN.

There could be multiple criteria.

We can have Interface numbers, or Port number. System with similar port can be part.

Of MAC address we can provide access to that mac or IP address etc...

Its based on what method we obtain.

Stations are configured in three ways:

- Manually
- Semi-Automatically
- Automatically.

Manually: we have software in this case and we chose users manually to give them access.

Semi-Automatically: is btw manual and automatic here we can define our rules etc.

Automatically: In which software is that will do all part. If port number of changes for any user it will automatically changes.

Communication Between Switches:

In Multi-switched backbone, each switch must know.

- Which station belongs to which VLAN
- Memberships of the stations that is connected to VLAN

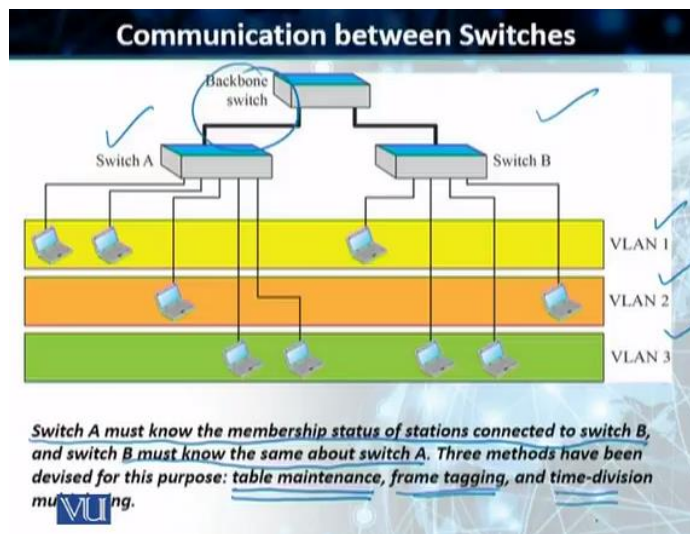
Here in this case

Here switch A must know the membership status of Station B and vice versa

Table maintenance: is a technique to maintain switches. When broadcast is done that switch make entry in table.

Frame tagging: frame is tagging btw the switches an extra header gets added to MAC frame to define the VLAN.

Time Division Multiplex: connection btw switches is divided through Time shared channel.



Advantage of VLANS

- 1) Cost and Time Reduction
- 2) Creating Virtual Workgroups: Can be used in short distance and Long distance as we..
- 3) Security: user will in one particular VLAN will have access to that particular VLAN only.

Comparison of Modern Access Technologies.

- 1) Telco
- 2) HFC
- 3) FTTx

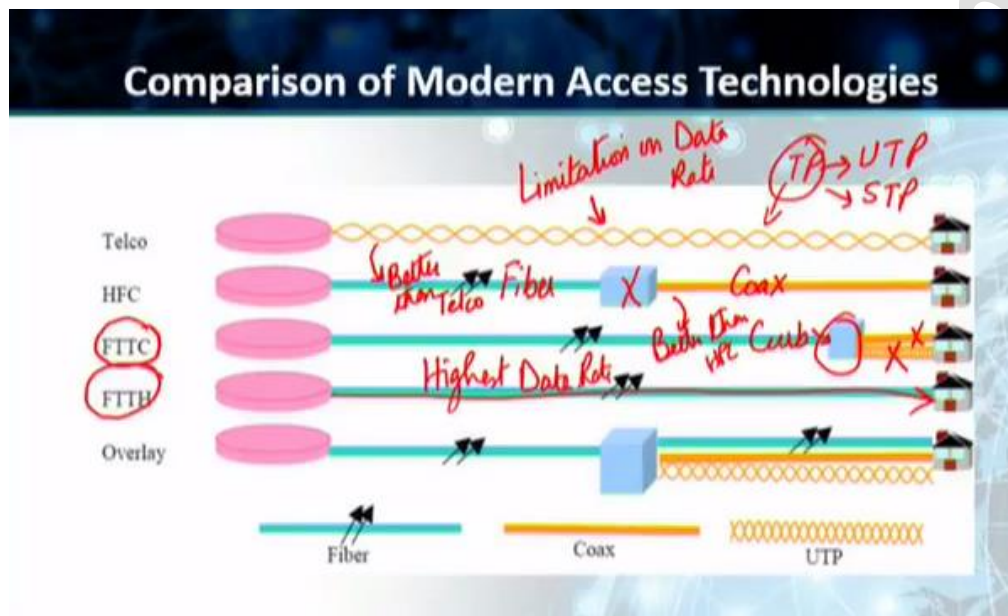
Telco: used in Telecommunication all the way provided from provider to home its all the way twisted pair. As its twisted pair we have limitation in data rate.

HFC: in we using fiber to connection and from that connection to home we uses Coax have better data rate than telco

FTTC: in which Fiber goes all the way to curb and at the last point we used to have twisted pair to our home.

FTTH: in which the Fiber is being used from Provided office all the way to home. It provide the fastest speed and fastest data rate.

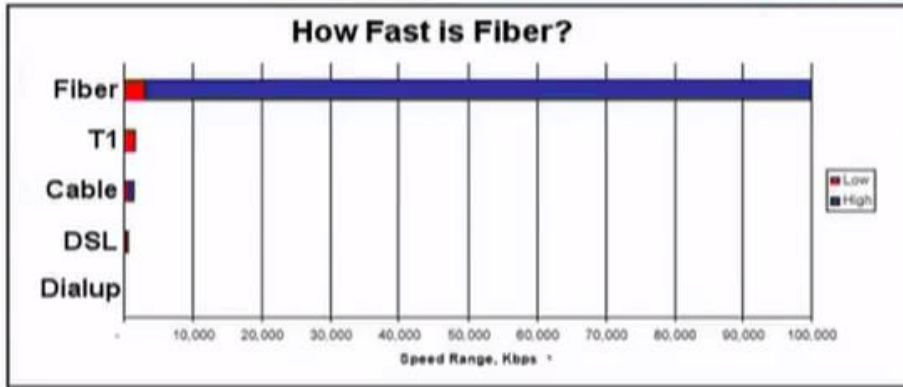
Overlay:



Fiber is future of Internet

Its speed can be considered here. Other technologies no where near to fiber.

Fiber – The Medium of the Future!



Fiber to The Curb (FTTC)

In which Fiber is used for a long part but not entire link from provider to end user.

From Optical to electrical signals conversion takes place somewhere near the end user.

Terminal segment of FTTC network is usually twisted pair or coaxial cable.

Final Optical Receiver in FTTC network typically serves several customers. Mean at the end point of fiber there is a receiver that is providing connection to multiple users.

Here we have Optical Fiber unit ONU

We have fiber and then we have ONU and that onu further give connection to users but connection till the ONU is fiber

ONU is providing data connection for both telephone and for data so we can separate connection by using a splitter and connect both data and phone at a time.

Fiber to The Home FTTH:

To get High Speed Data, Reliable voice and High quality Video FTTH is introduced.

Home (FTTH)

- Need: High-speed data, reliable voice and high-quality video
- Problems:
 - ✓ How to get high speed lines out to each customer?
 - ✓ How to future-proof the architecture?
- Solution: FTTH

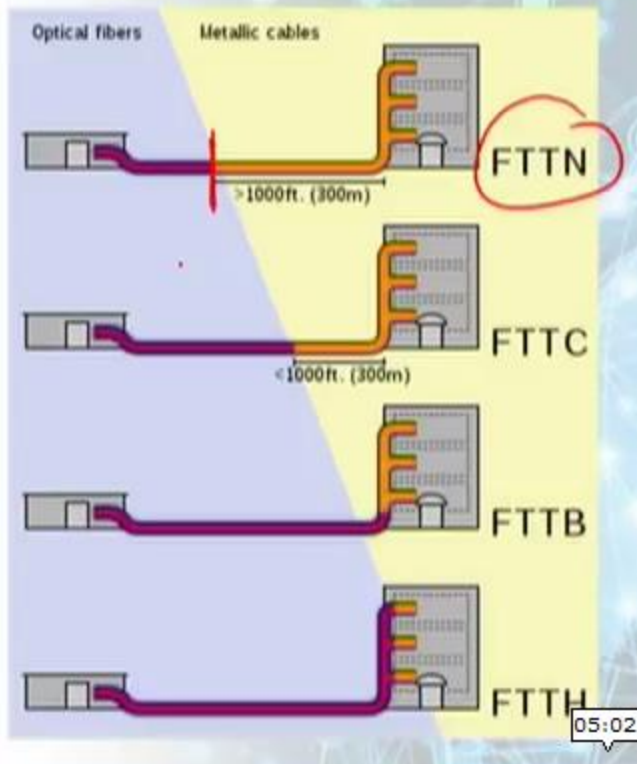
FTTH is installation of fiber all the way from provider to the subscriber.

It is the one of the latest access technologies

FTTH is also referred to as fiber to the building FTTB:

FTTH:

Fiber To The Home (FTTH)



We got fiber to the Node in the start:

Fiber is used within the 1000 ft if it extend then we prefer to use metallic cables (twisted or coax)

FTTC: in FTTC Fiber is used lesser than 1000ft then we lesser than 300m that is covered by twisted of coax

FTTB: in which we have Fiber all the way to building and then we use metallic within the building

FTTH: in which we got Fiber all the way to the home. It could be separated if we use anytime of wireless or we use any other wire...

Prepared by Hamza

Prepared by Hamza